

Electronic Supplementary Information (ESI)

Machine learning-guided design and synthesis of benzimidazole-based HDAC inhibitors:
prospective validation of a fine-tuned pretrained graph transformer

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S1 HPLC purity of the final compounds

The purity of the six final hydroxamic acids IXa–IXf was determined by reversed-phase HPLC on a Shimadzu chromatograph running LabSolutions, fitted with an RP-8 column (250×4.6 mm, $5 \mu\text{m}$) and a photodiode-array detector set to 280 nm (4 nm bandwidth). The mobile phase was methanol/water (70:30, v/v), delivered isocratically at 1.0 mL min^{-1} . The injection volume was 10 μL , except for IXa (5 μL). Purity is reported as the percentage peak area at 280 nm. All six compounds exceeded 99% purity.

Sample codes throughout this document are the internal identifiers printed on the instrument reports. TA2-A, TA2-B and TA2-C correspond to IXa, IXb and IXc, and TA1-A, TA1-B and TA1-C to IXd, IXe and IXf. This correspondence is the corrected one established by the HSQC, HMBC and ROESY spectra (Sections S6–S8). It reverses the regiochemistry originally assigned to the two sets of samples.

Table S1. HPLC purity and retention times of the six series IX final compounds, determined by reversed-phase HPLC with photodiode-array detection at 280 nm. Purity is the percentage of total peak area. Sample codes are the identifiers used in the raw instrument reports.

Compound	Sample code	Purity (% area)	t_{R} (min)
IXa	TA2-A	99.64	6.19
IXb	TA2-B	99.63	6.39
IXc	TA2-C	99.95	6.26
IXd	TA1-A	99.87	6.28
IXe	TA1-B	99.11	6.43
IXf	TA1-C	99.50	6.32

In every chromatogram the product elutes as a single dominant peak at 6.19–6.43 min, accompanied by one minor peak at 3.96–4.12 min accounting for 0.05–0.89% of the total area. This minor peak appears at a comparable retention time in all six runs.

The individual analysis reports, each comprising the chromatogram and the integrated peak table, are reproduced in Figures S1–S6.

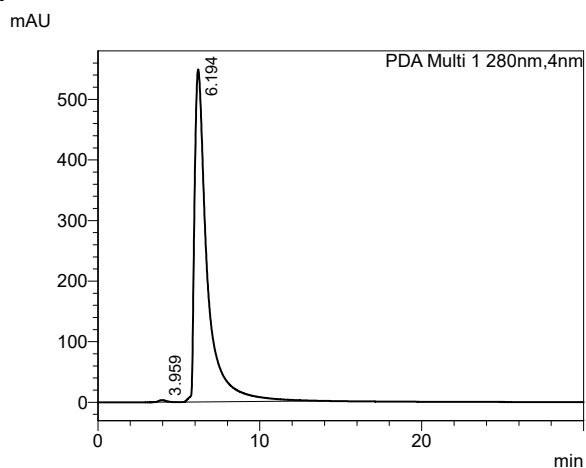


Analysis Report

<Sample Information>

Sample Name	: TA2-A	Sample Type	: Unknown
Sample ID	: TA2-A	Acquired by	: System Administrator
Data Filename	: TA2-A 2.lcd	Processed by	: System Administrator
Method Filename	: test tap TA-MD.lcm		
Batch Filename	: 260722-TA.lcb		
Vial #	: 1-5		
Injection Volume	: 5 uL		
Date Acquired	: 7/22/2026 5:43:56 PM		
Date Processed	: 7/22/2026 6:13:57 PM		

<Chromatogram>



<Peak Table>

PDA Ch1 280nm							
Peak#	Ret. Time	Area	Height	Conc.	Unit	Mark	Name
1	3.959	109195	3447	0.361			
2	6.194	30141282	548863	99.639			
Total		30250477	552310				

E:\co thuan\MD\TA2-A 2.lcd

Figure S1. HPLC analysis report for IXa (sample code TA2-A): 99.64% purity, $t_R = 6.19$ min.



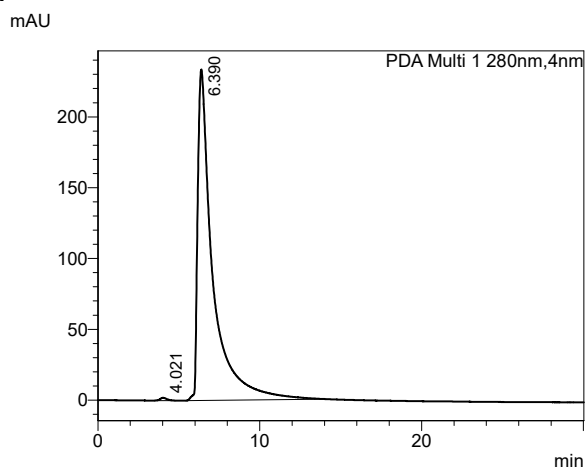
Analysis Report

<Sample Information>

Sample Name : TA2-B
 Sample ID : TA2-B
 Data Filename : TA2-B.lcd
 Method Filename : test tap TA-MD.lcm
 Batch Filename : 260722-TA.lcb
 Vial # : 1-6
 Injection Volume : 10 uL
 Date Acquired : 7/22/2026 3:42:28 PM
 Date Processed : 7/22/2026 5:23:37 PM

Sample Type : Unknown
 Acquired by : System Administrator
 Processed by : System Administrator

<Chromatogram>



<Peak Table>

PDA Ch1 280nm

Peak#	Ret. Time	Area	Height	Conc.	Unit	Mark	Name
1	4.021	56000	1952	0.369			
2	6.390	15135111	233680	99.631		S	
Total		15191111	235633				

E:\co thuan\MD\TA2-B.lcd

Figure S2. HPLC analysis report for IXb (sample code TA2-B): 99.63% purity, $t_R = 6.39$ min.



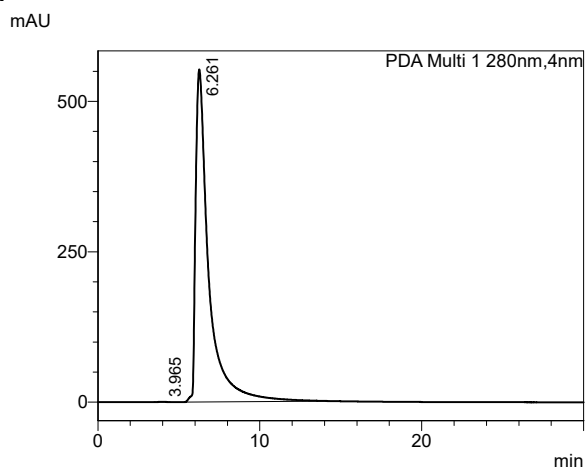
Analysis Report

<Sample Information>

Sample Name : TA2-C
 Sample ID : TA2-C
 Data Filename : TA2-C.lcd
 Method Filename : test tap TA-MD.lcm
 Batch Filename : 260722-TA.lcb
 Vial # : 1-7
 Injection Volume : 10 uL
 Date Acquired : 7/22/2026 4:12:55 PM
 Date Processed : 7/22/2026 4:42:57 PM

Sample Type : Unknown
 Acquired by : System Administrator
 Processed by : System Administrator

<Chromatogram>



<Peak Table>

PDA Ch1 280nm							
Peak#	Ret. Time	Area	Height	Conc.	Unit	Mark	Name
1	3.965	16489	597	0.053			
2	6.261	31096815	552752	99.947		S	
Total		31113304	553349				

E:\co thuan\MD\TA2-C.lcd

Figure S3. HPLC analysis report for IXc (sample code TA2-C): 99.95% purity, $t_R = 6.26$ min.



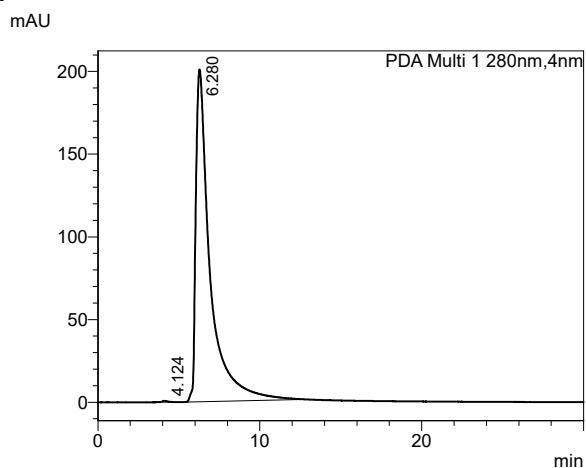
Analysis Report

<Sample Information>

Sample Name : TA1-A
 Sample ID : TA1-A
 Data Filename : TA1-A.lcd
 Method Filename : test tap TA-MD.lcm
 Batch Filename : 260722-TA.lcb
 Vial # : 1-2
 Injection Volume : 10 uL
 Date Acquired : 7/22/2026 1:10:15 PM
 Date Processed : 7/22/2026 5:16:51 PM

Sample Type : Unknown
 Acquired by : System Administrator
 Processed by : System Administrator

<Chromatogram>



<Peak Table>

PDA Ch1 280nm

Peak#	Ret. Time	Area	Height	Conc.	Unit	Mark	Name
1	4.124	16329	767	0.132			
2	6.280	12333415	200905	99.868		S	
Total		12349744	201672				

E:\co thuan\MD\TA1-A.lcd

Figure S4. HPLC analysis report for IXd (sample code TA1-A): 99.87% purity, $t_R = 6.28$ min.



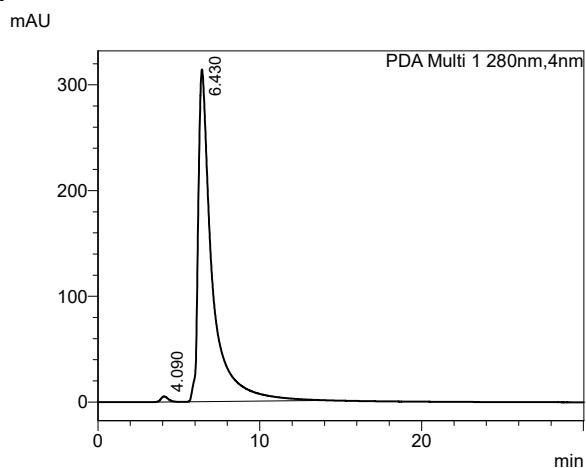
Analysis Report

<Sample Information>

Sample Name : TA1-B
 Sample ID : TA1-B
 Data Filename : TA1-B.lcd
 Method Filename : test tap TA-MD.lcm
 Batch Filename : 260722-TA.lcb
 Vial # : 1-3
 Injection Volume : 10 uL
 Date Acquired : 7/22/2026 1:40:42 PM
 Date Processed : 7/22/2026 5:18:41 PM

Sample Type : Unknown
 Acquired by : System Administrator
 Processed by : System Administrator

<Chromatogram>



<Peak Table>

PDA Ch1 280nm

Peak#	Ret. Time	Area	Height	Conc.	Unit	Mark	Name
1	4.090	170068	5343	0.892			
2	6.430	18891108	314217	99.108			
Total		19061176	319560				

E:\co thuan\MD\TA1-B.lcd

Figure S5. HPLC analysis report for IXe (sample code TA1-B): 99.11% purity, $t_R = 6.43$ min.



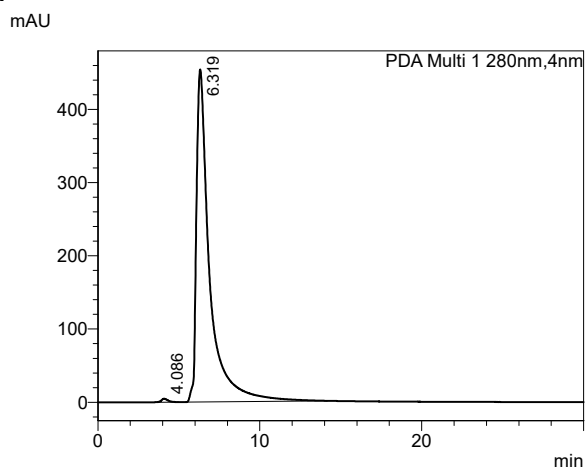
Analysis Report

<Sample Information>

Sample Name : TA1-C
 Sample ID : TA1-C
 Data Filename : TA1-C.lcd
 Method Filename : test tap TA-MD.lcm
 Batch Filename : 260722-TA.lcb
 Vial # : 1-4
 Injection Volume : 10 uL
 Date Acquired : 7/22/2026 2:11:08 PM
 Date Processed : 7/22/2026 5:19:48 PM

Sample Type : Unknown
 Acquired by : System Administrator
 Processed by : System Administrator

<Chromatogram>



<Peak Table>

PDA Ch1 280nm

Peak#	Ret. Time	Area	Height	Conc.	Unit	Mark	Name
1	4.086	130298	4717	0.503			
2	6.319	25776529	454228	99.497			
Total		25906826	458945				

E:\co thuan\MD\TA1-C.lcd

Figure S6. HPLC analysis report for IXf (sample code TA1-C): 99.50% purity, $t_R = 6.32$ min.

S2 HRMS spectra

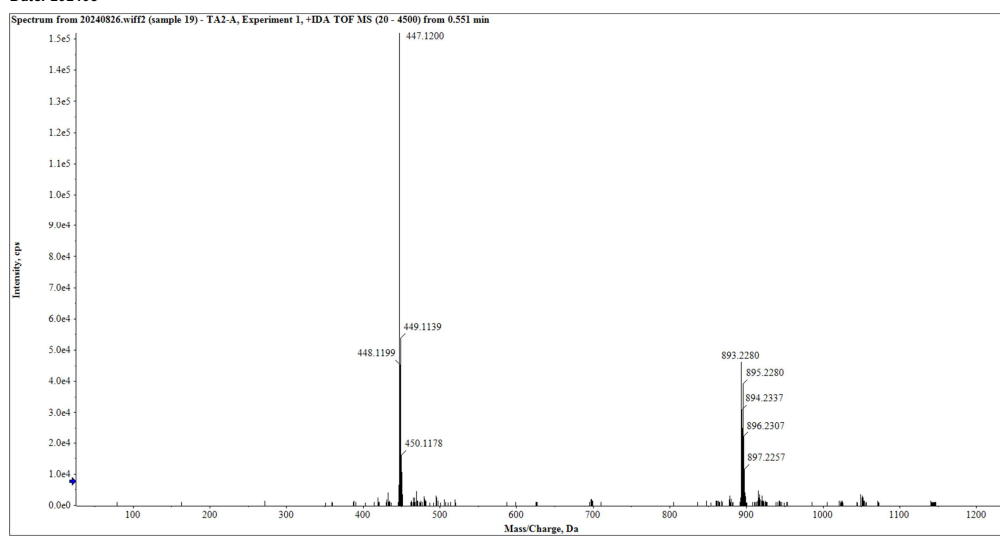
High-resolution mass spectra were recorded on a SCIEX X500 QTOF instrument in positive-ion mode using an information-dependent acquisition TOF MS/MS method over m/z 20–4500. The full-scan spectrum of each compound is reproduced below; in every case the base peak corresponds to the $[M+H]^+$ ion quoted in the characterisation data. Only the survey scan is reproduced here. Sample codes are as in Table [S1](#).

Sample name: TA2-A

Operator: Hằng VHH

Method: +IDA TOF MS/MS

Date: 202408



Device Model: SCIEX X500 QTOF

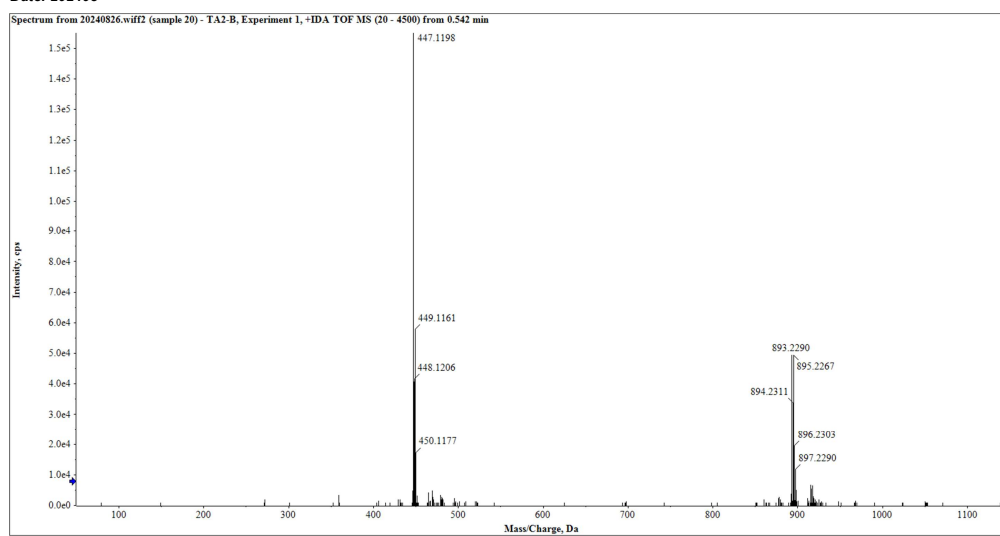
Figure S7. HRMS spectrum of IXa (sample code TA2-A); $[M+H]^+$ found 447.1200.

Sample name: TA2-B

Operator: Hằng VHH

Method: +IDA TOF MS/MS

Date: 202408



Device Model: SCIEX X500 QTOF

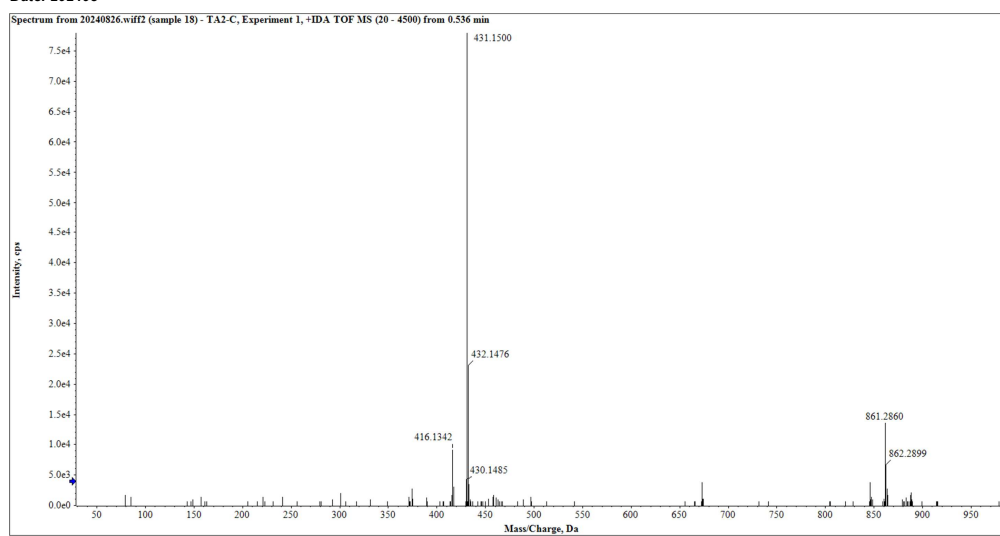
Figure S8. HRMS spectrum of IXb (sample code TA2-B); $[M+H]^+$ found 447.1198.

Sample name: TA2-C

Operator: Hằng VHH

Method: +IDA TOF MS/MS

Date: 202408



Device Model: SCIEX X500 QTOF

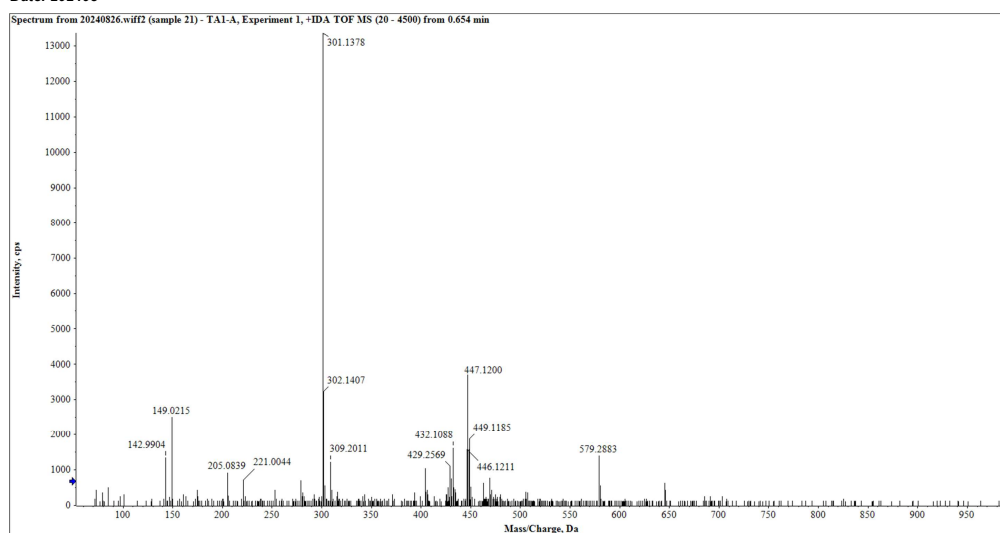
Figure S9. HRMS spectrum of IXc (sample code TA2-C); $[M+H]^+$ found 431.1500.

Sample name: TA1-A

Operator: Hằng VHH

Method: +IDA TOF MS/MS

Date: 202408



Device Model: SCIEX X500 QTOF

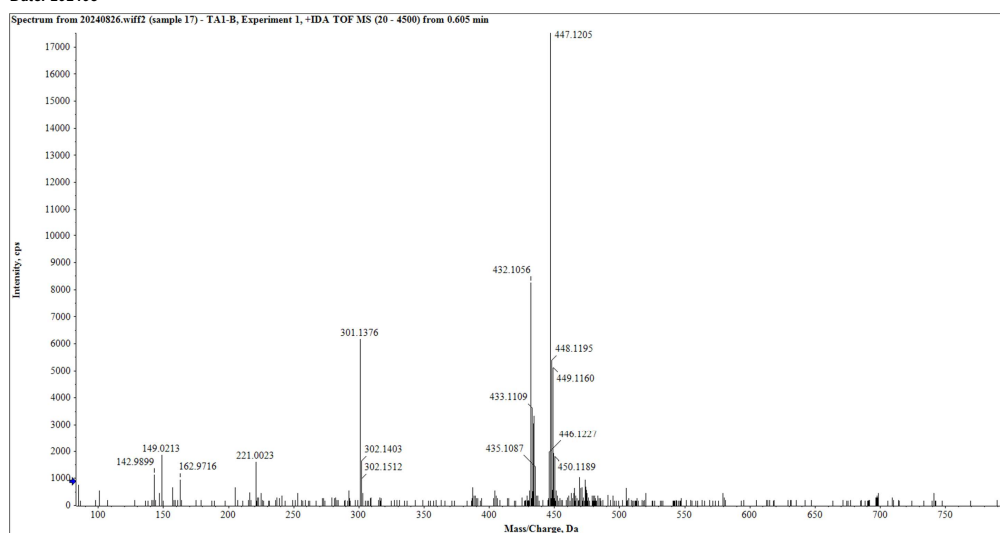
Figure S10. HRMS spectrum of IXd (sample code TA1-A); $[M+H]^+$ found 447.1200.

Sample name: TA1-B

Operator: Hằng VHH

Method: +IDA TOF MS/MS

Date: 202408



Device Model: SCIEX X500 QTOF

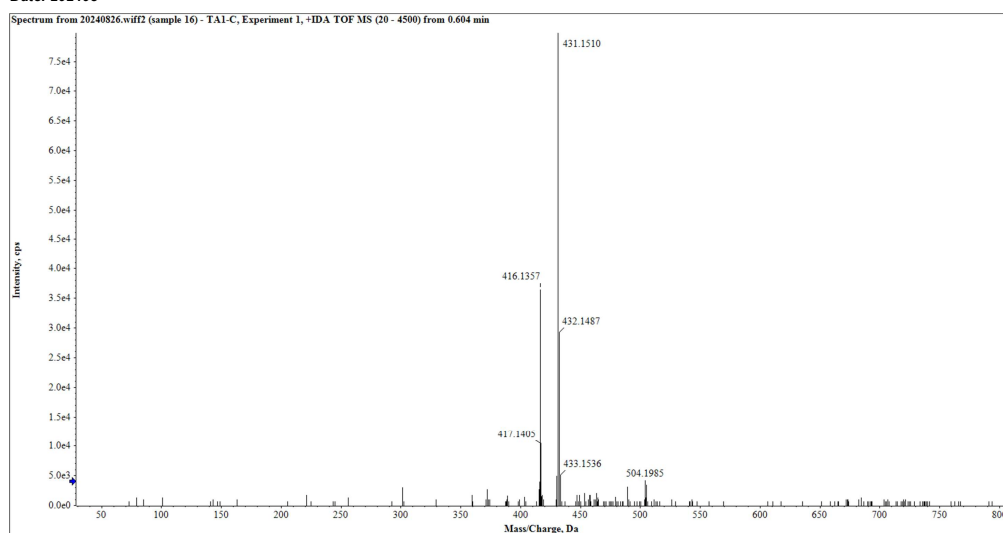
Figure S11. HRMS spectrum of IXe (sample code TA1-B); $[M+H]^+$ found 447.1205.

Sample name: TA1-C

Operator: Hằng VHH

Method: +IDA TOF MS/MS

Date: 202408



Device Model: SCIEX X500 QTOF

Figure S12. HRMS spectrum of IXf (sample code TA1-C); $[M+H]^+$ found 431.1510.

S3 Infrared spectra

IR spectra were recorded in transmission mode over 400–4000 cm^{-1} on a Shimadzu IRAffinity-1S spectrometer (20 scans, resolution 4 cm^{-1}), with the samples ground and pressed into potassium bromide discs. Each figure reproduces the instrument report. Sample codes are as in Table [S1](#).

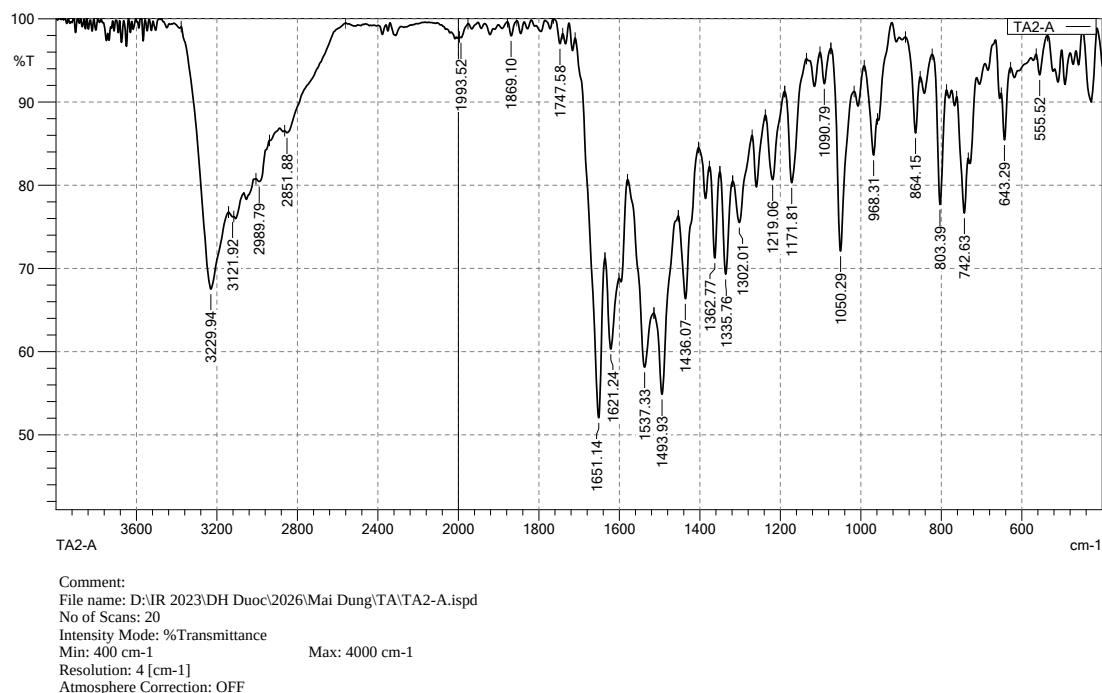


Figure S13. IR spectrum (KBr) of IXa (sample code TA2-A).

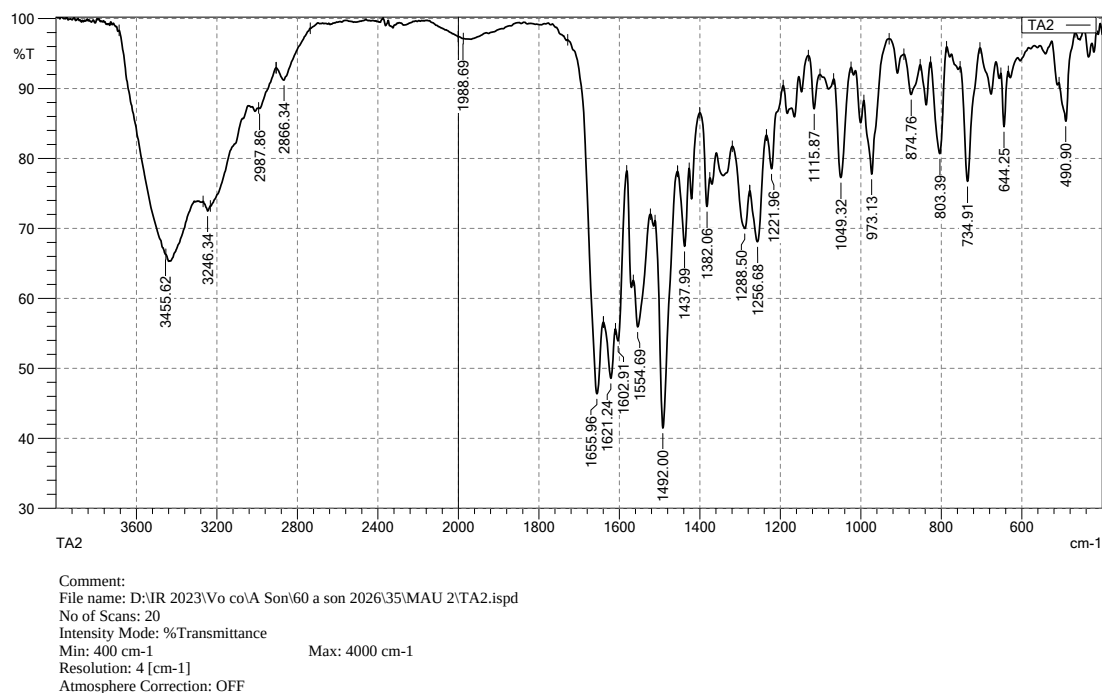


Figure S14. IR spectrum (KBr) of IXb (sample code TA2-B).

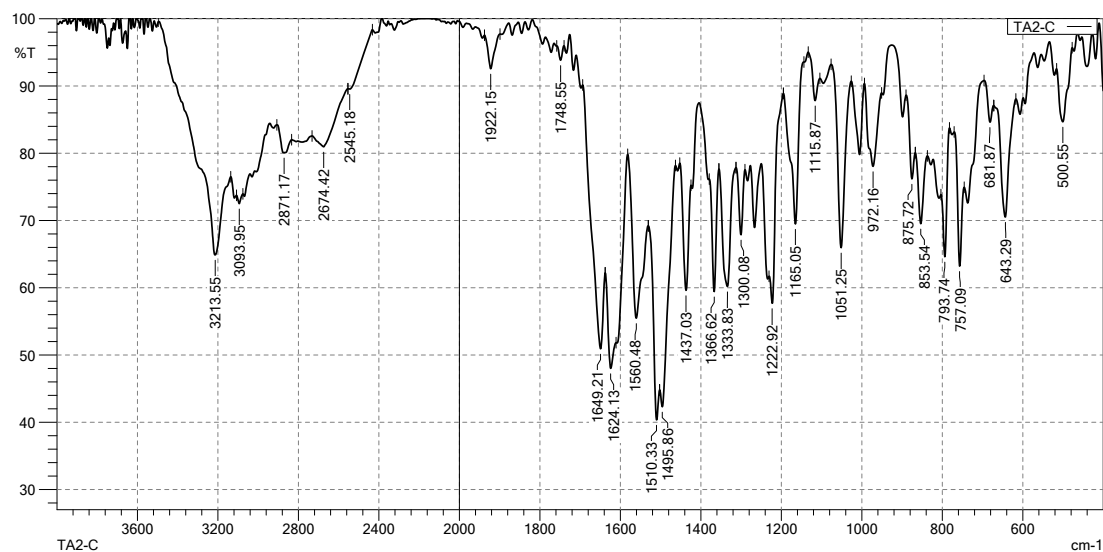


Figure S15. IR spectrum (KBr) of IXc (sample code TA2-C).

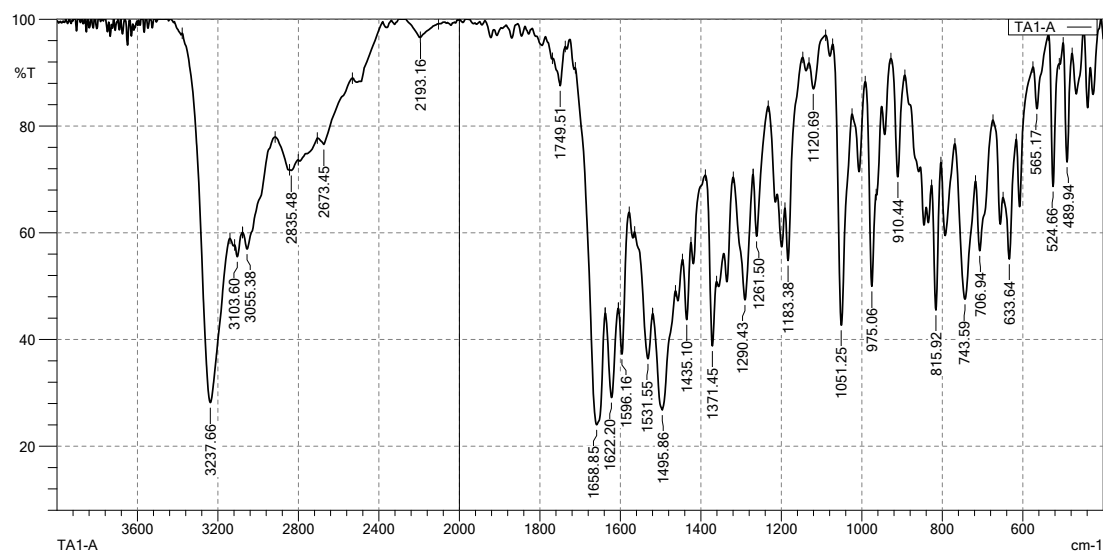


Figure S16. IR spectrum (KBr) of IXd (sample code TA1-A).

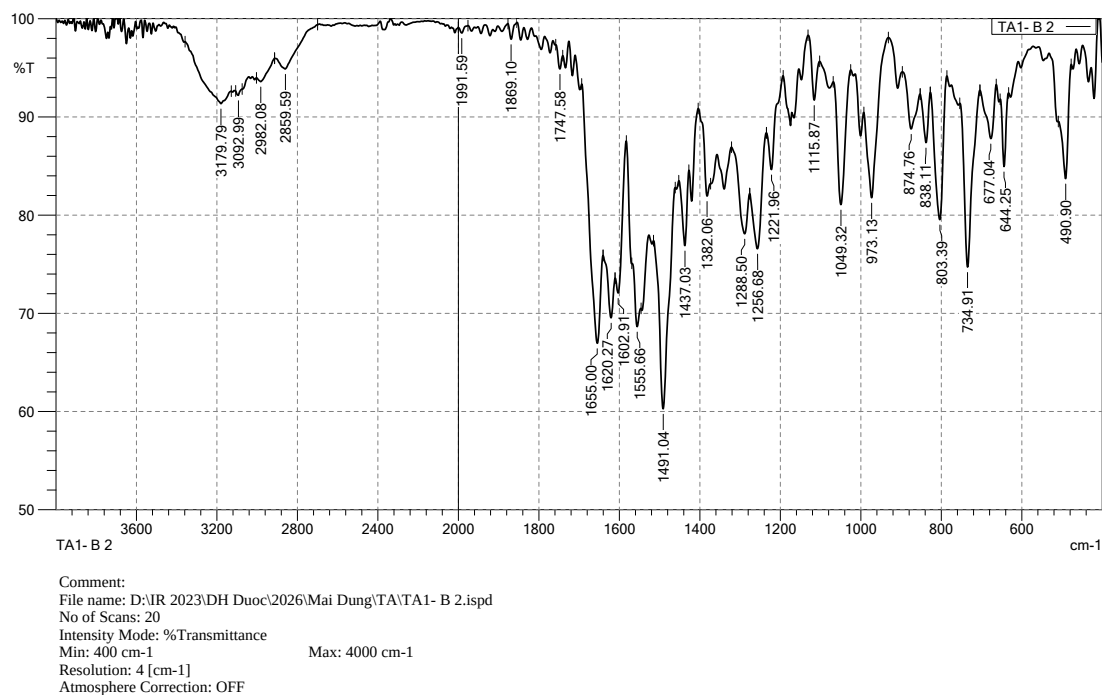


Figure S17. IR spectrum (KBr) of IXe (sample code TA1-B).

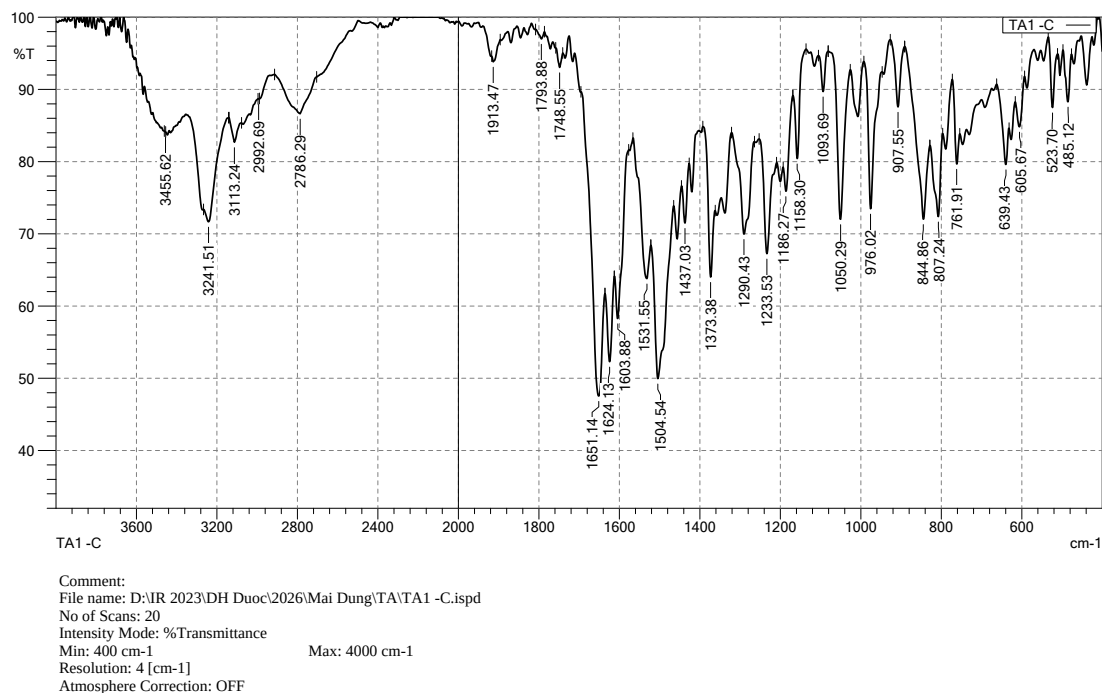


Figure S18. IR spectrum (KBr) of IXf (sample code TA1-C).

S4 ^1H NMR spectra

^1H NMR spectra were recorded in $\text{DMSO-}d_6$ on a Bruker AVANCE NEO spectrometer at 600 MHz, with chemical shifts referenced to the residual solvent signal. The spectra are reproduced in the order IXa–IXf. Locants follow IUPAC numbering, with N1 carrying the benzyl group. The sample code printed on each trace is the internal identifier (Table [S1](#)).

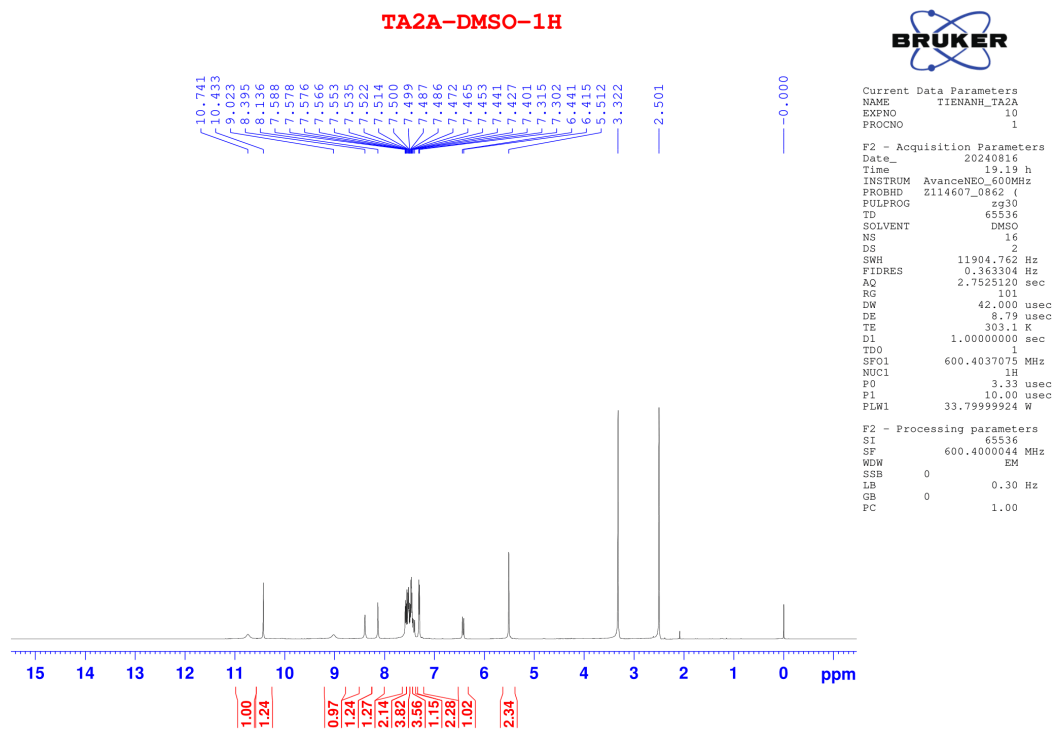


Figure S19. ^1H NMR spectrum (600 MHz, $\text{DMSO}-d_6$) of IXa (sample code TA2-A).

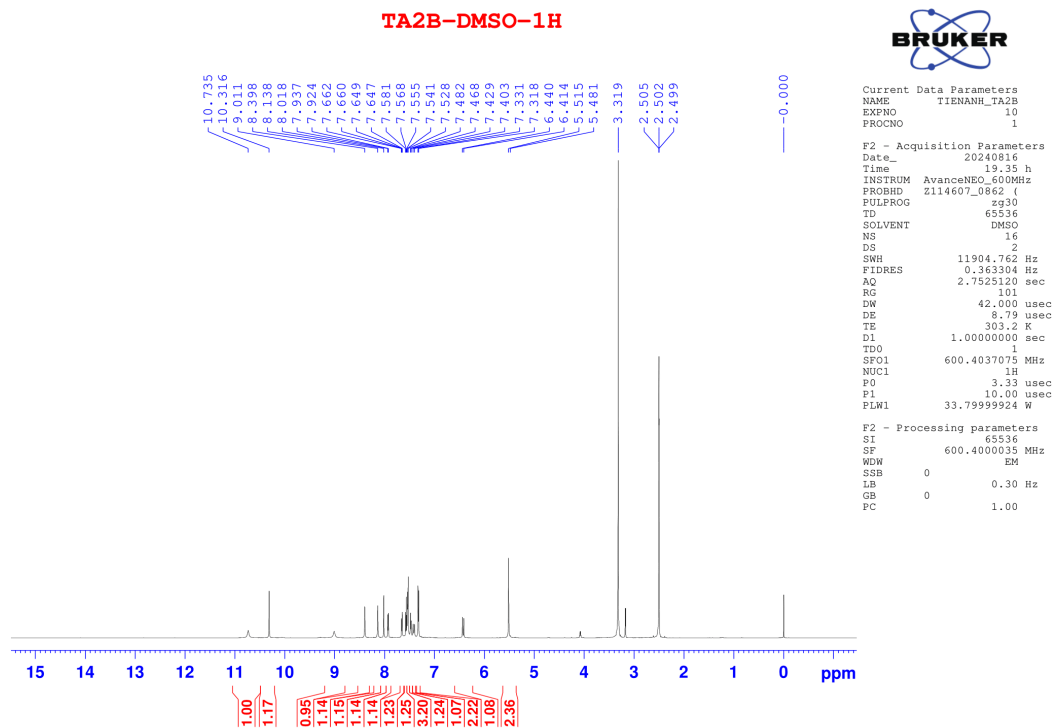


Figure S20. ^1H NMR spectrum (600 MHz, $\text{DMSO}-d_6$) of IXb (sample code TA2-B).

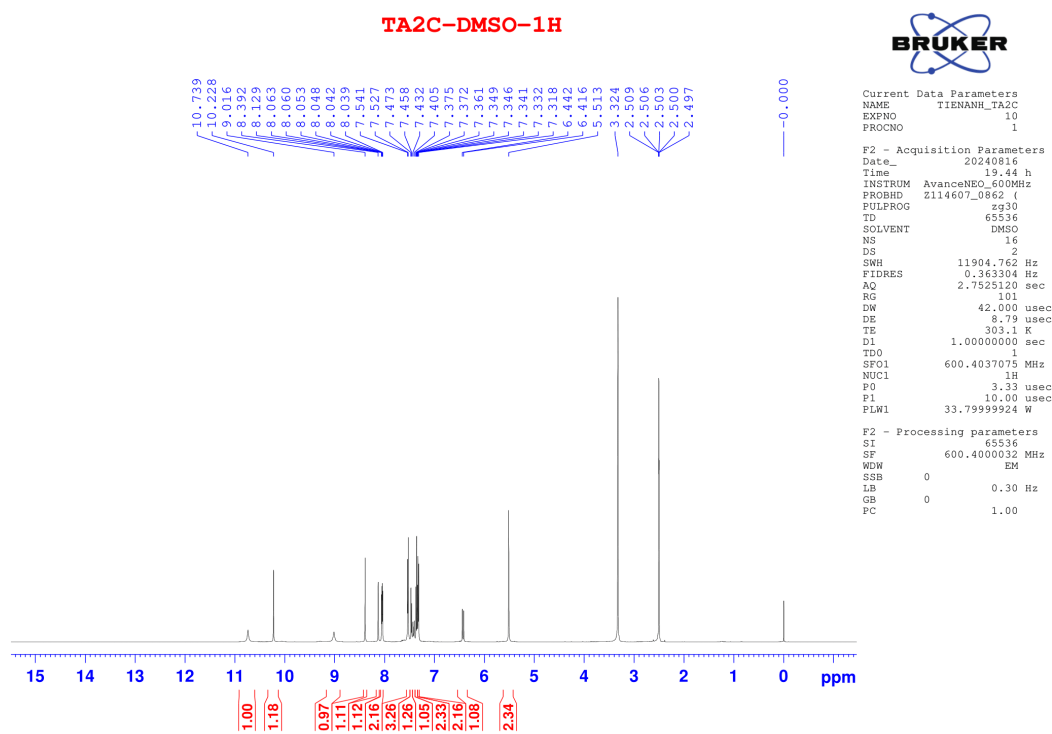


Figure S21. ^1H NMR spectrum (600 MHz, $\text{DMSO-}d_6$) of IXc (sample code TA2-C).

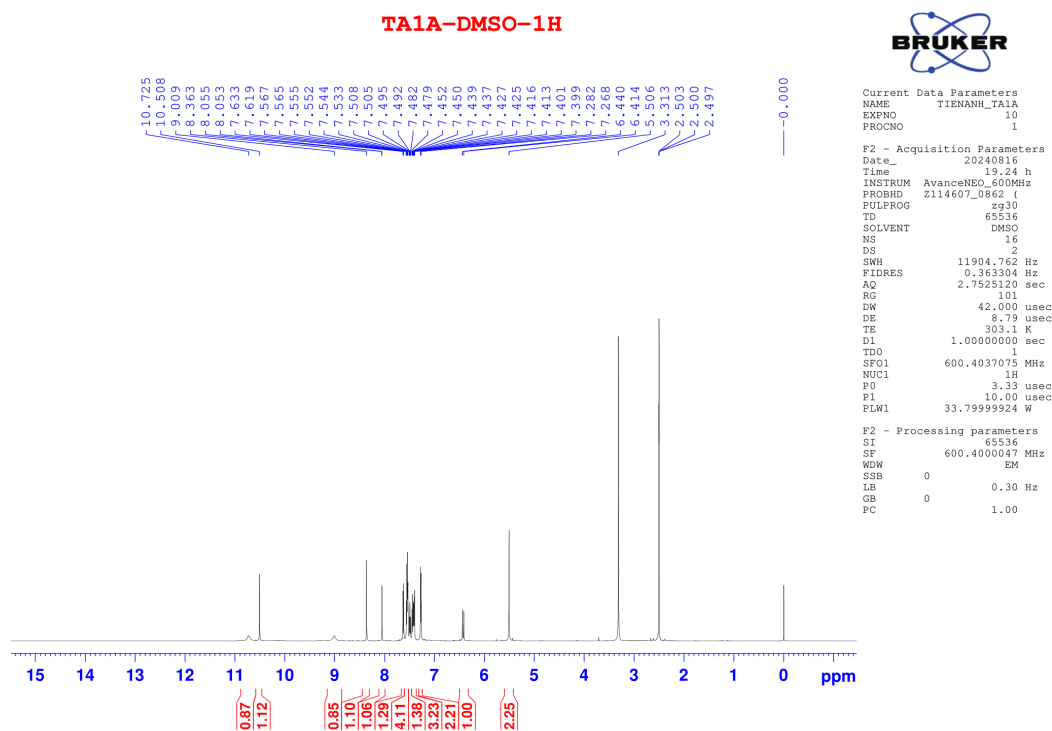


Figure S22. ^1H NMR spectrum (600 MHz, $\text{DMSO-}d_6$) of IXd (sample code TA1-A).

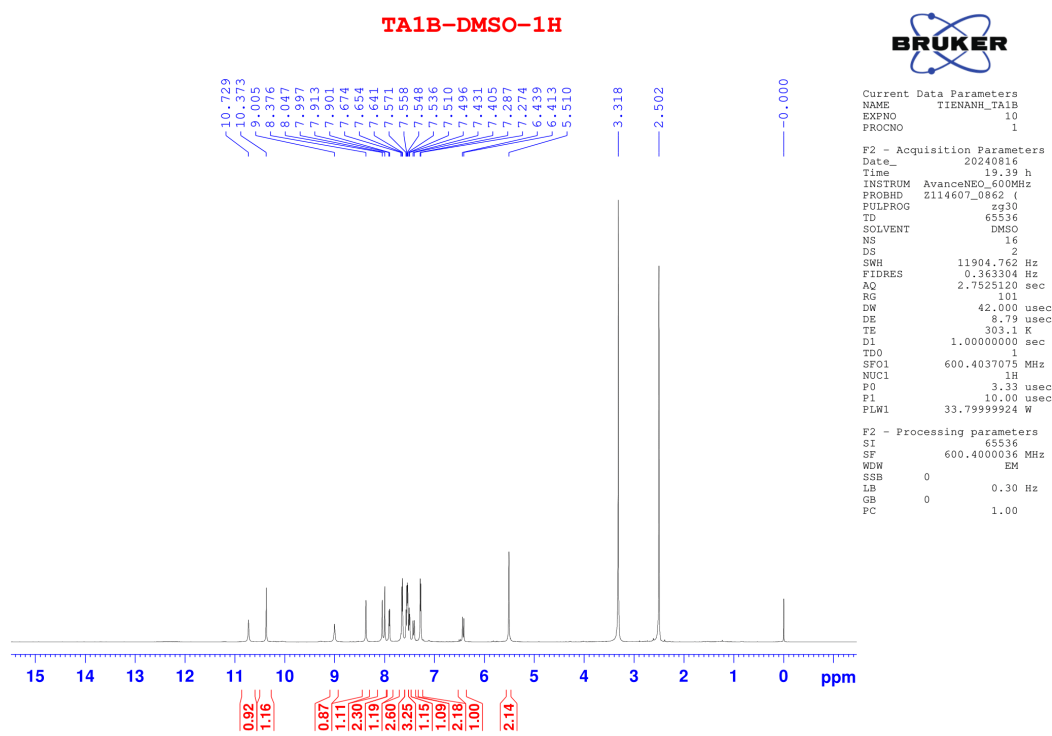


Figure S23. ^1H NMR spectrum (600 MHz, $\text{DMSO-}d_6$) of IXe (sample code TA1-B).

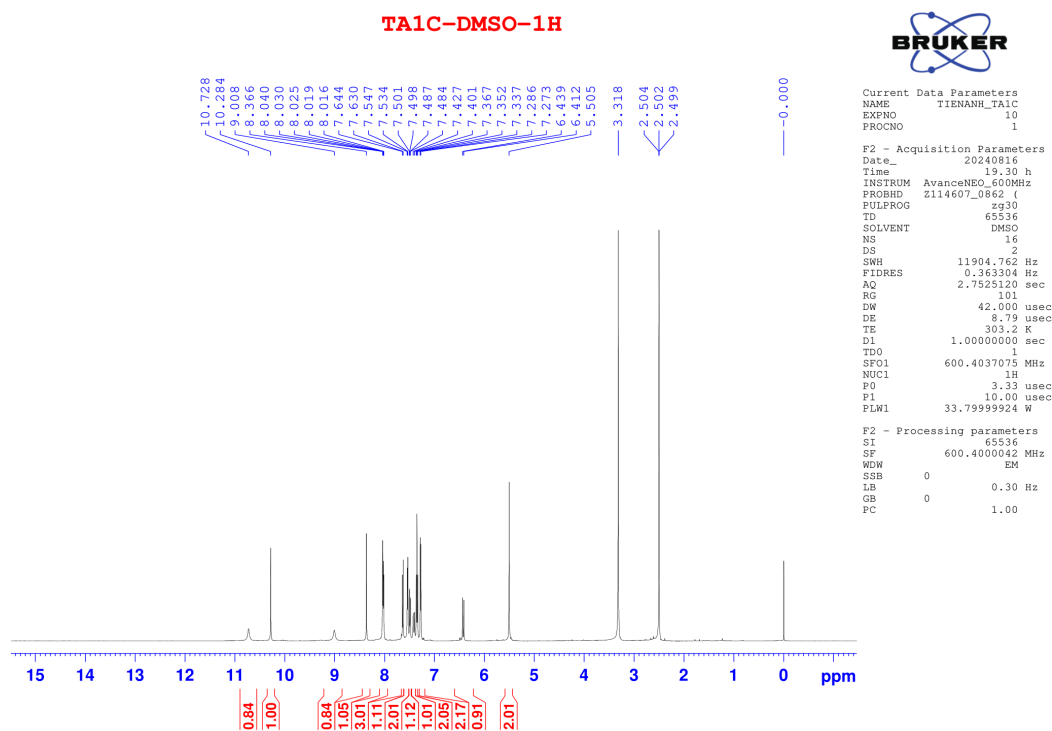


Figure S24. ^1H NMR spectrum (600 MHz, $\text{DMSO-}d_6$) of IXf (sample code TA1-C).

S5 ^{13}C NMR spectra

^{13}C NMR spectra were recorded in DMSO- d_6 at 151 MHz on the same instrument, except those of IXb and IXe, recorded at 126 MHz on a Bruker 500 MHz spectrometer. All were acquired with composite-pulse decoupling (CPD).

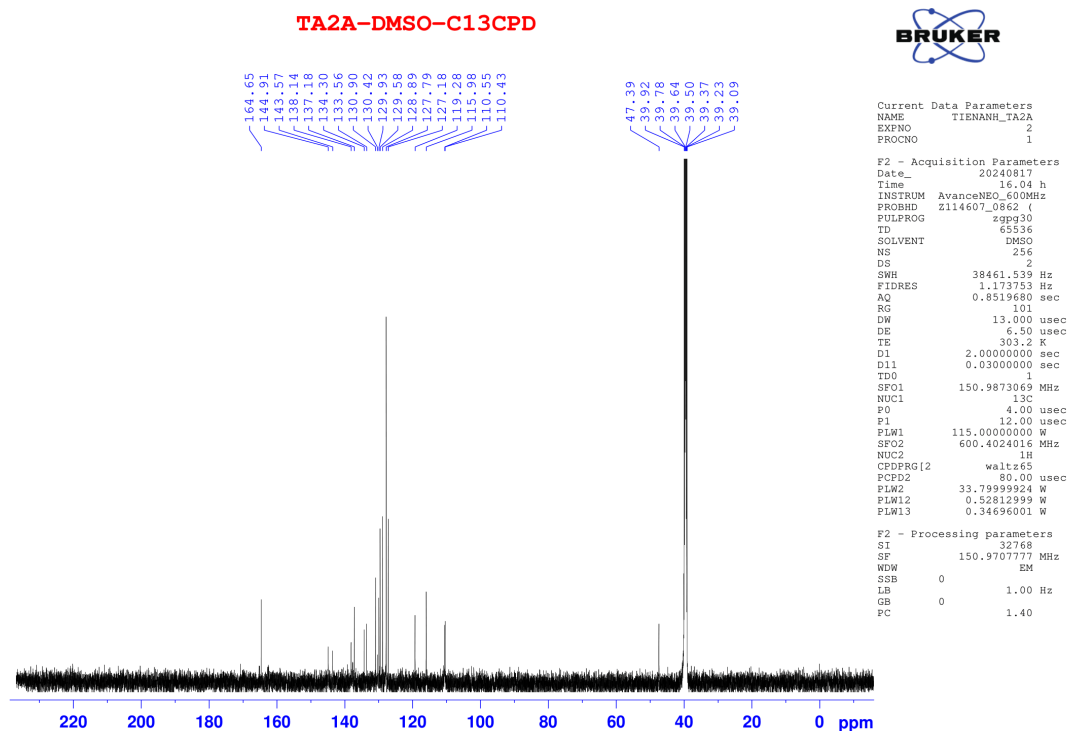


Figure S25. ^{13}C NMR spectrum (151 MHz, $\text{DMSO}-d_6$) of IXa (sample code TA2-A).

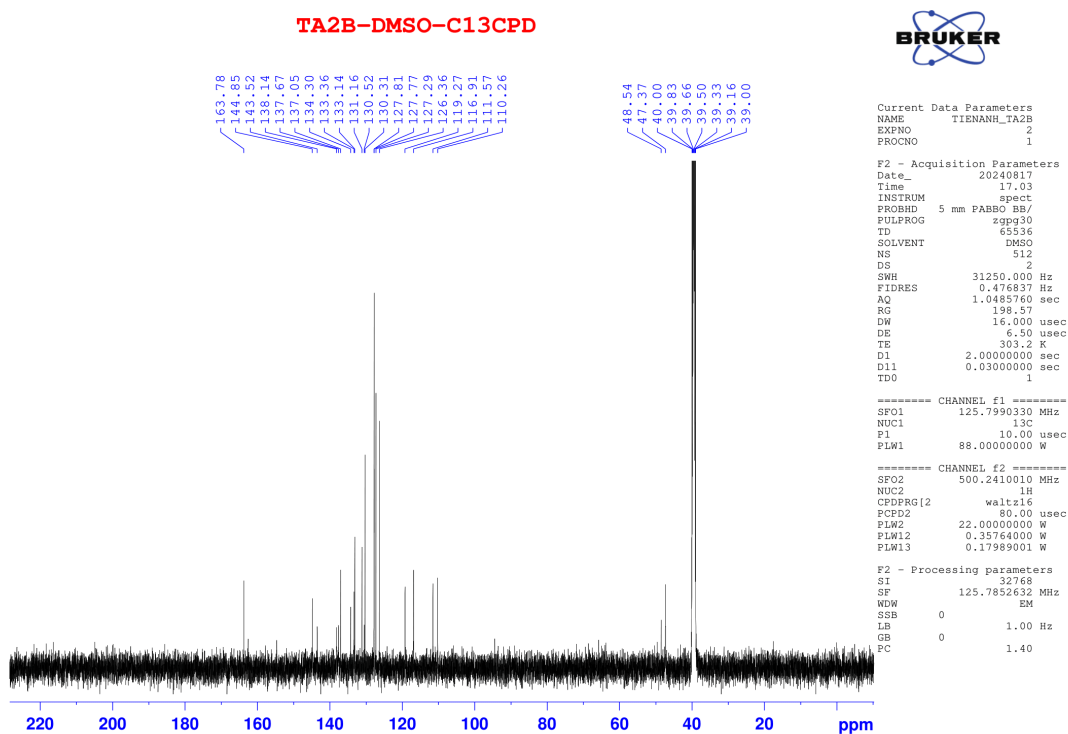


Figure S26. ^{13}C NMR spectrum (126 MHz, $\text{DMSO}-d_6$) of IXb (sample code TA2-B).

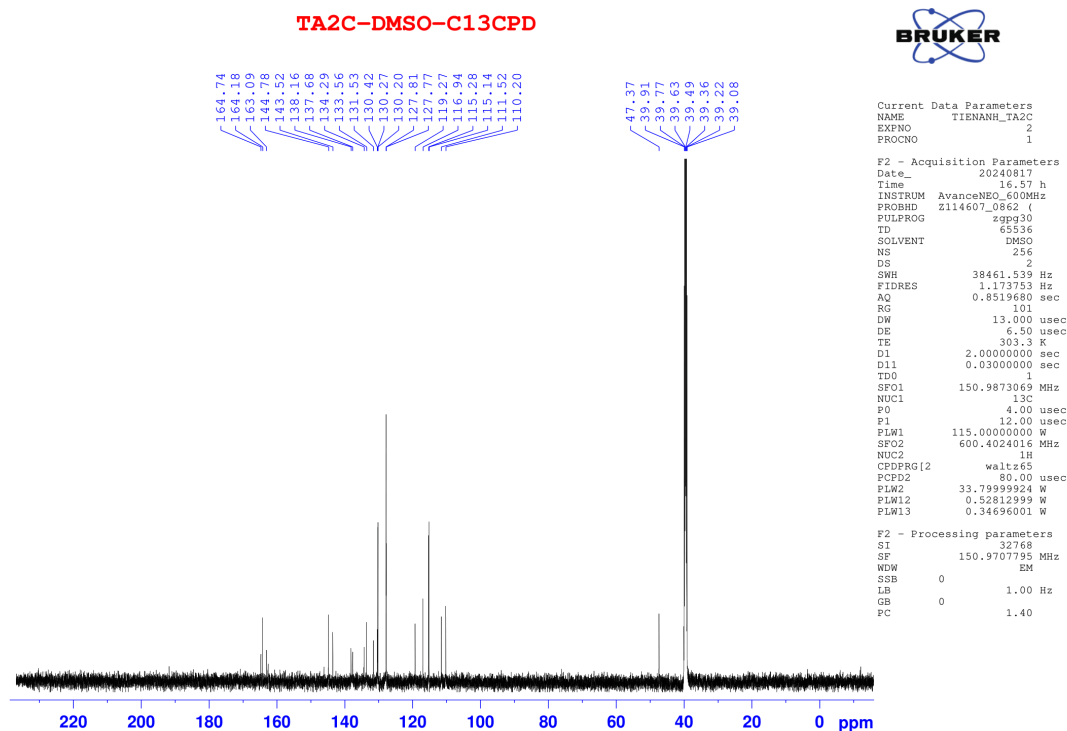


Figure S27. ^{13}C NMR spectrum (151 MHz, $\text{DMSO-}d_6$) of IXc (sample code TA2-C).

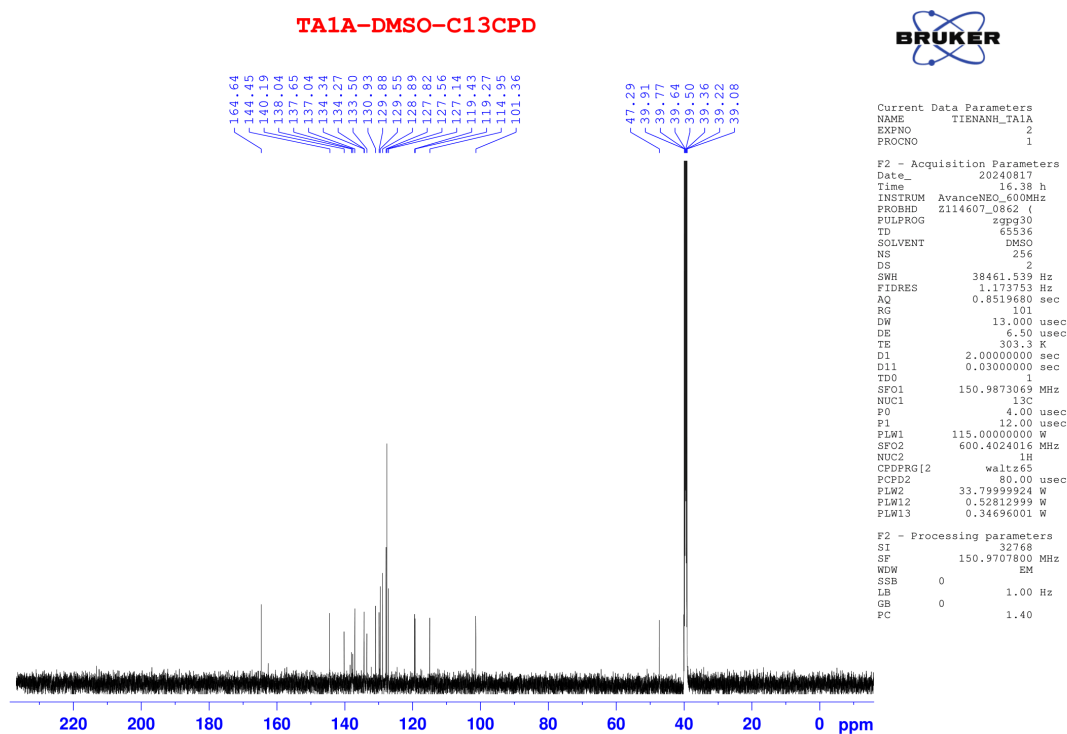


Figure S28. ^{13}C NMR spectrum (151 MHz, $\text{DMSO-}d_6$) of IXd (sample code TA1-A).

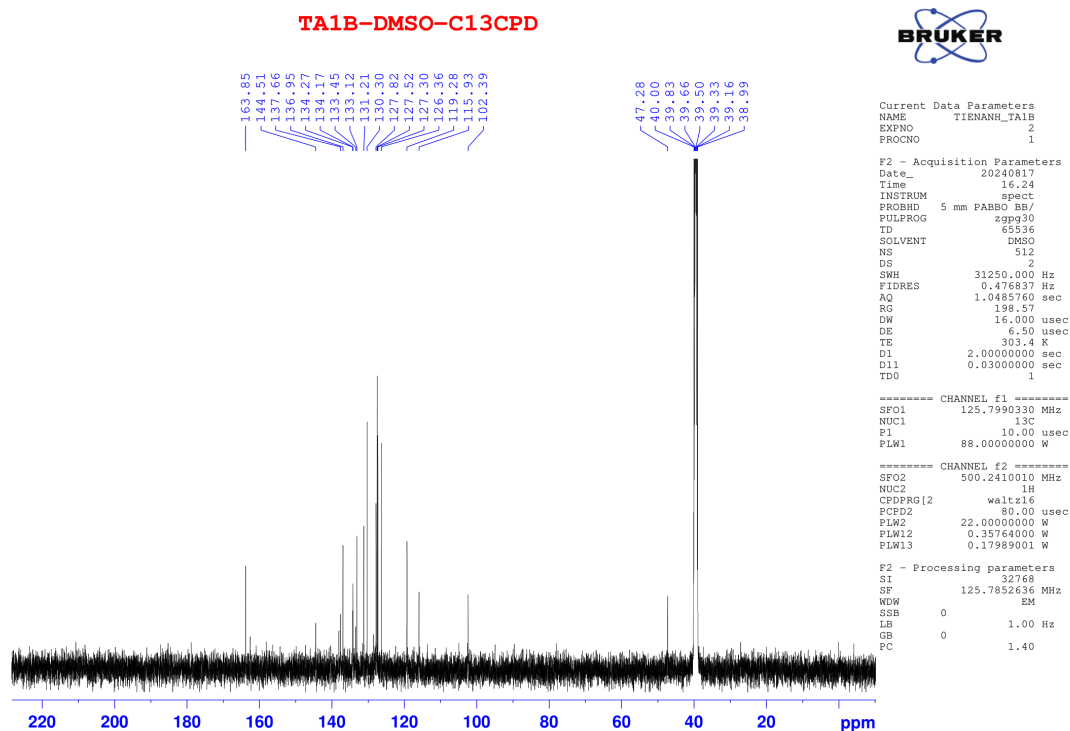


Figure S29. ^{13}C NMR spectrum (126 MHz, $\text{DMSO-}d_6$) of IXe (sample code TA1-B).

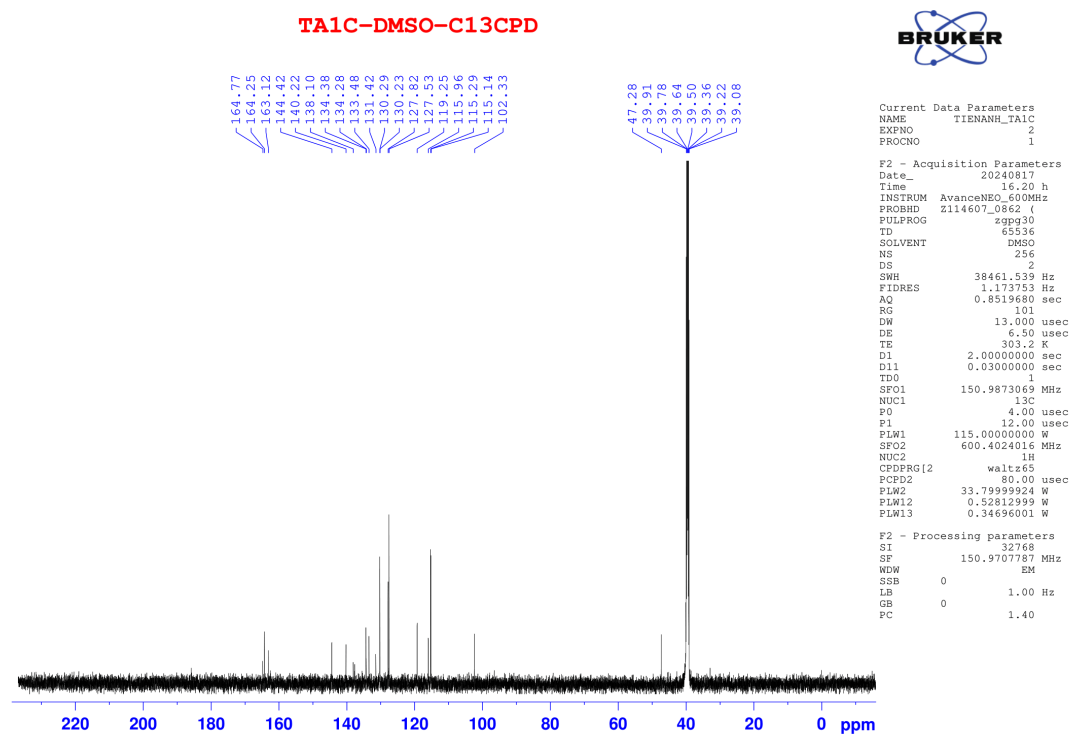


Figure S30. ^{13}C NMR spectrum (151 MHz, $\text{DMSO-}d_6$) of IXf (sample code TA1-C).

S6 HSQC spectra

Multiplicity-edited HSQC spectra (pulse programme `hsqcedetgpsisp2.3`) were recorded in DMSO- d_6 at 500 MHz, except those of IXc and IXf, recorded at 600 MHz. For each compound the full spectrum is followed by the expansions printed with it. HSQC places the carbon of the isolated benzo-ring singlet at δ 110.55–111.57 in IXa–IXc, where the singlet is H-4, and at δ 101.36–102.39 in IXd–IXf, where it is H-7.

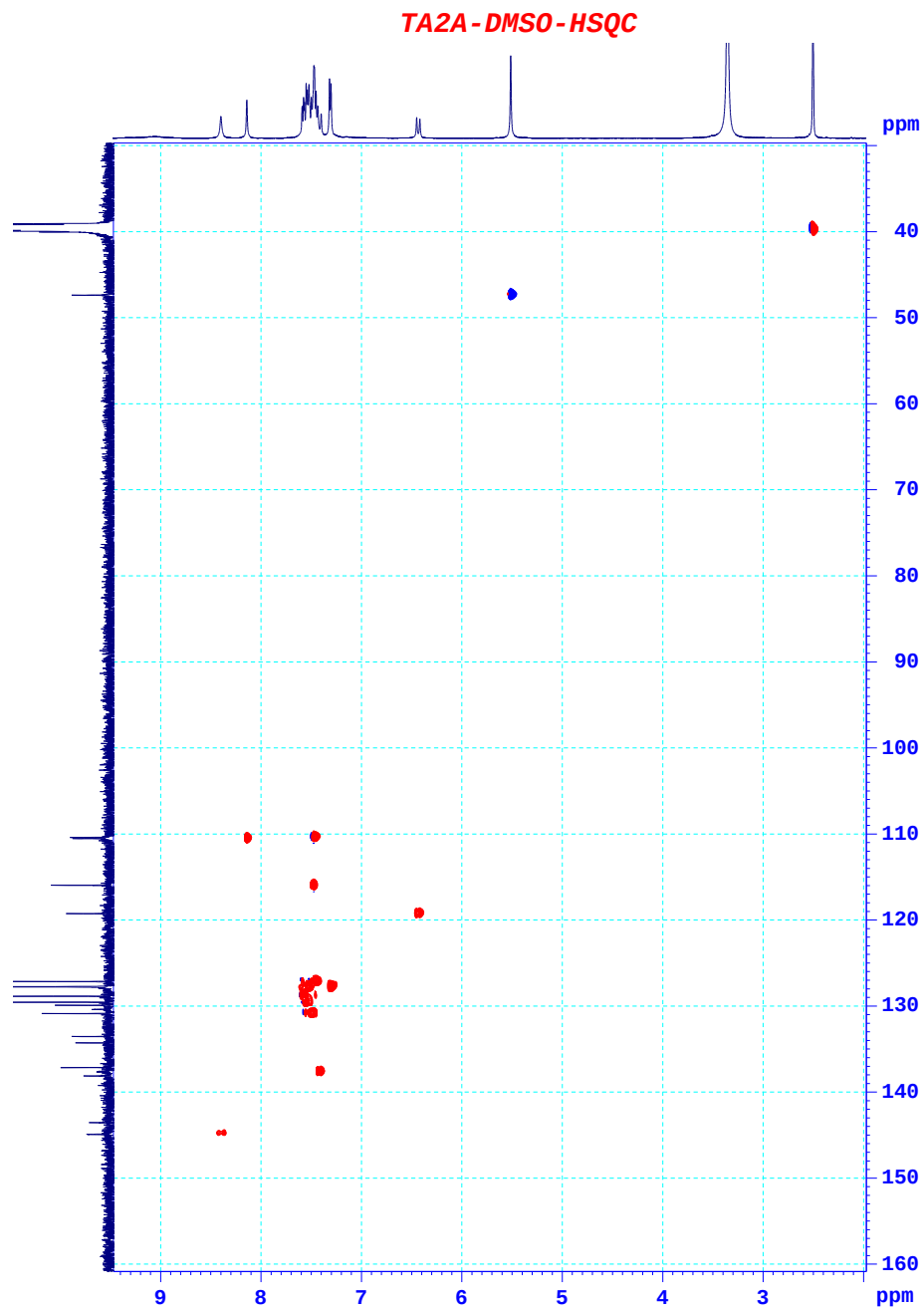


Figure S31. Multiplicity-edited HSQC spectrum (500 MHz, DMSO- d_6) of IXa (sample code TA2-A): full spectrum.

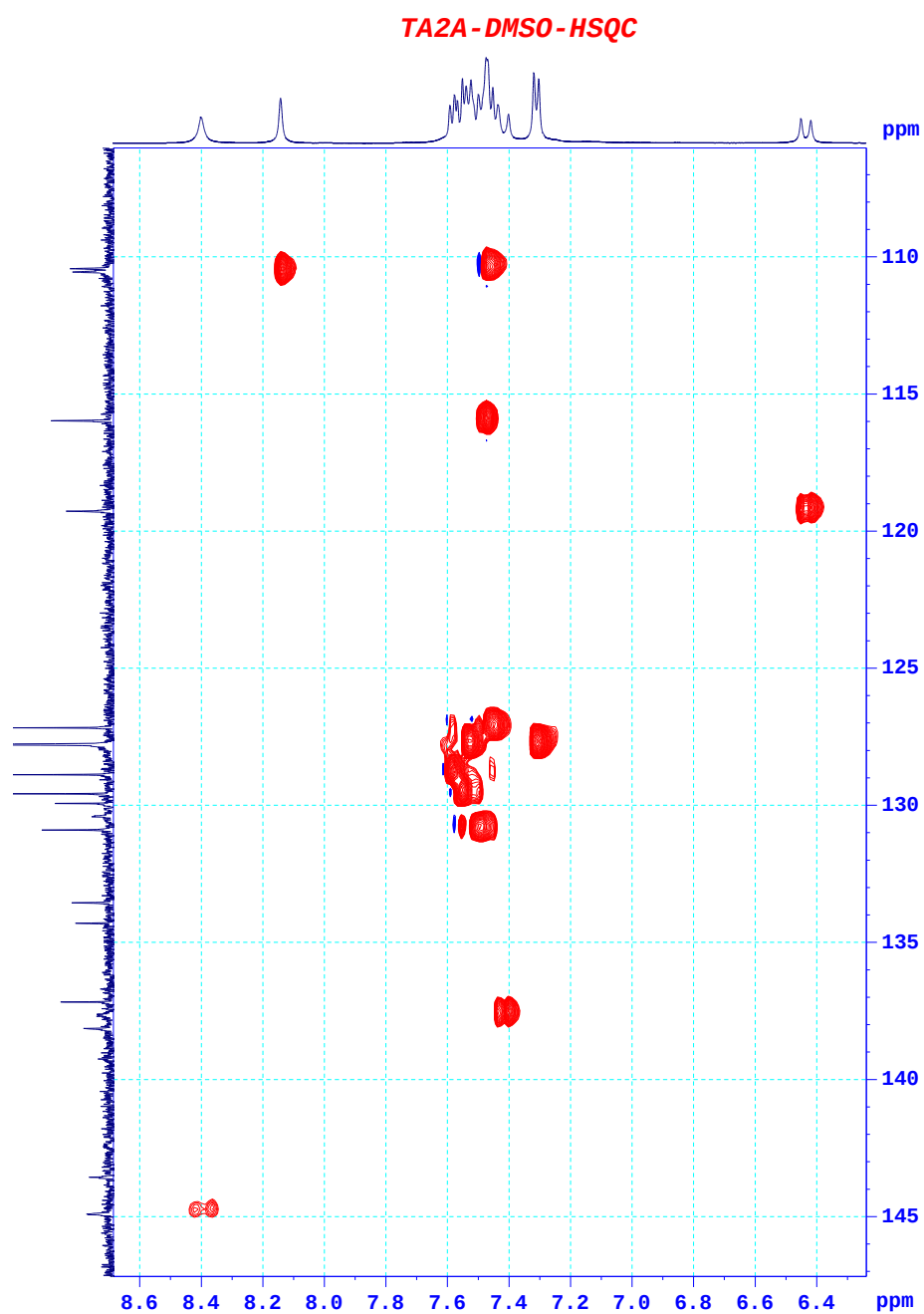


Figure S31. HSQC spectrum of IXa, continued: expansion.

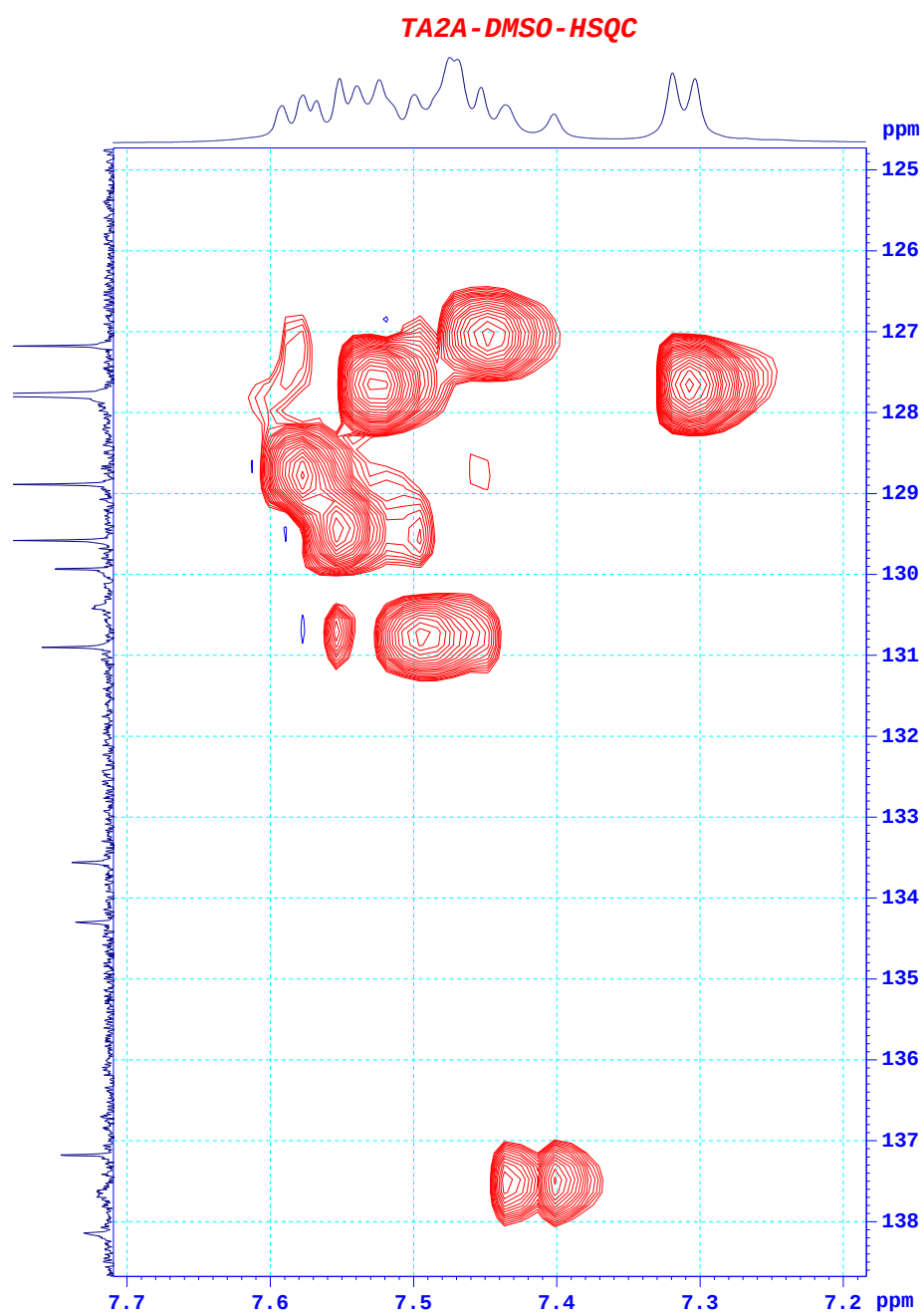


Figure S31. HSQC spectrum of IXa, continued: expansion.

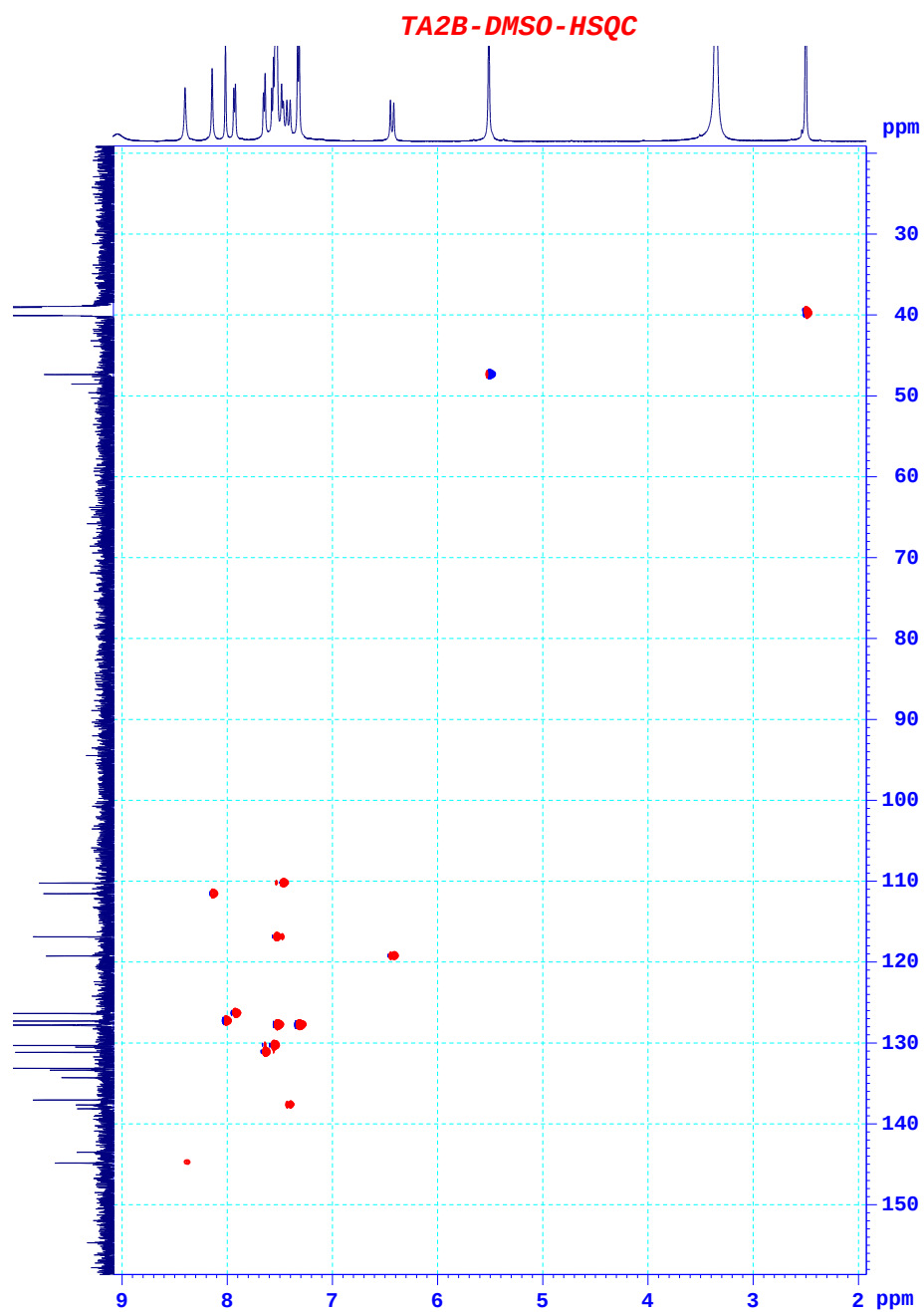


Figure S32. Multiplicity-edited HSQC spectrum (500 MHz, DMSO- d_6) of IXb (sample code TA2-B): full spectrum.

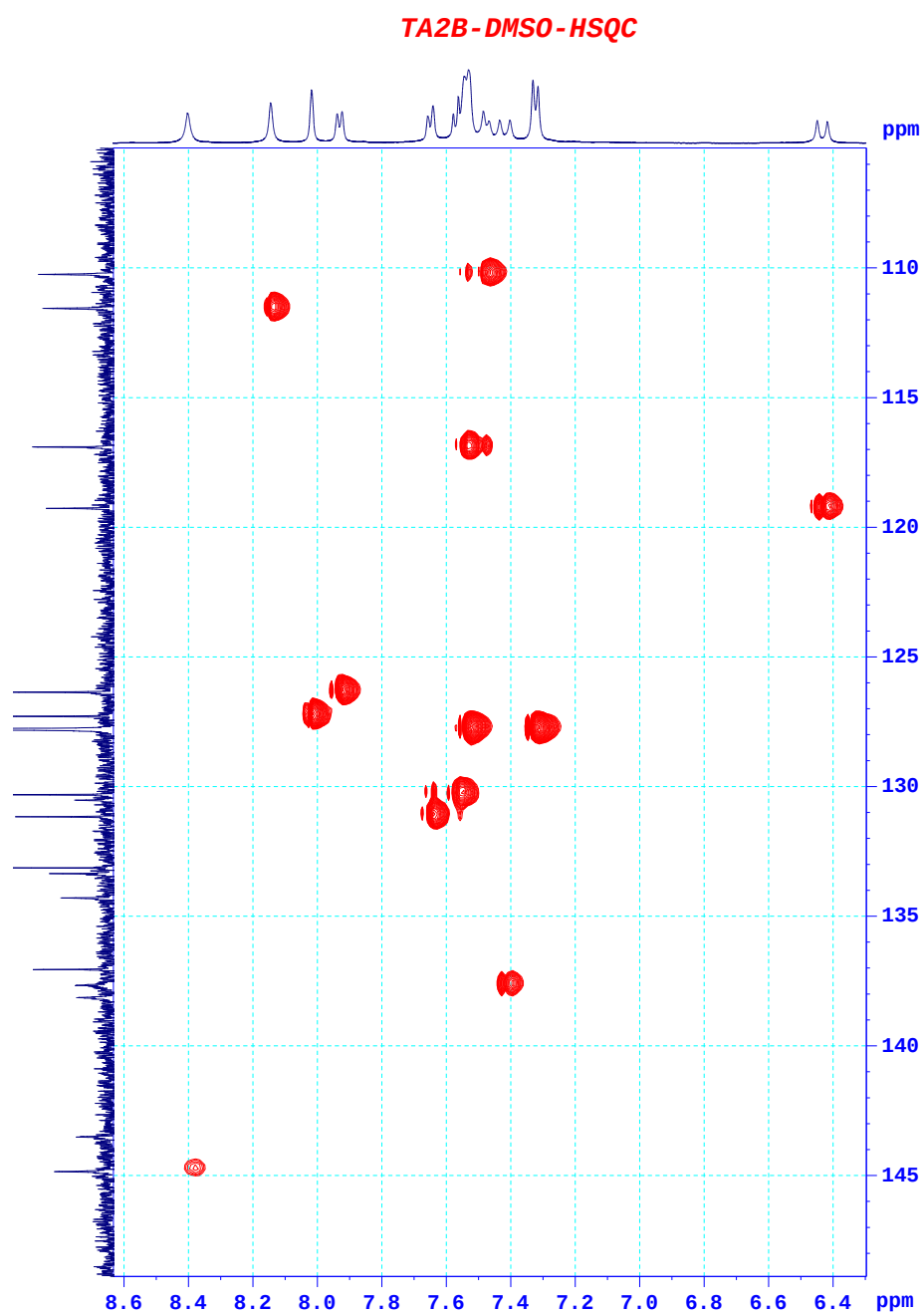


Figure S32. HSQC spectrum of IXb, continued: expansion.

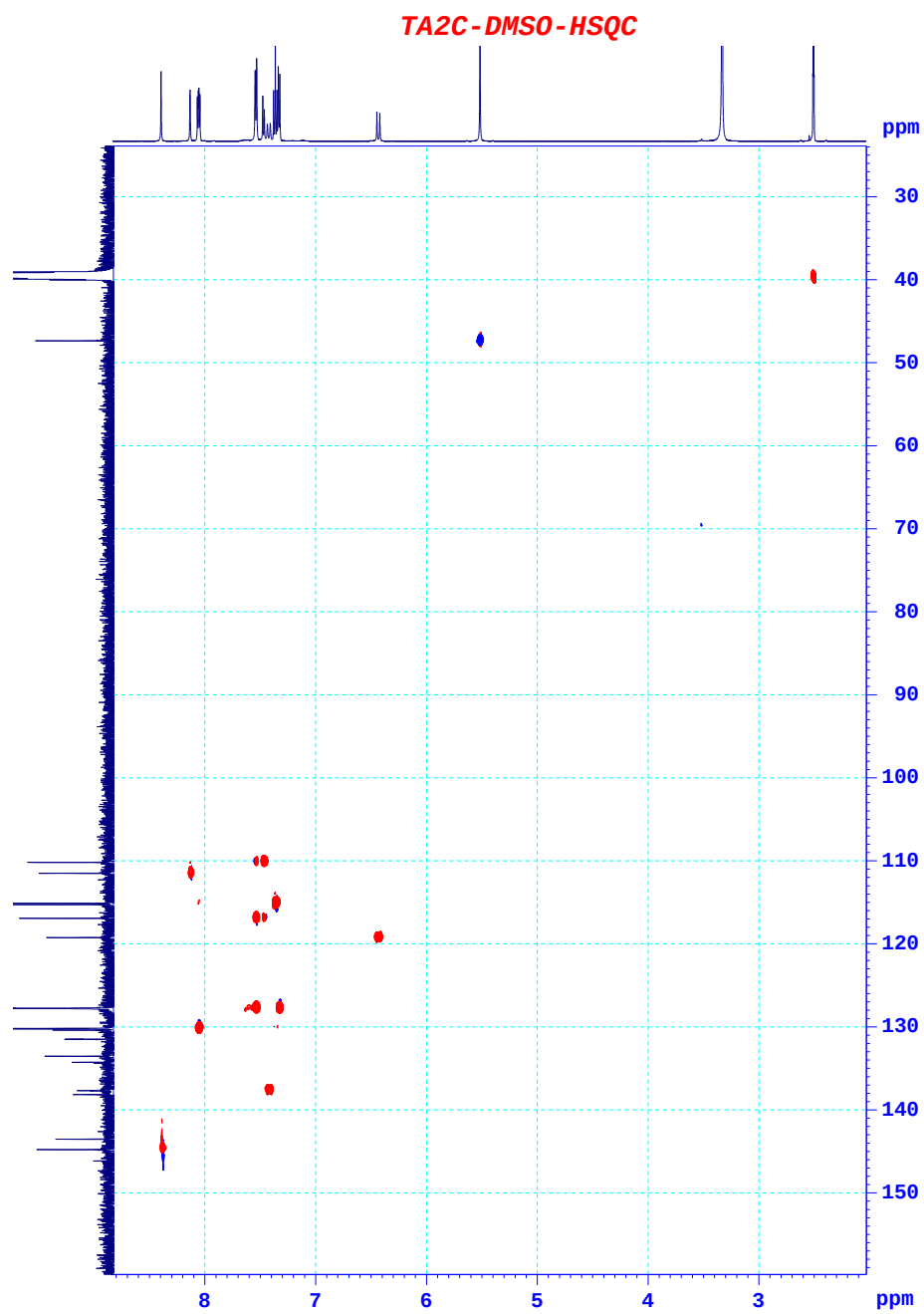


Figure S33. Multiplicity-edited HSQC spectrum (600 MHz, DMSO- d_6) of IXc (sample code TA2-C): full spectrum.

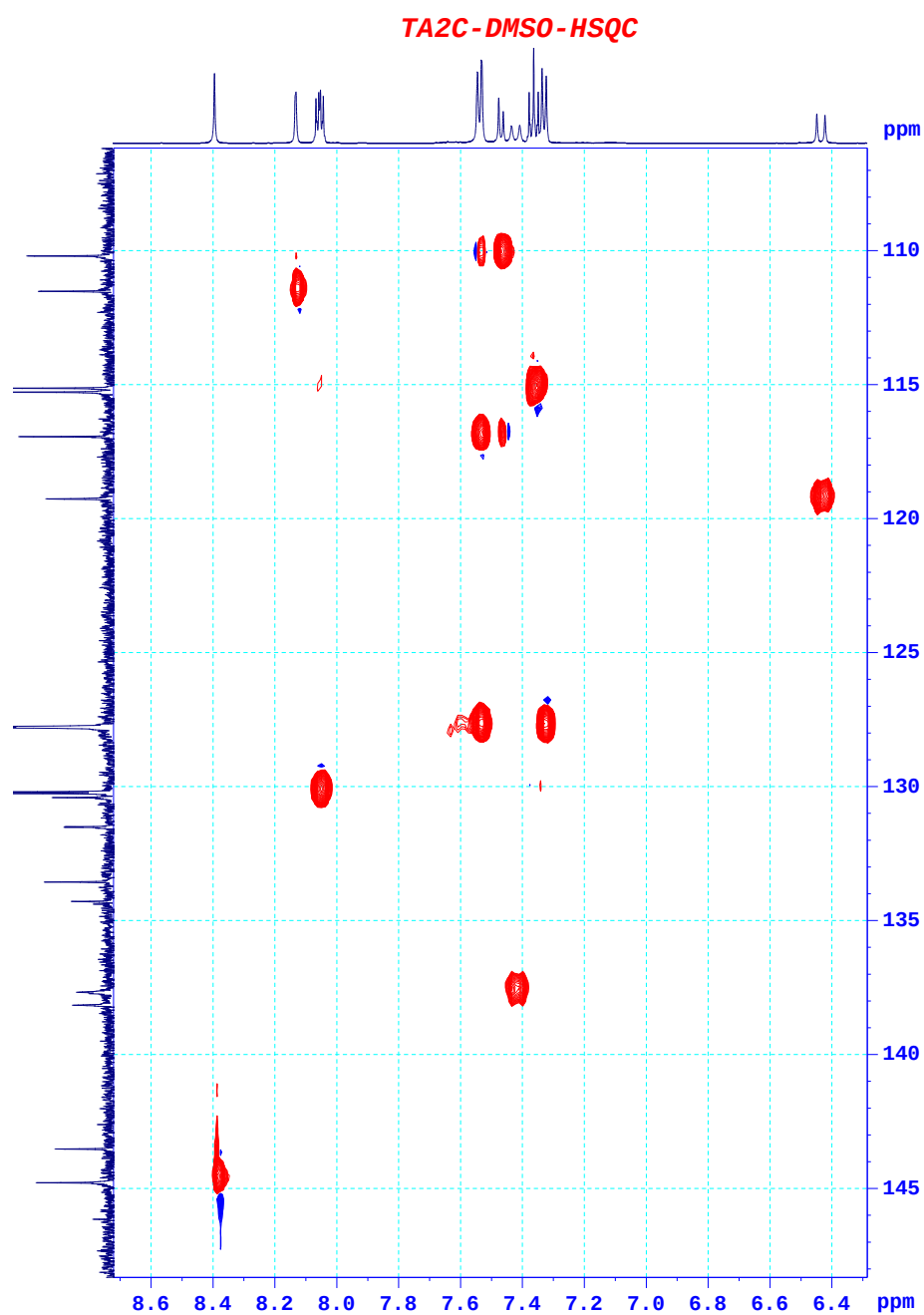


Figure S33. HSQC spectrum of IXc, continued: expansion.

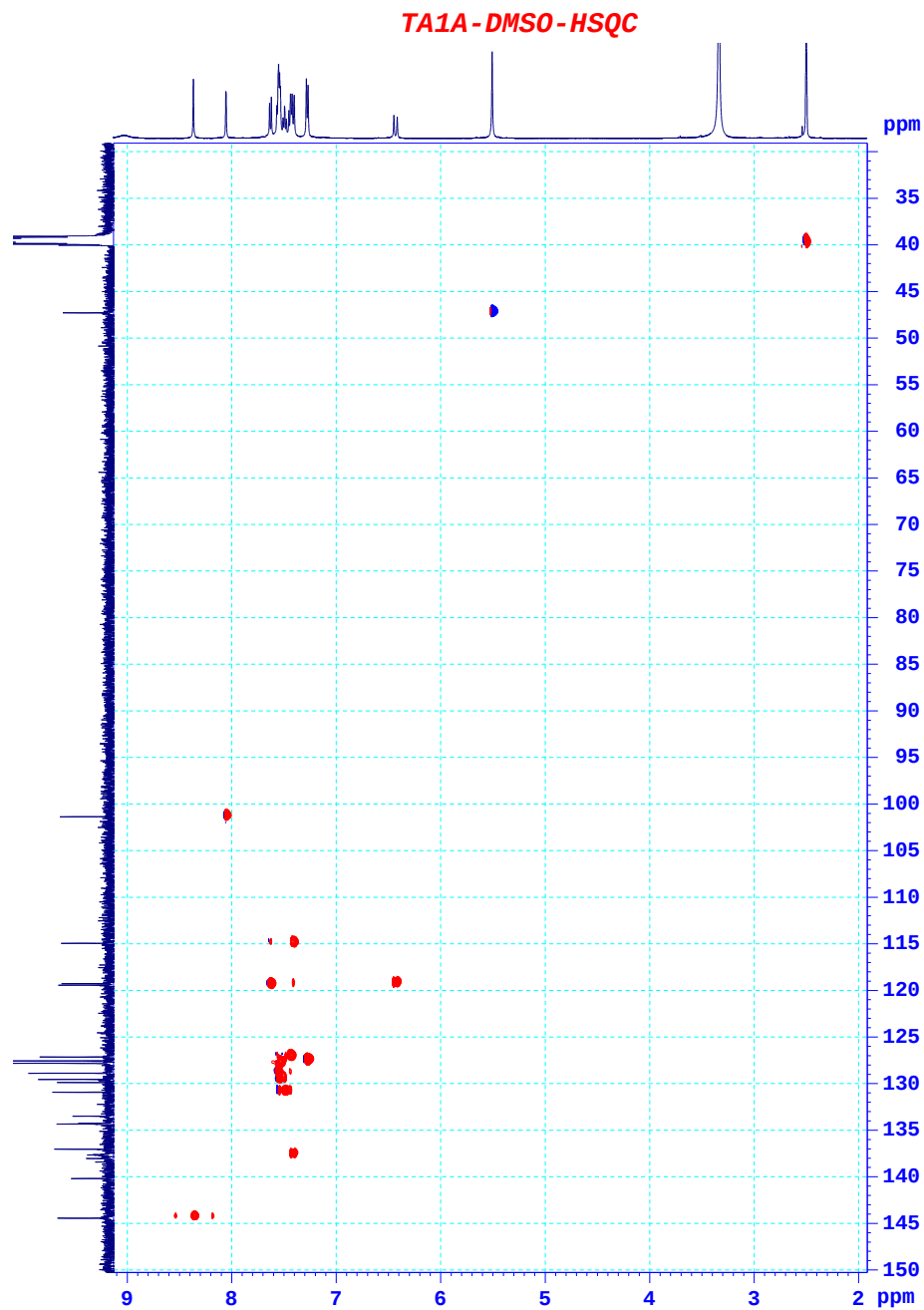


Figure S34. Multiplicity-edited HSQC spectrum (500 MHz, DMSO- d_6) of IXd (sample code TA1-A): full spectrum.

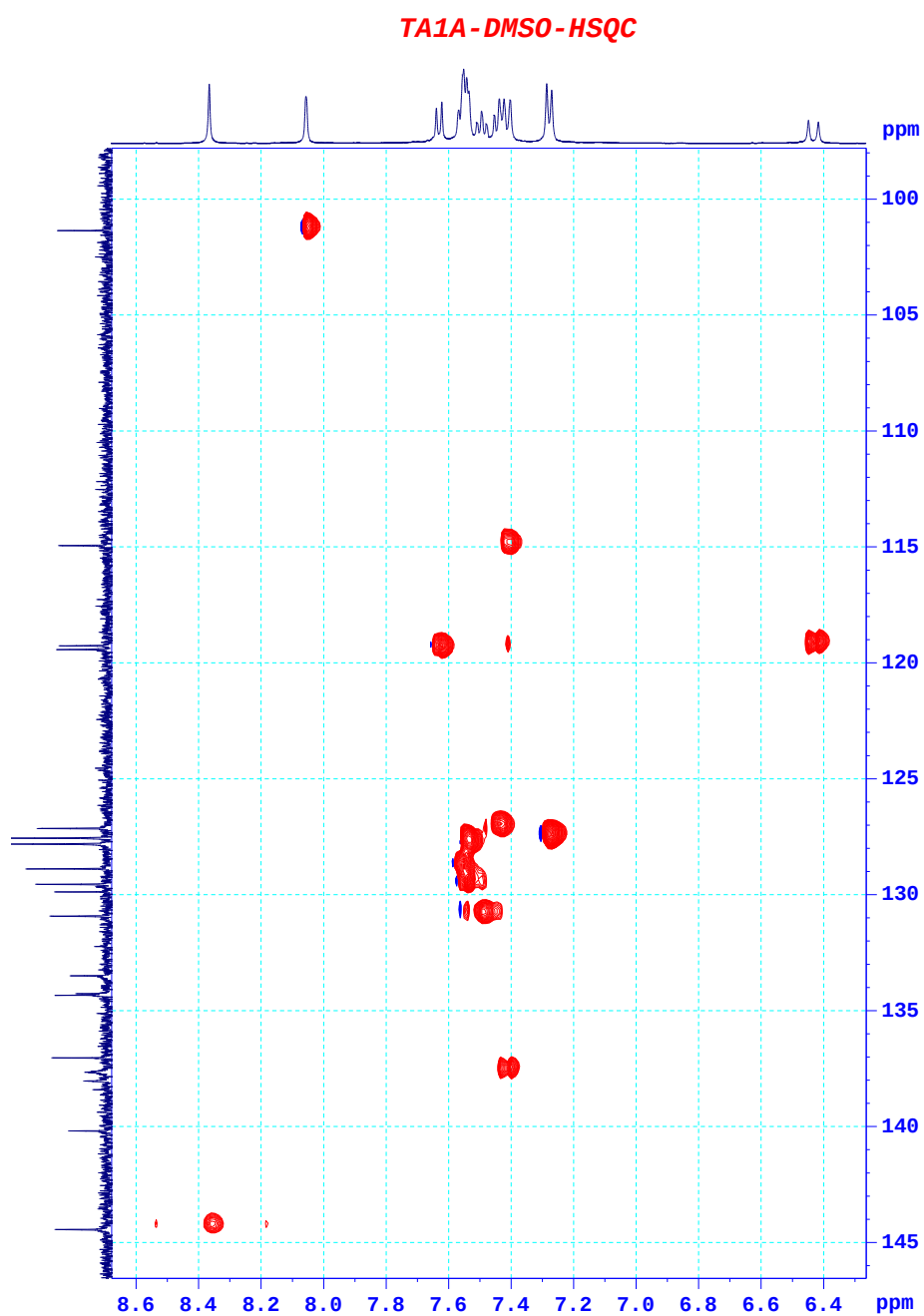


Figure S34. HSQC spectrum of IXd, continued: expansion.

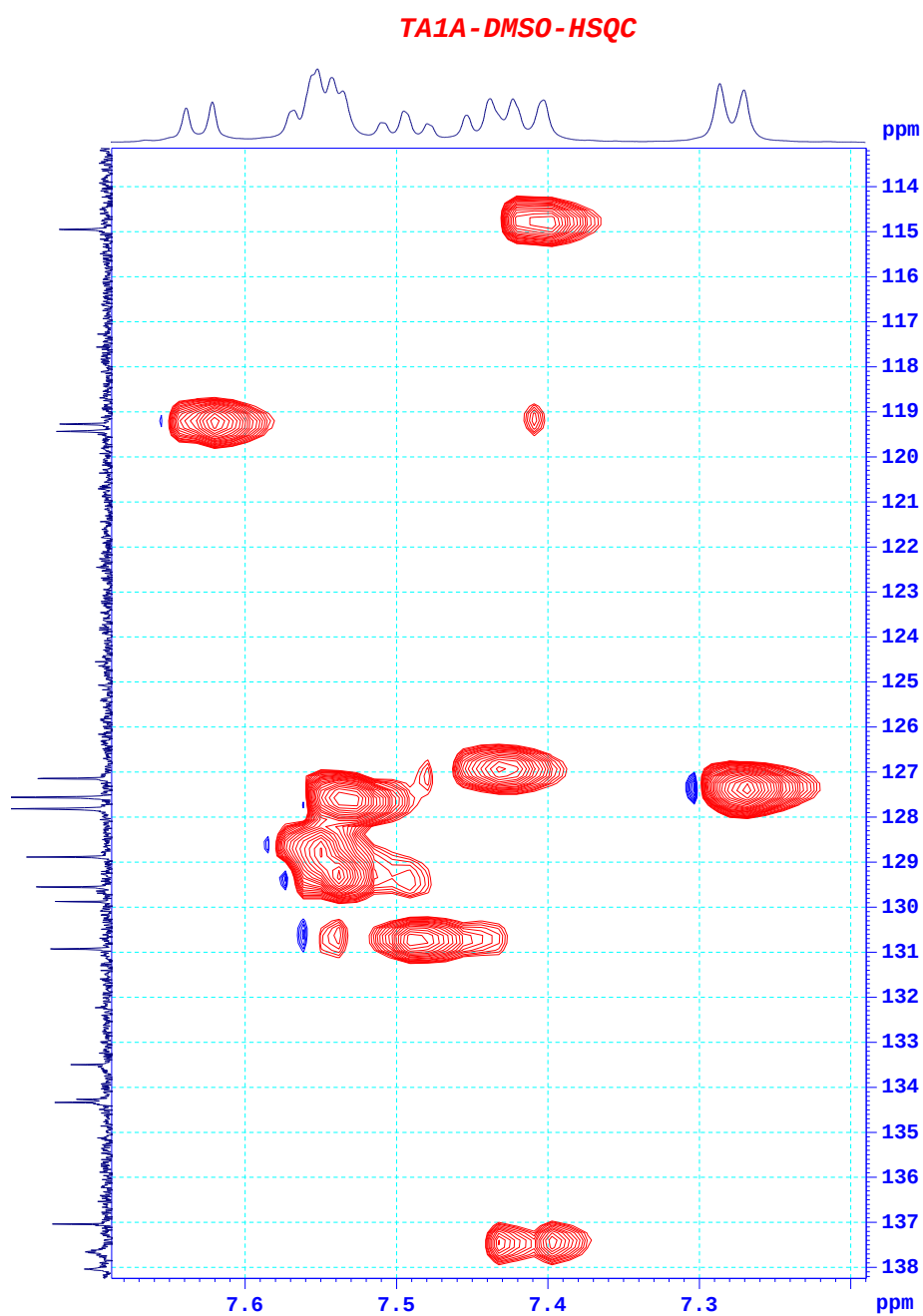


Figure S34. HSQC spectrum of IXd, continued: expansion.

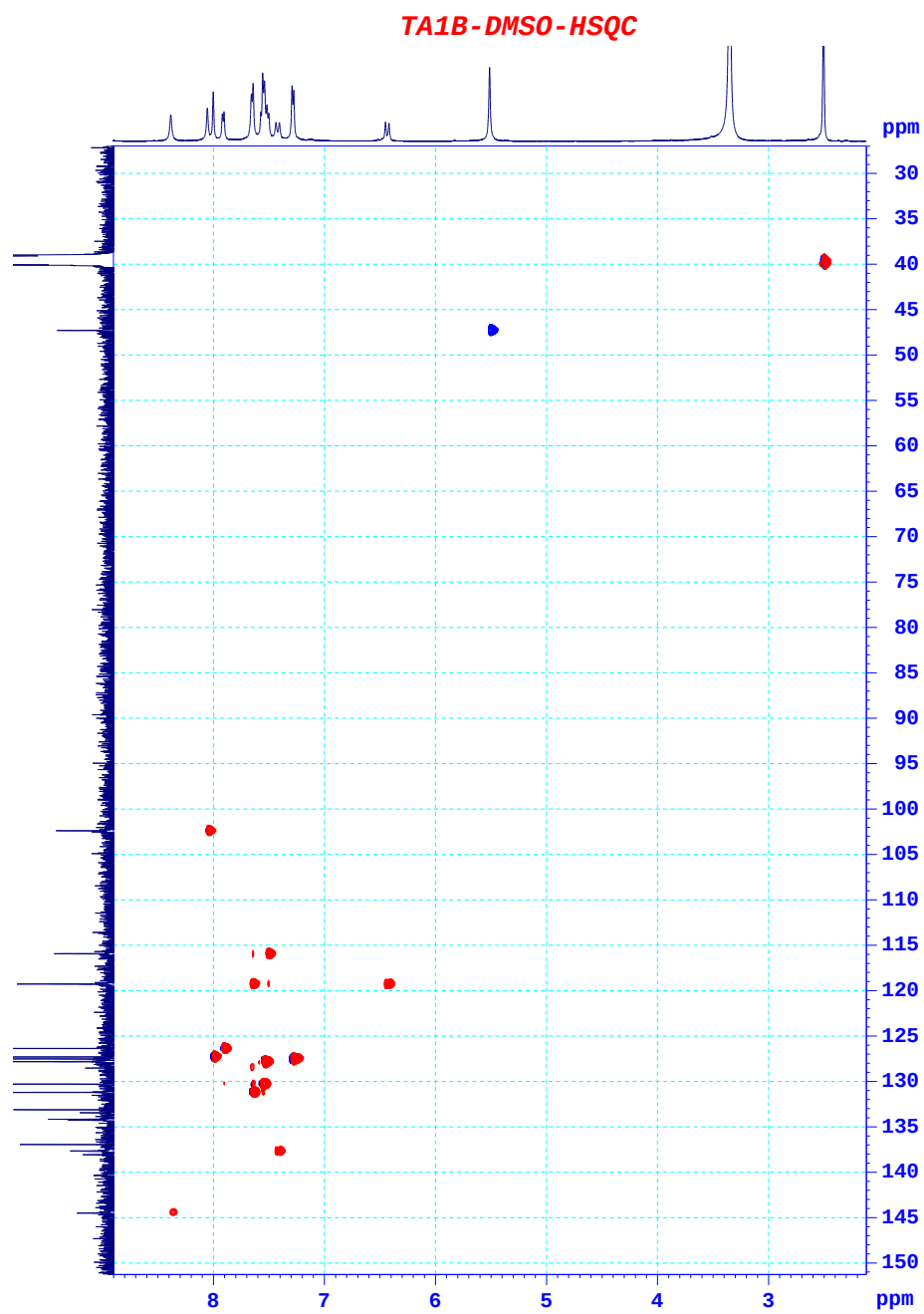


Figure S35. Multiplicity-edited HSQC spectrum (500 MHz, DMSO- d_6) of IXe (sample code TA1-B): full spectrum.

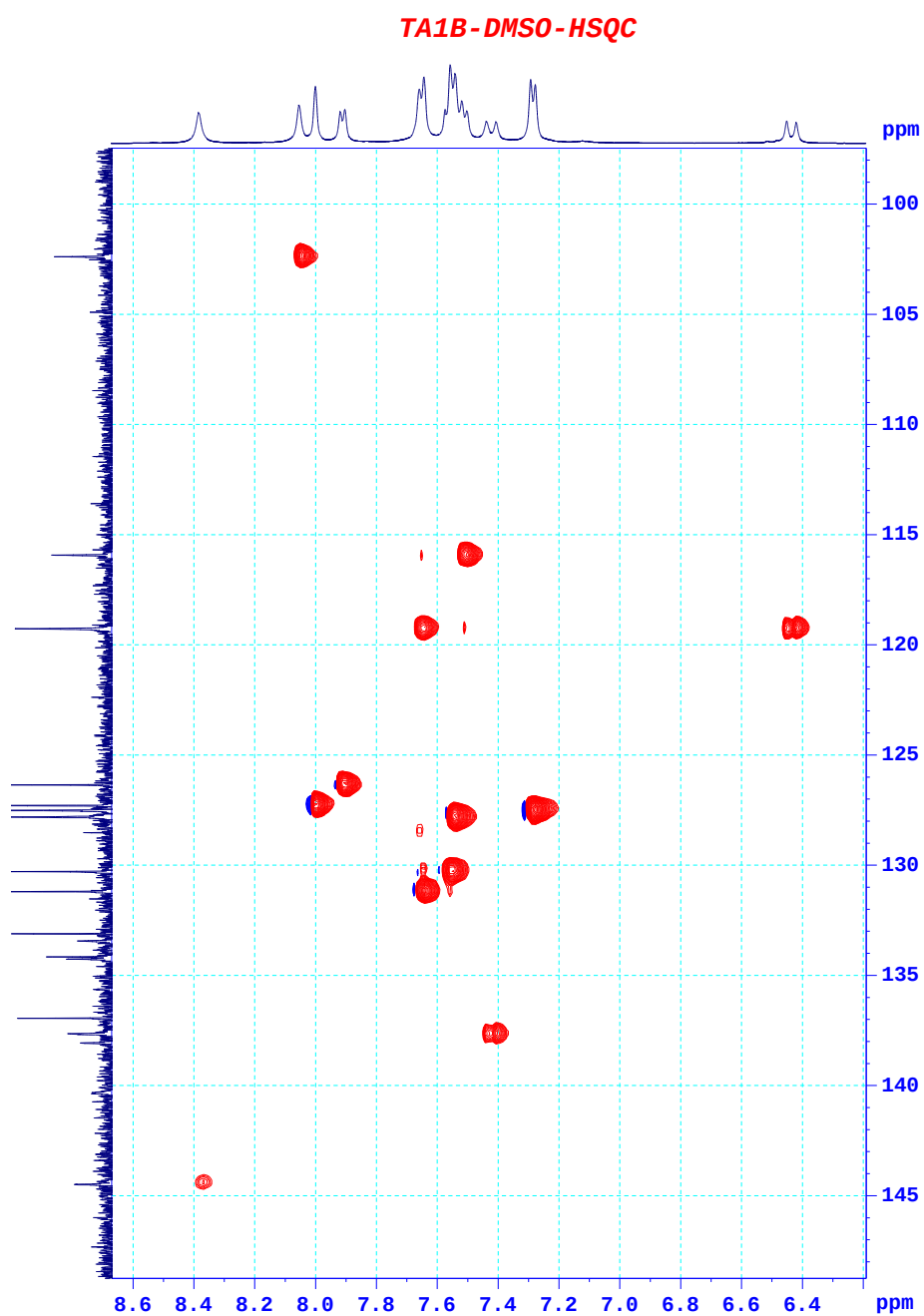


Figure S35. HSQC spectrum of IXe, continued: expansion.

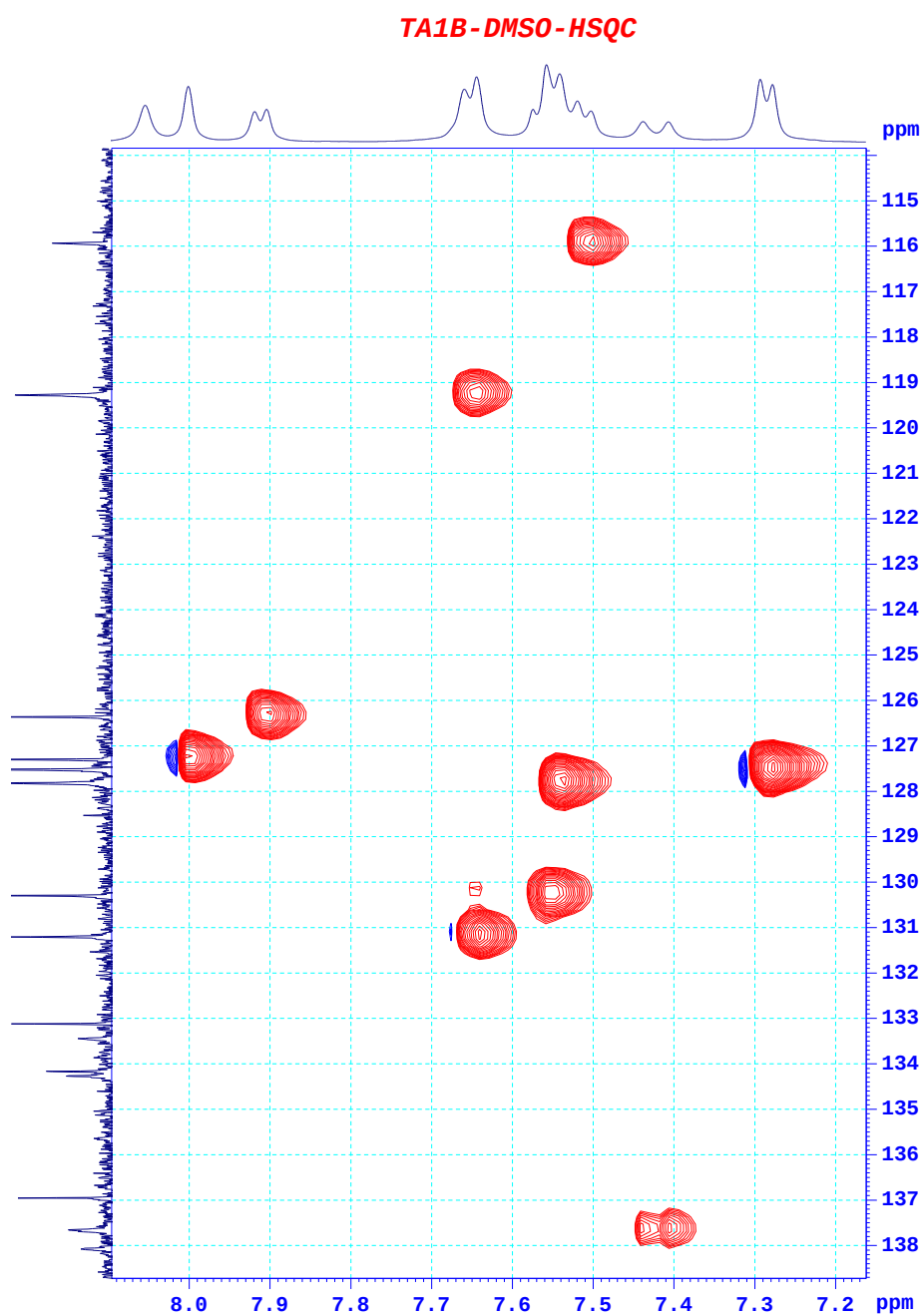


Figure S35. HSQC spectrum of IXe, continued: expansion.

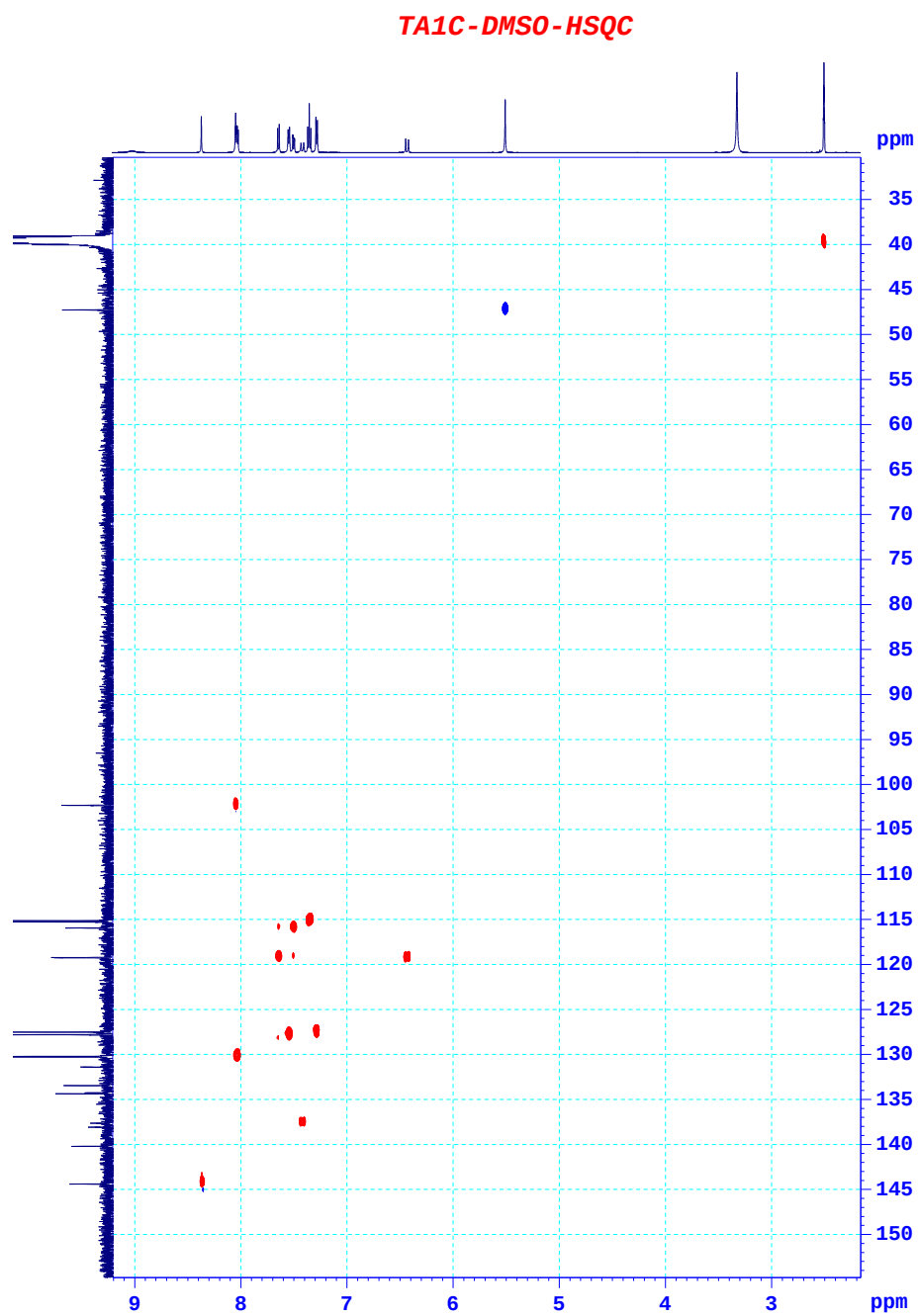


Figure S36. Multiplicity-edited HSQC spectrum (600 MHz, DMSO- d_6) of IXf (sample code TA1-C): full spectrum.

TA1C-DMSO-HSQC

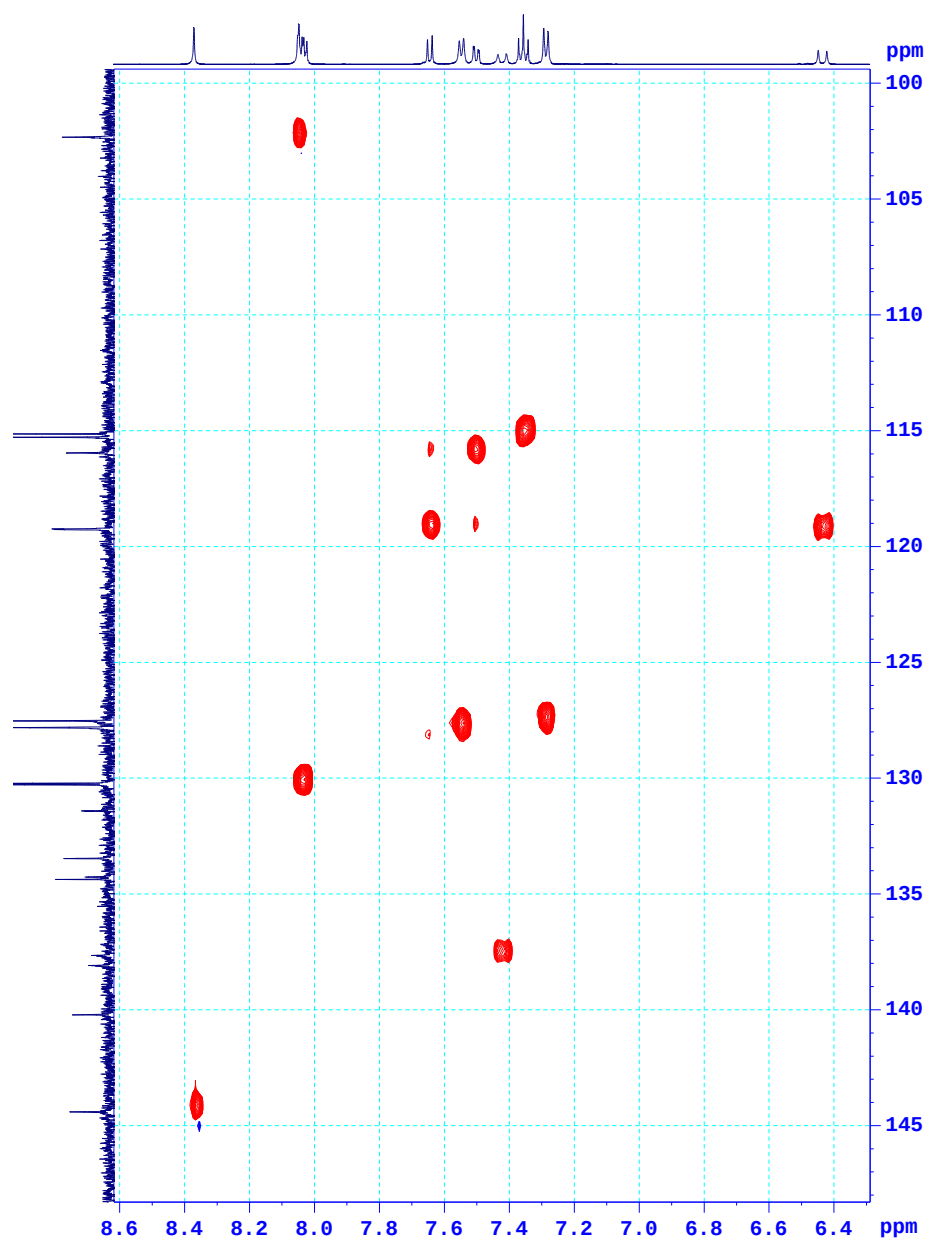


Figure S36. HSQC spectrum of IXf, continued: expansion.

S7 HMBC spectra

HMBC spectra (pulse programme `hmbcgplpndqf`) were recorded in DMSO- d_6 at 500 MHz, except those of IXc and IXf, recorded at 600 MHz. The full spectrum of each compound is followed by its expansions. The amide NH shows three-bond correlations to C-4 and C-6 in IXa-IXc, and to C-7 and C-5 in IXd-IXf.

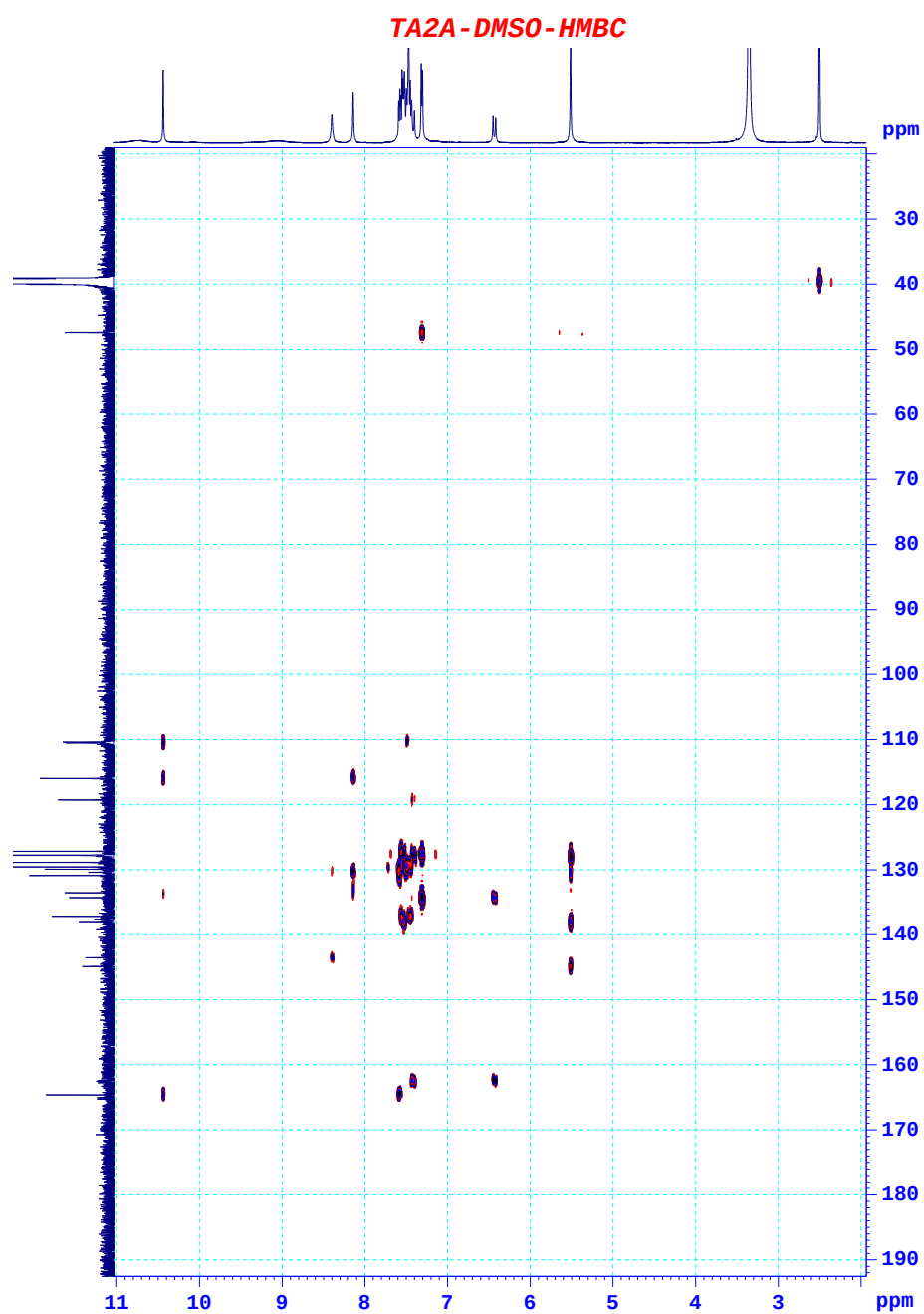


Figure S37. HMBC spectrum (500 MHz, $\text{DMSO}-d_6$) of IXa (sample code TA2-A): full spectrum.

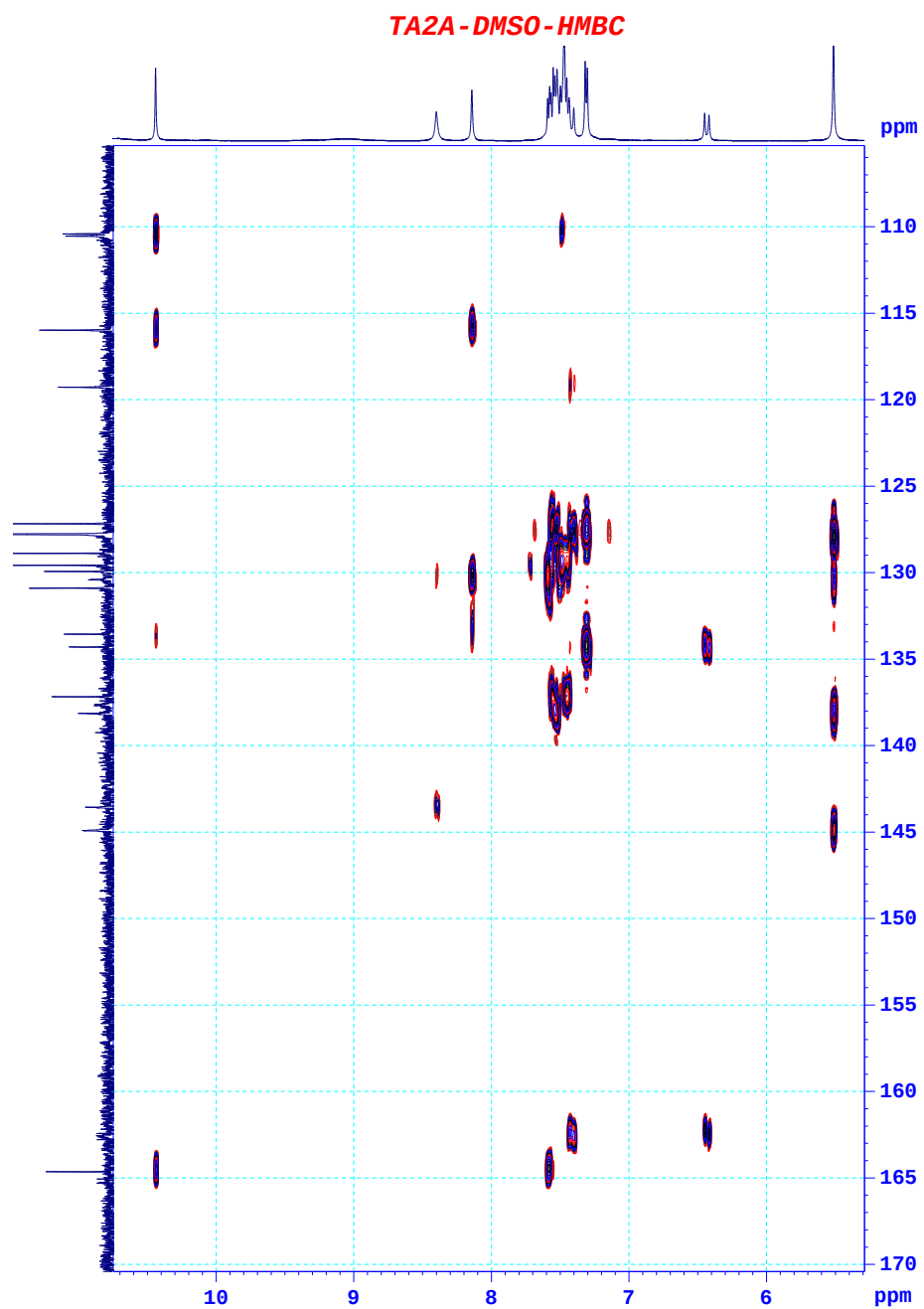


Figure S37. HMBC spectrum of IXa, continued: expansion.

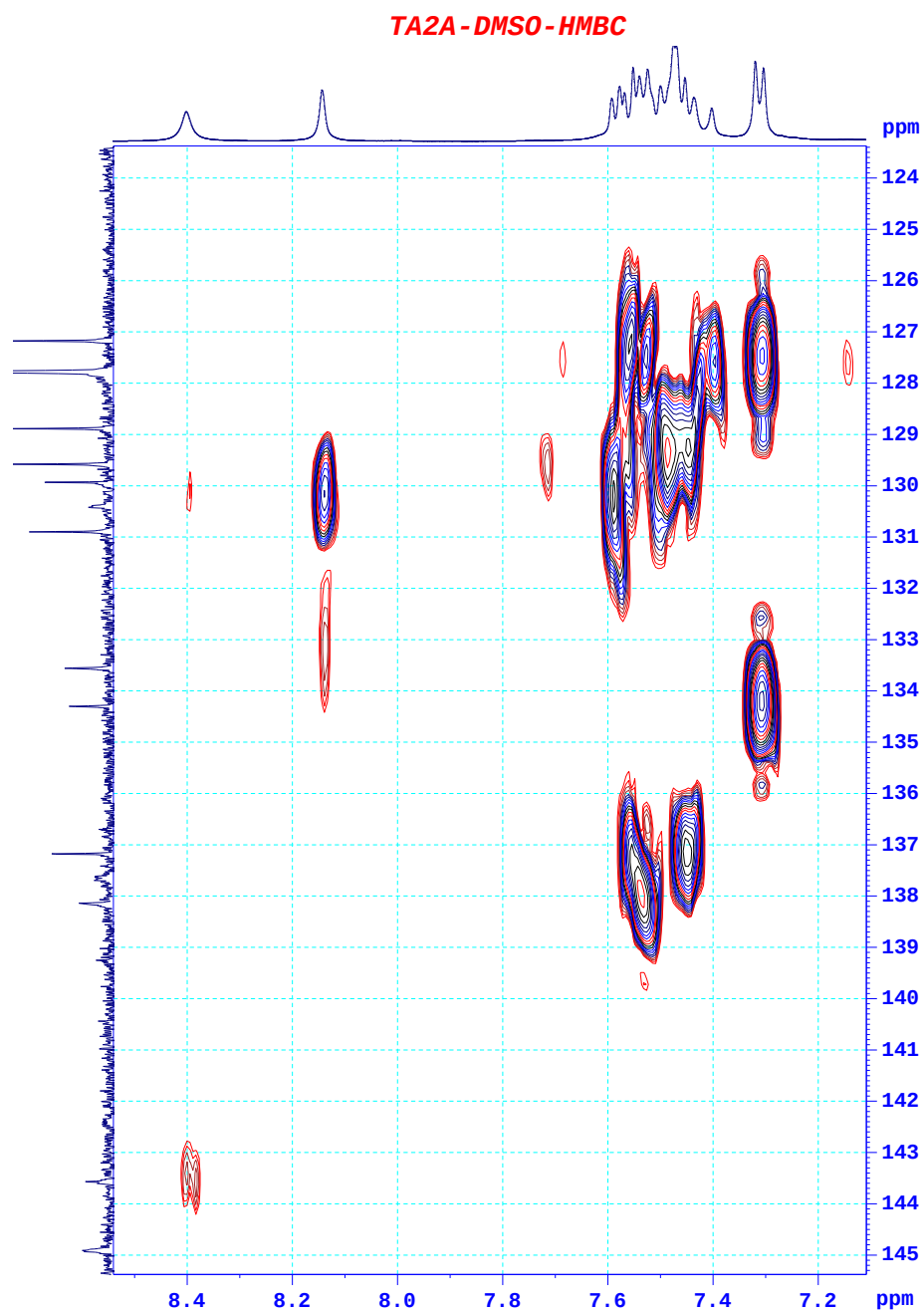


Figure S37. HMBC spectrum of IXa, continued: expansion.

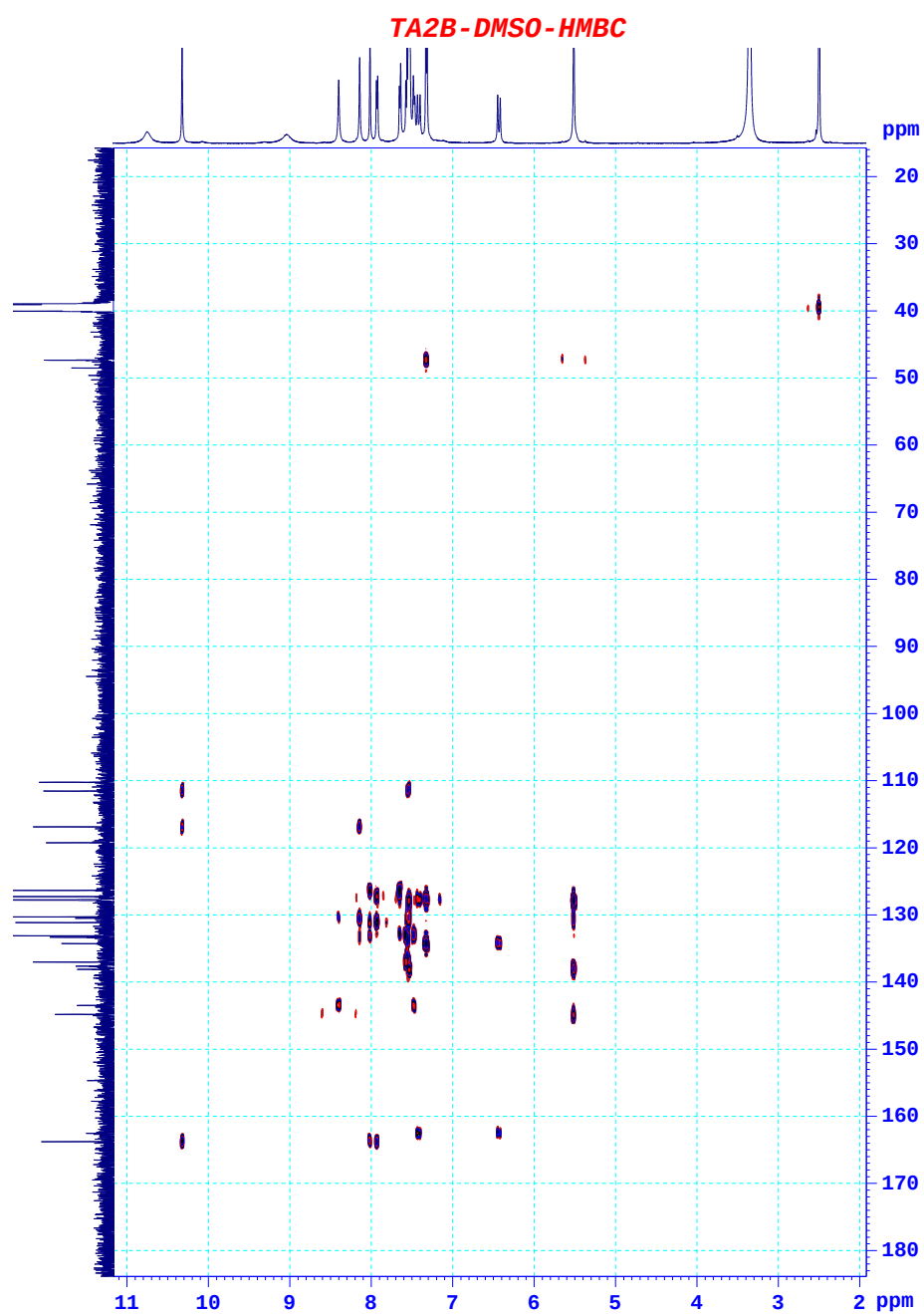


Figure S38. HMBC spectrum (500 MHz, DMSO- d_6) of IXb (sample code TA2-B): full spectrum.

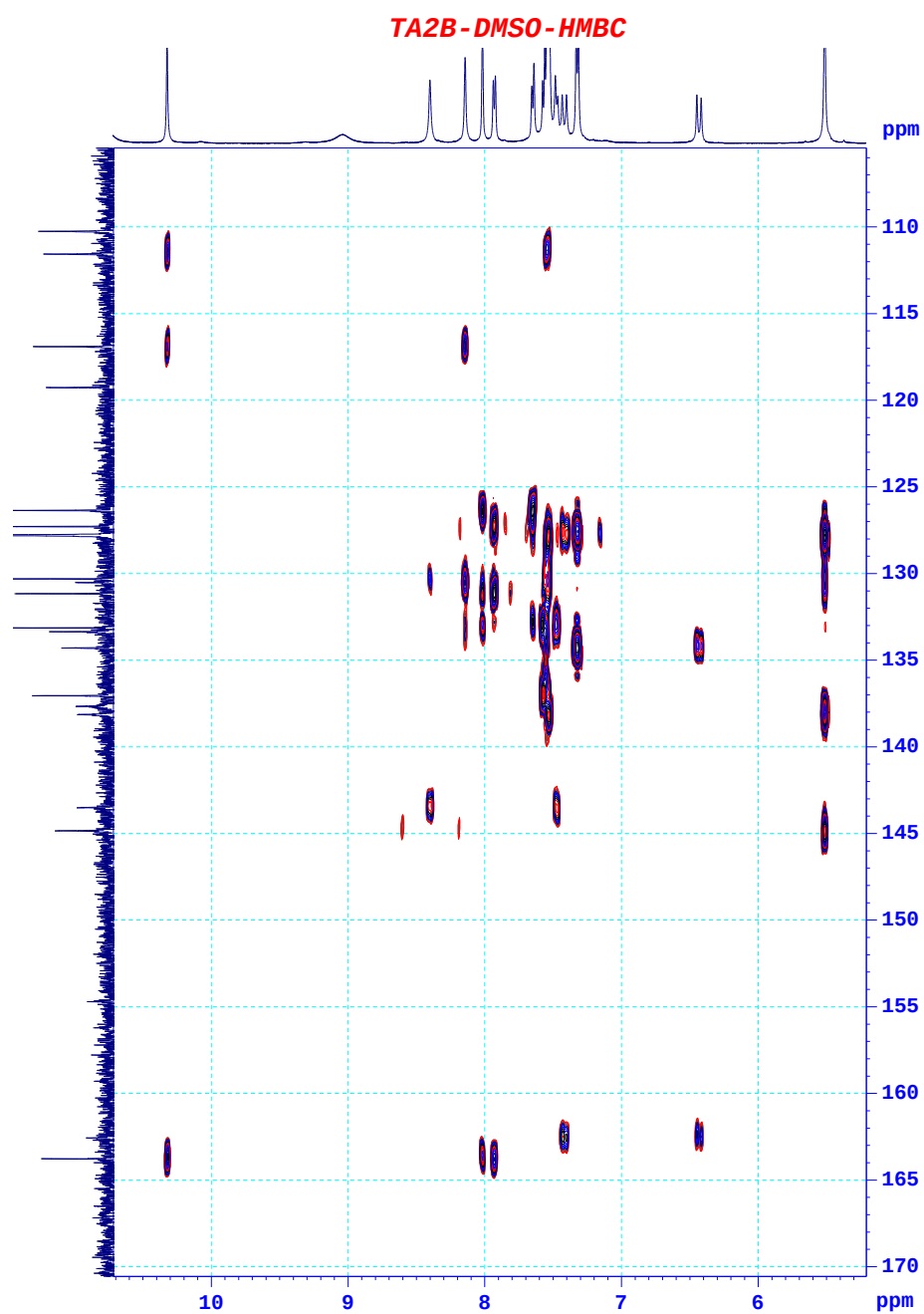


Figure S38. HMBC spectrum of IXb, continued: expansion.

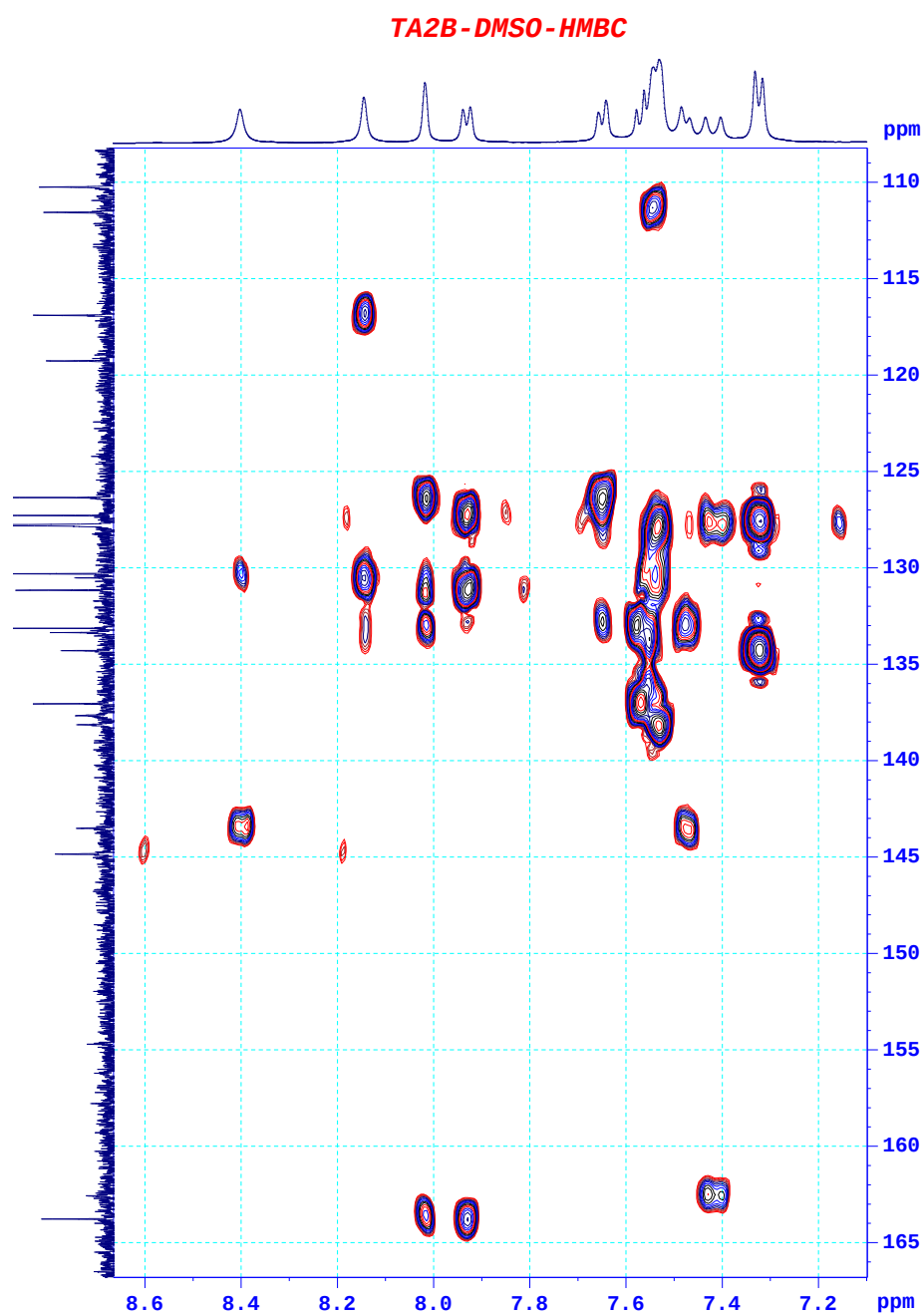


Figure S38. HMBC spectrum of IXb, continued: expansion.

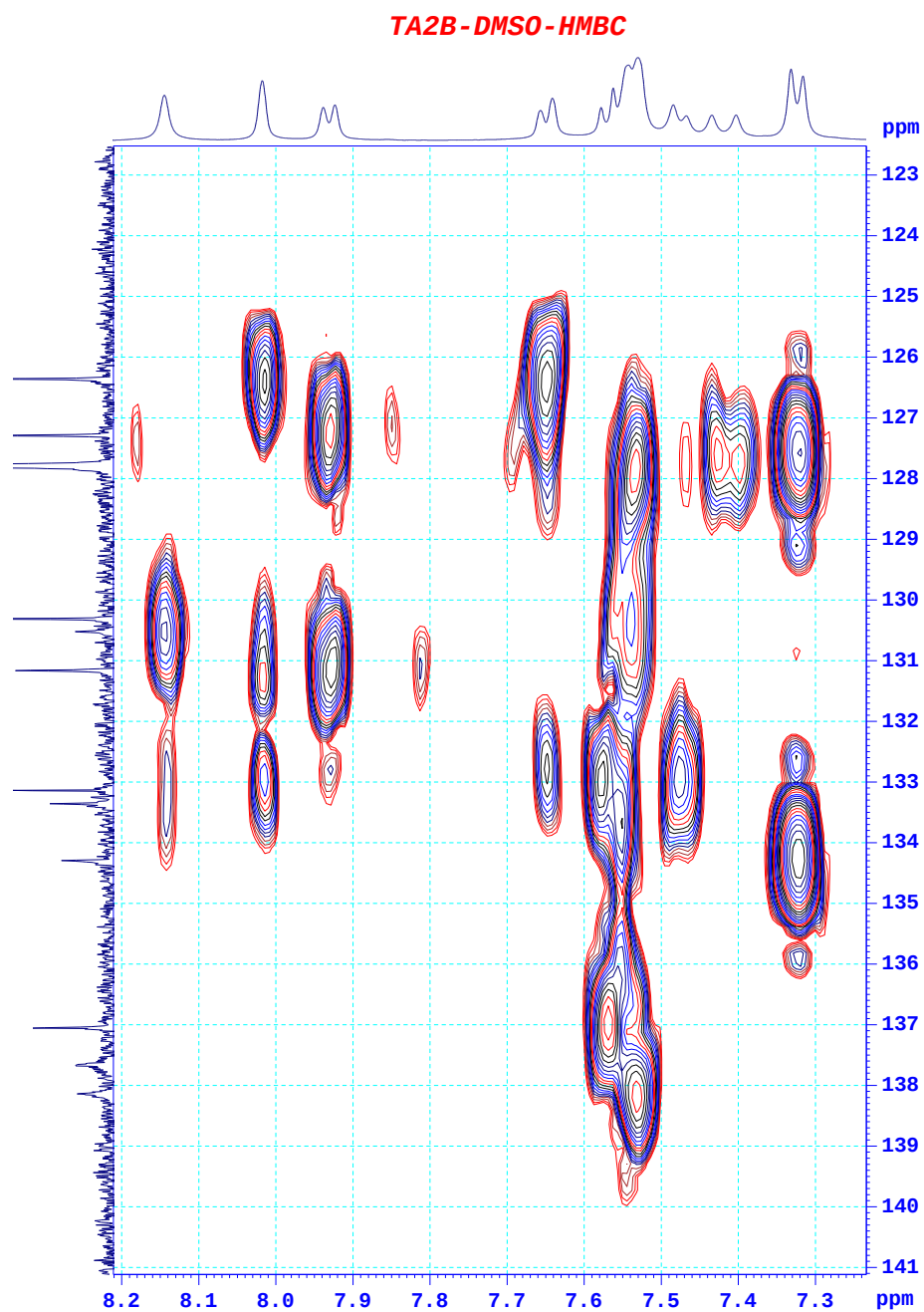


Figure S38. HMBC spectrum of IXb, continued: expansion.

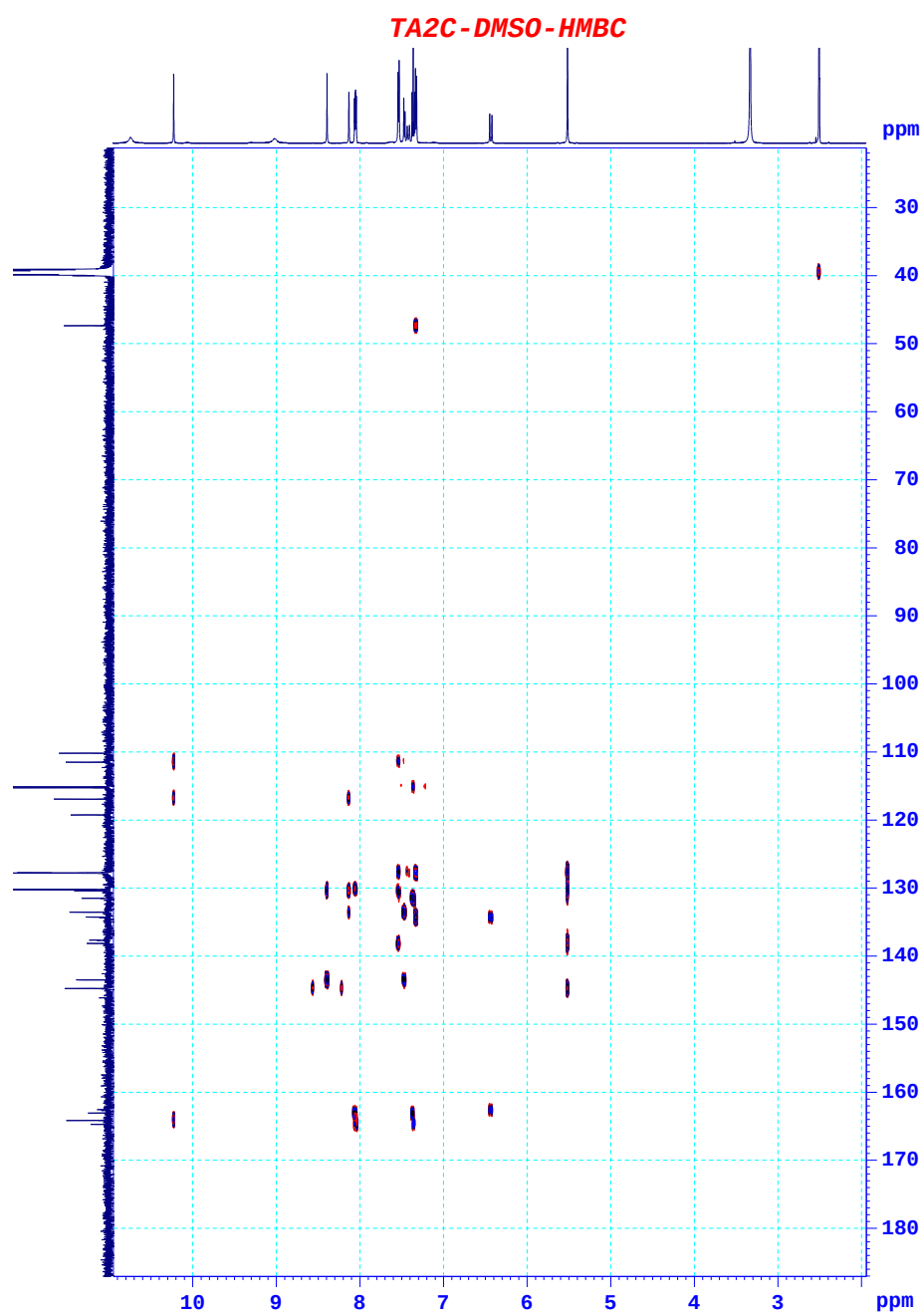


Figure S39. HMBC spectrum (600 MHz, $\text{DMSO}-d_6$) of IXc (sample code TA2-C): full spectrum.

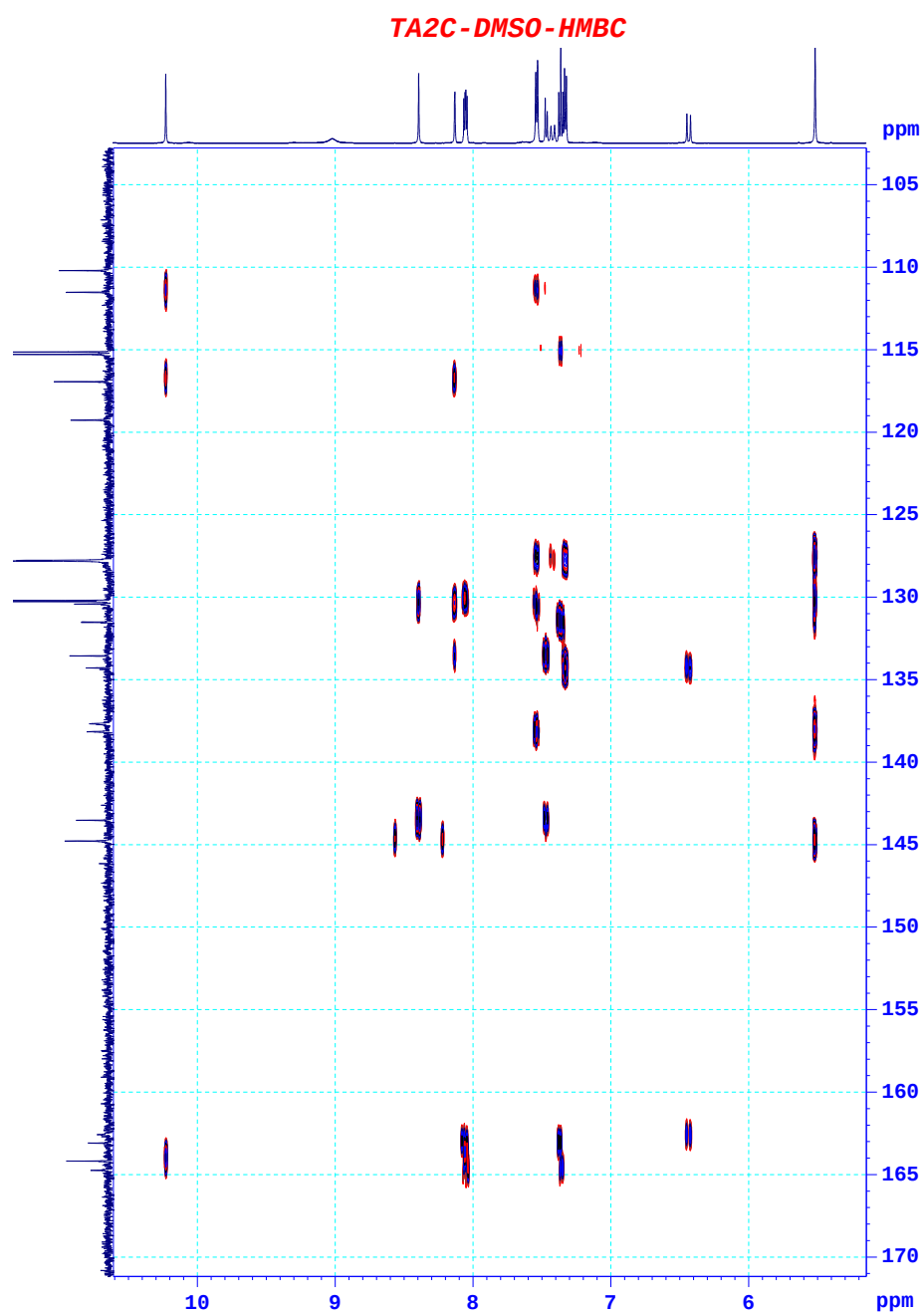


Figure S39. HMBC spectrum of IXc, continued: expansion.

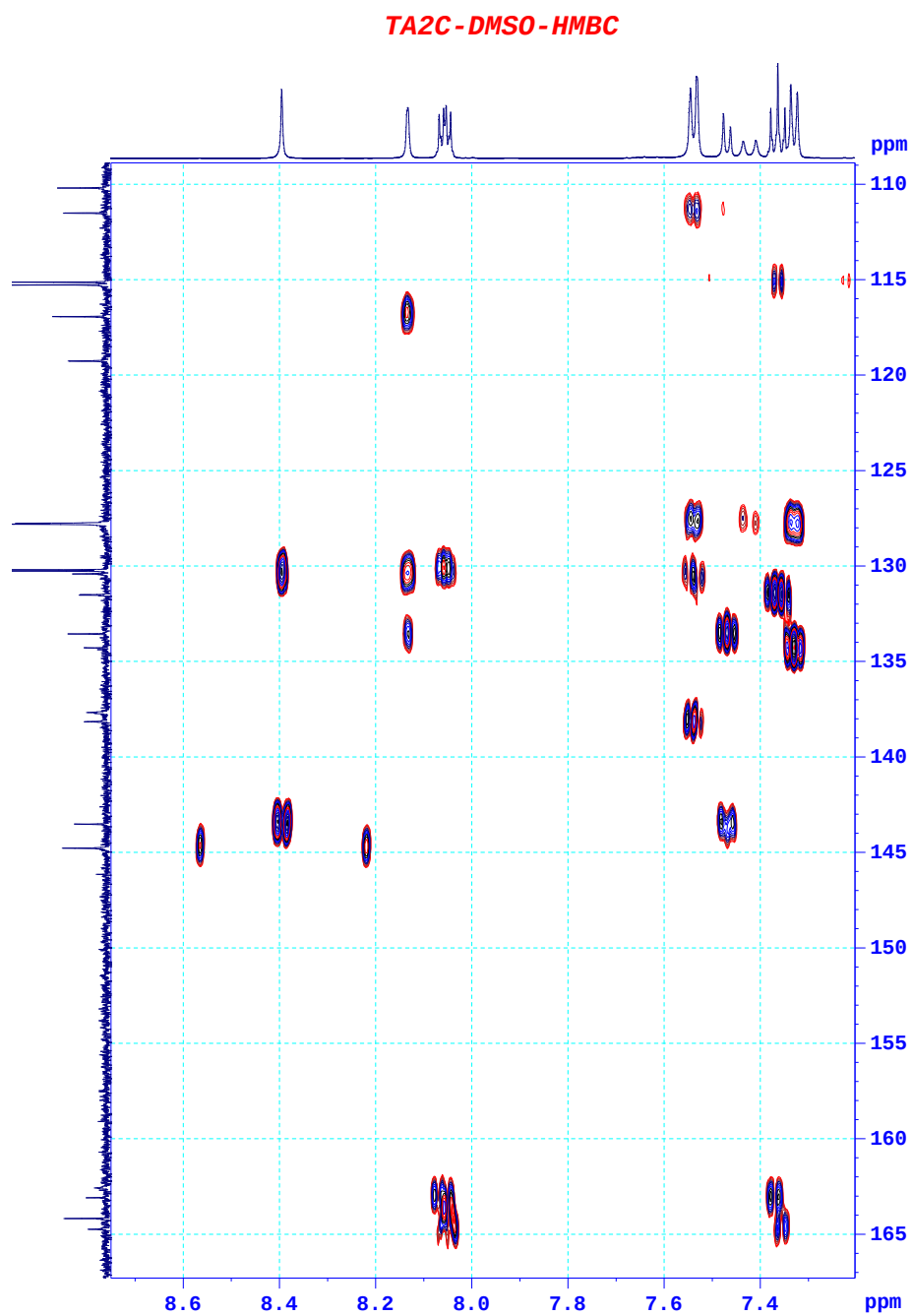


Figure S39. HMBC spectrum of IXc, continued: expansion.

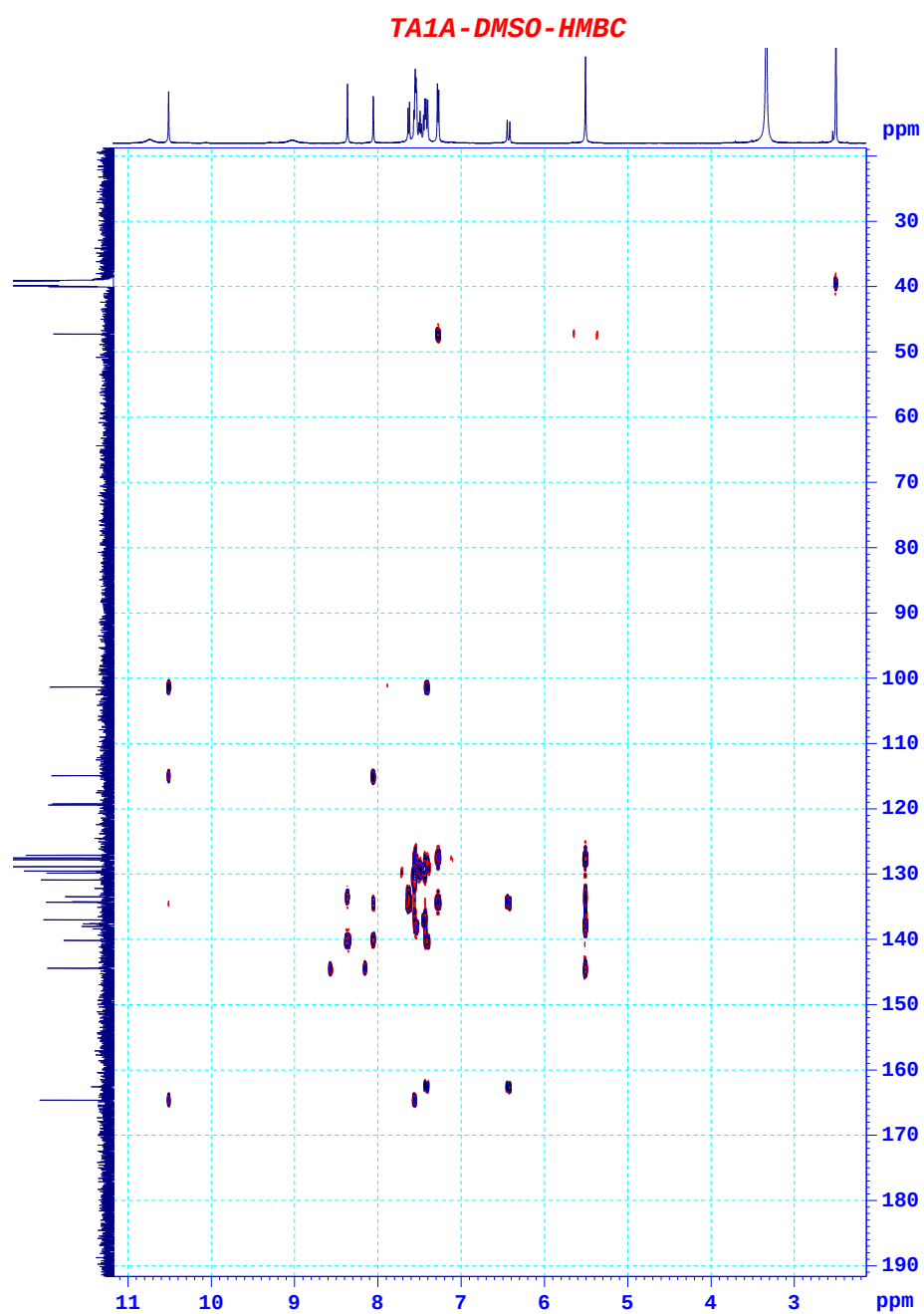


Figure S40. HMBC spectrum (500 MHz, DMSO- d_6) of IXd (sample code TA1-A): full spectrum.

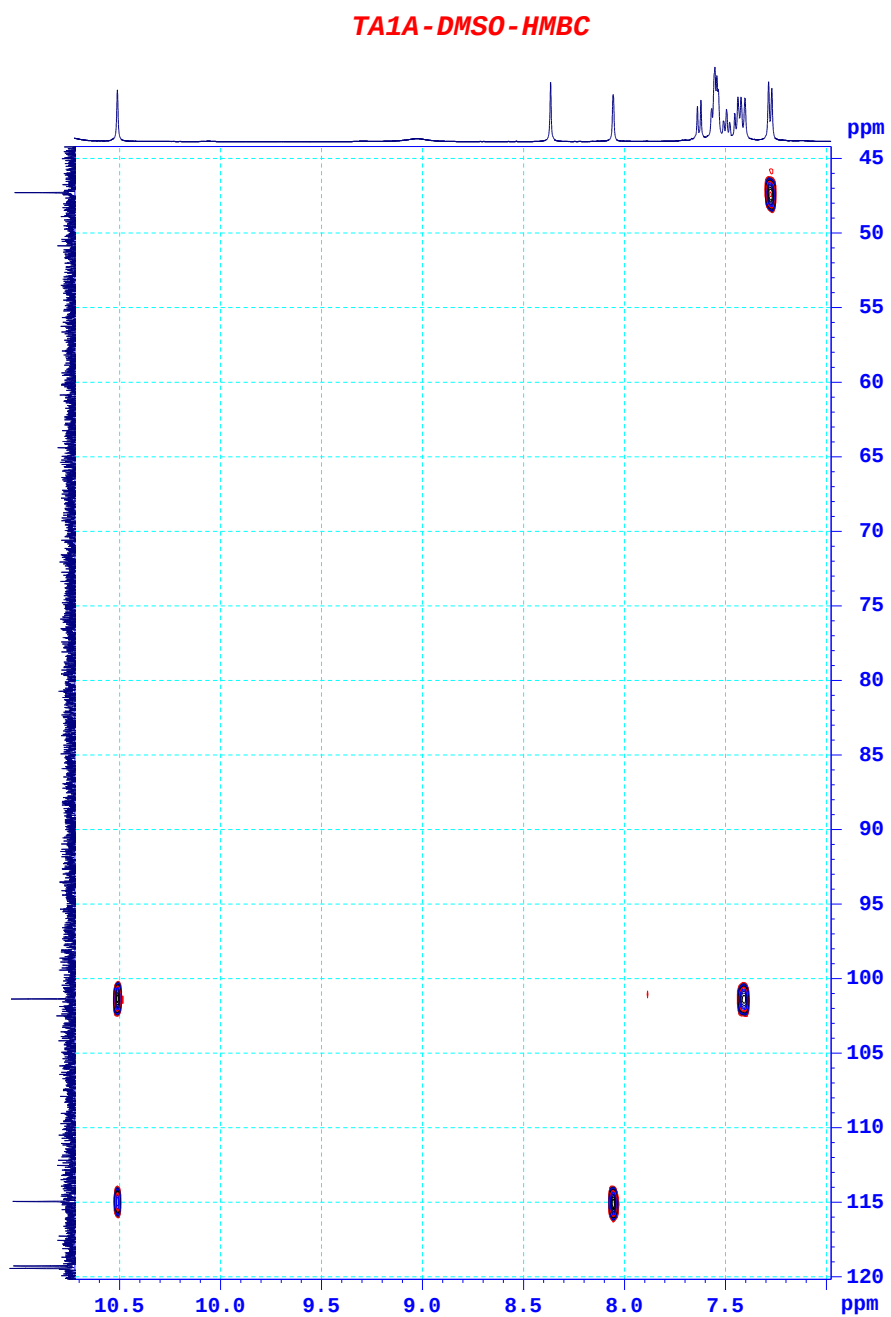


Figure S40. HMBC spectrum of IXd, continued: expansion.

TA1A-DMSO-HMBC

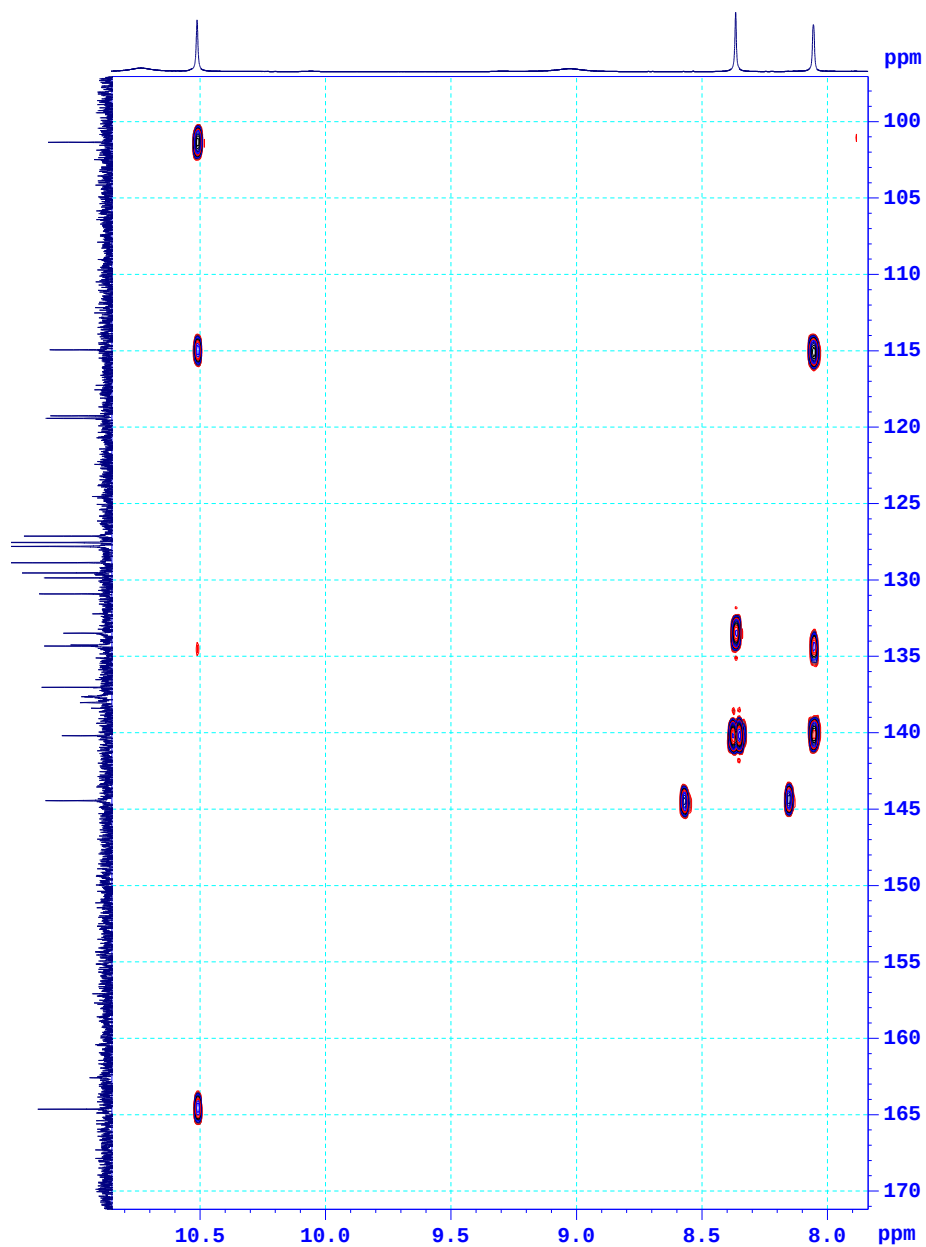


Figure S40. HMBC spectrum of IXd, continued: expansion.

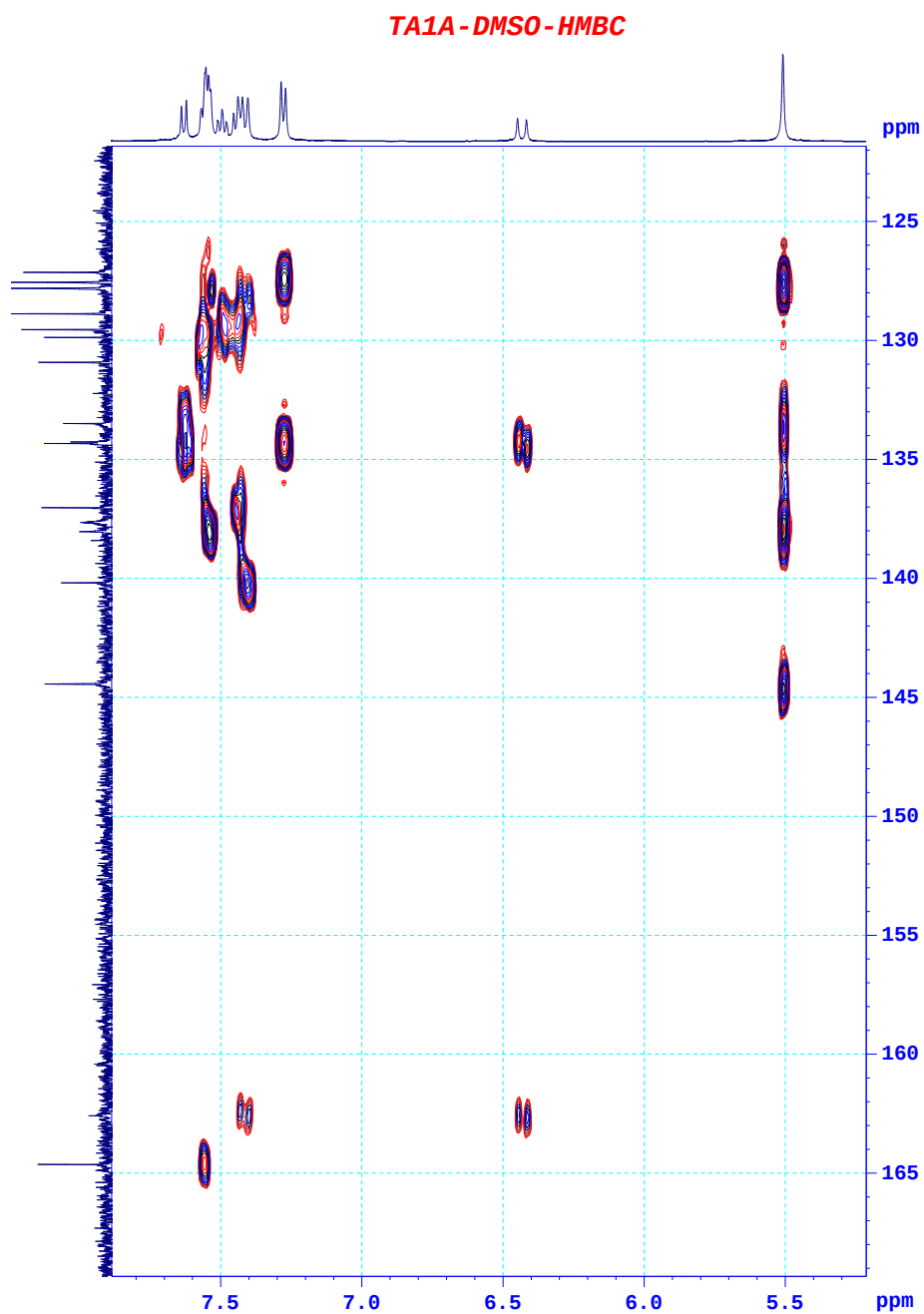


Figure S40. HMBC spectrum of IXd, continued: expansion.

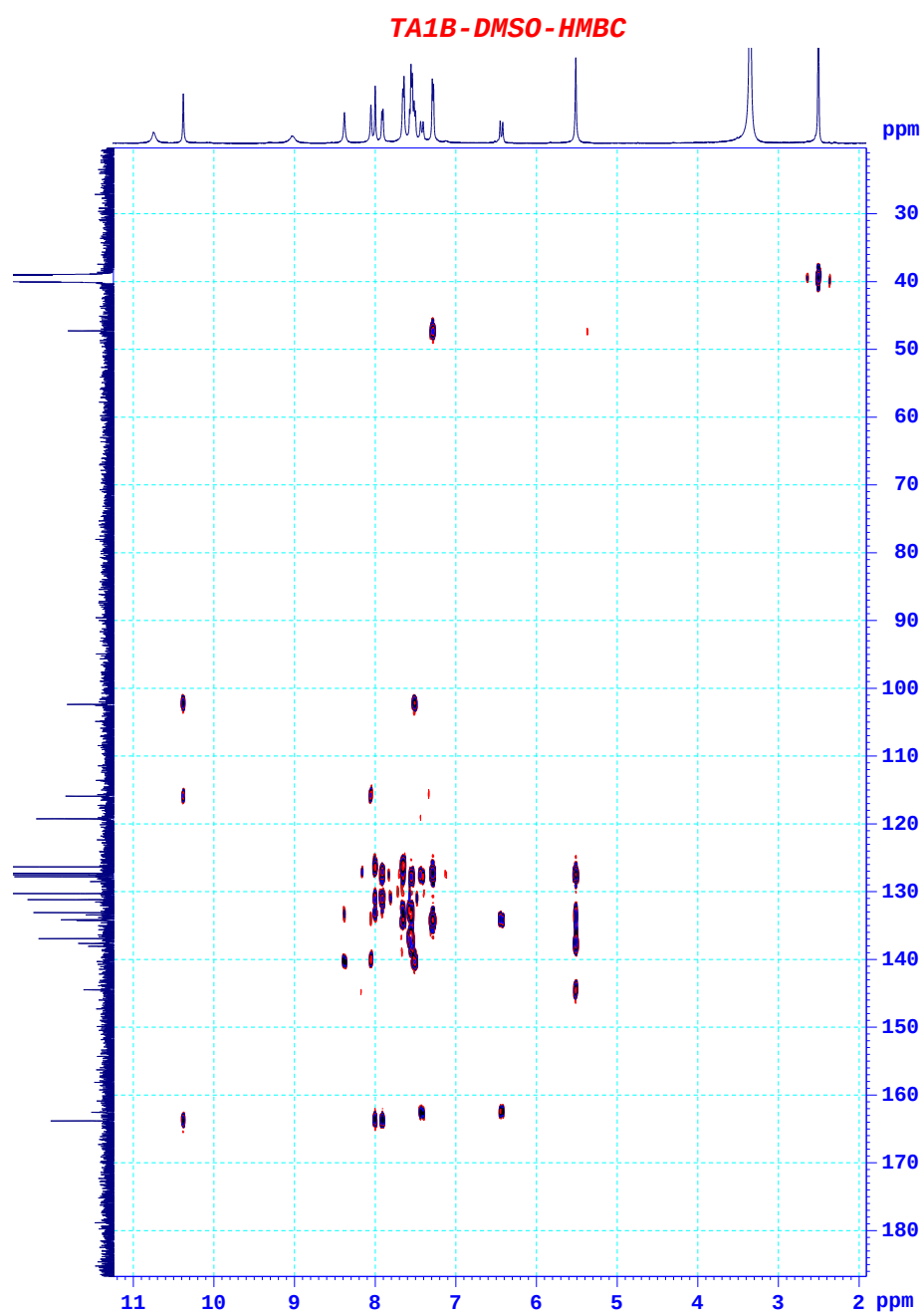


Figure S41. HMBC spectrum (500 MHz, DMSO- d_6) of IXe (sample code TA1-B): full spectrum.

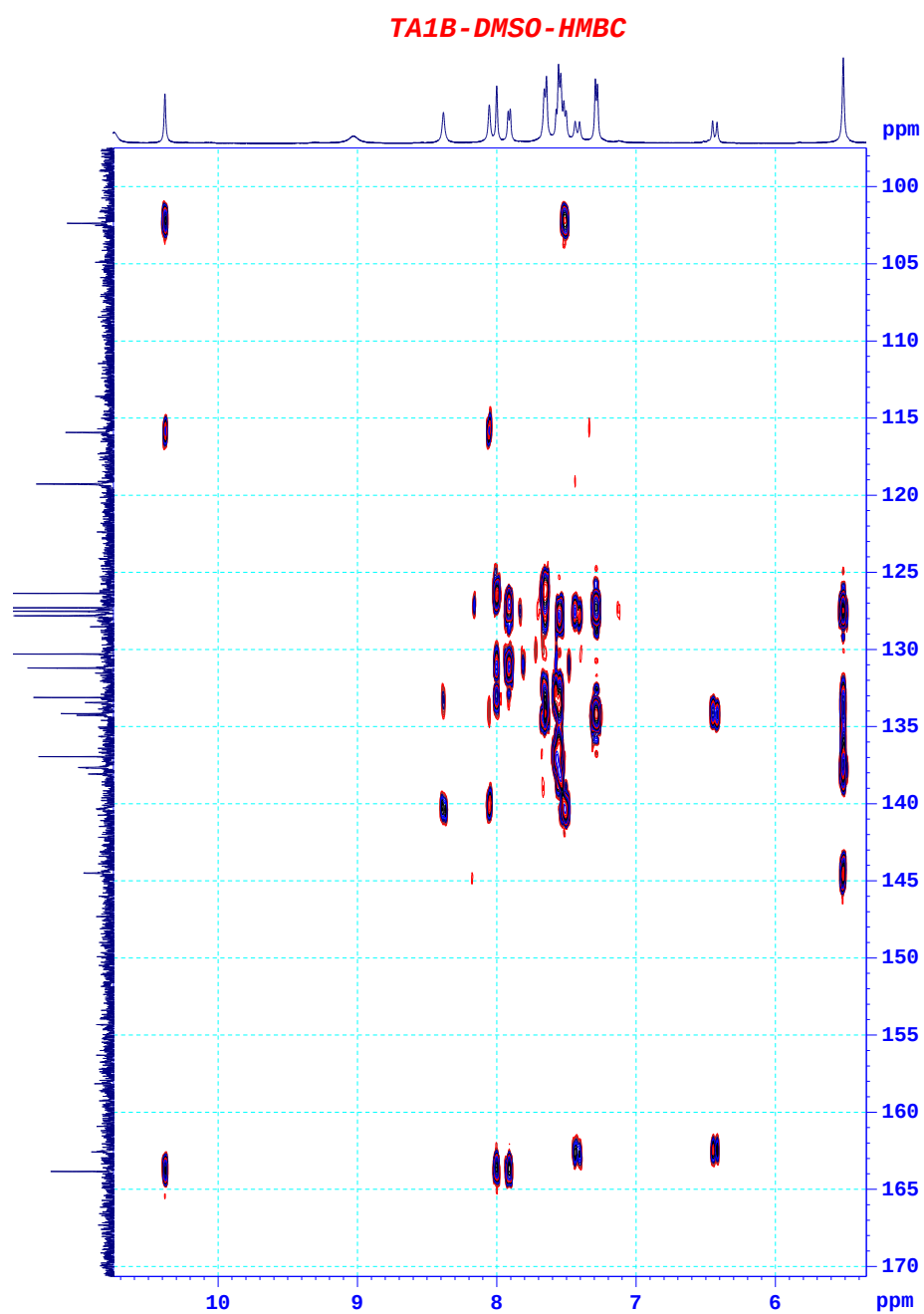


Figure S41. HMBC spectrum of IXe, continued: expansion.

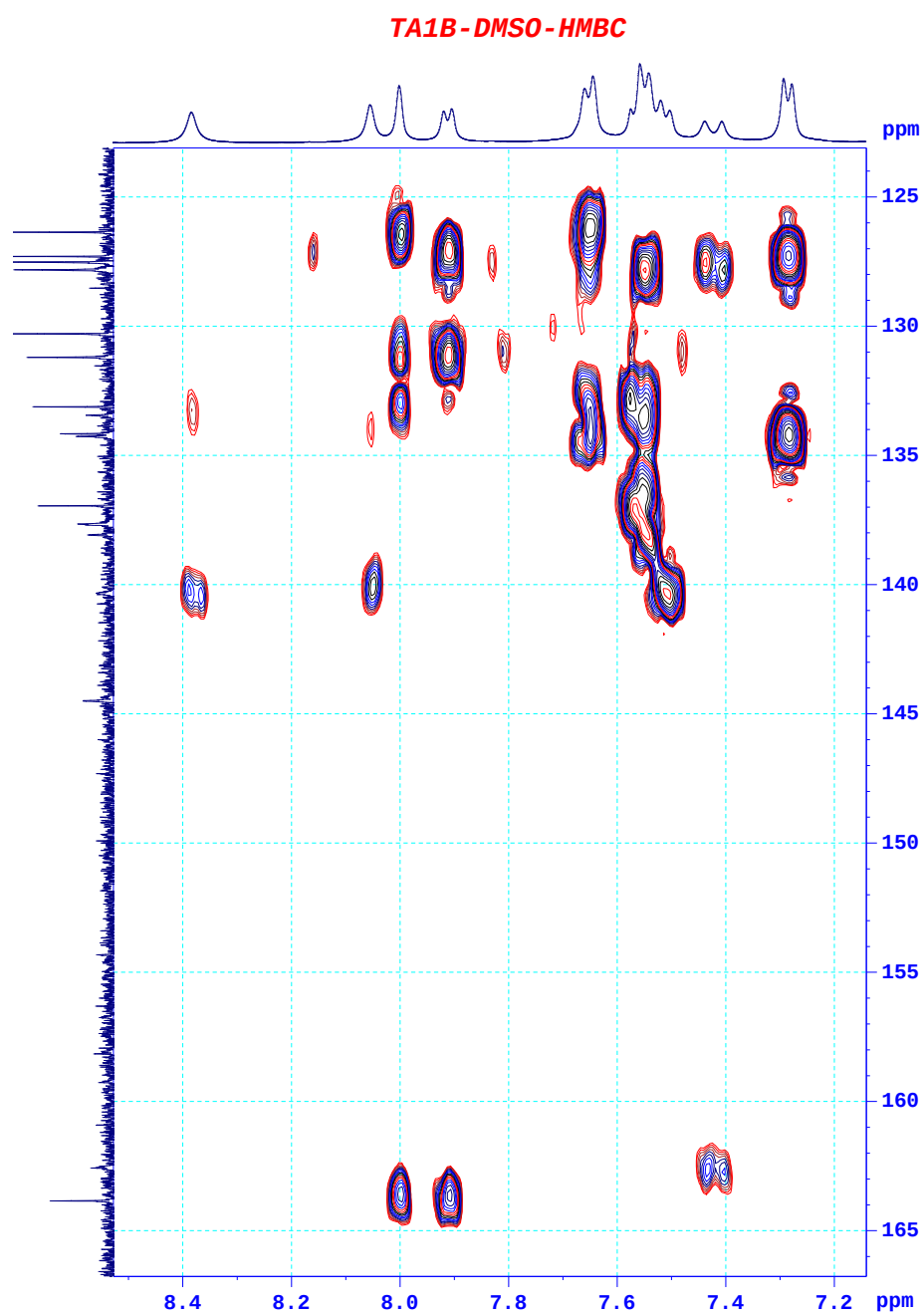


Figure S41. HMBC spectrum of IXe, continued: expansion.

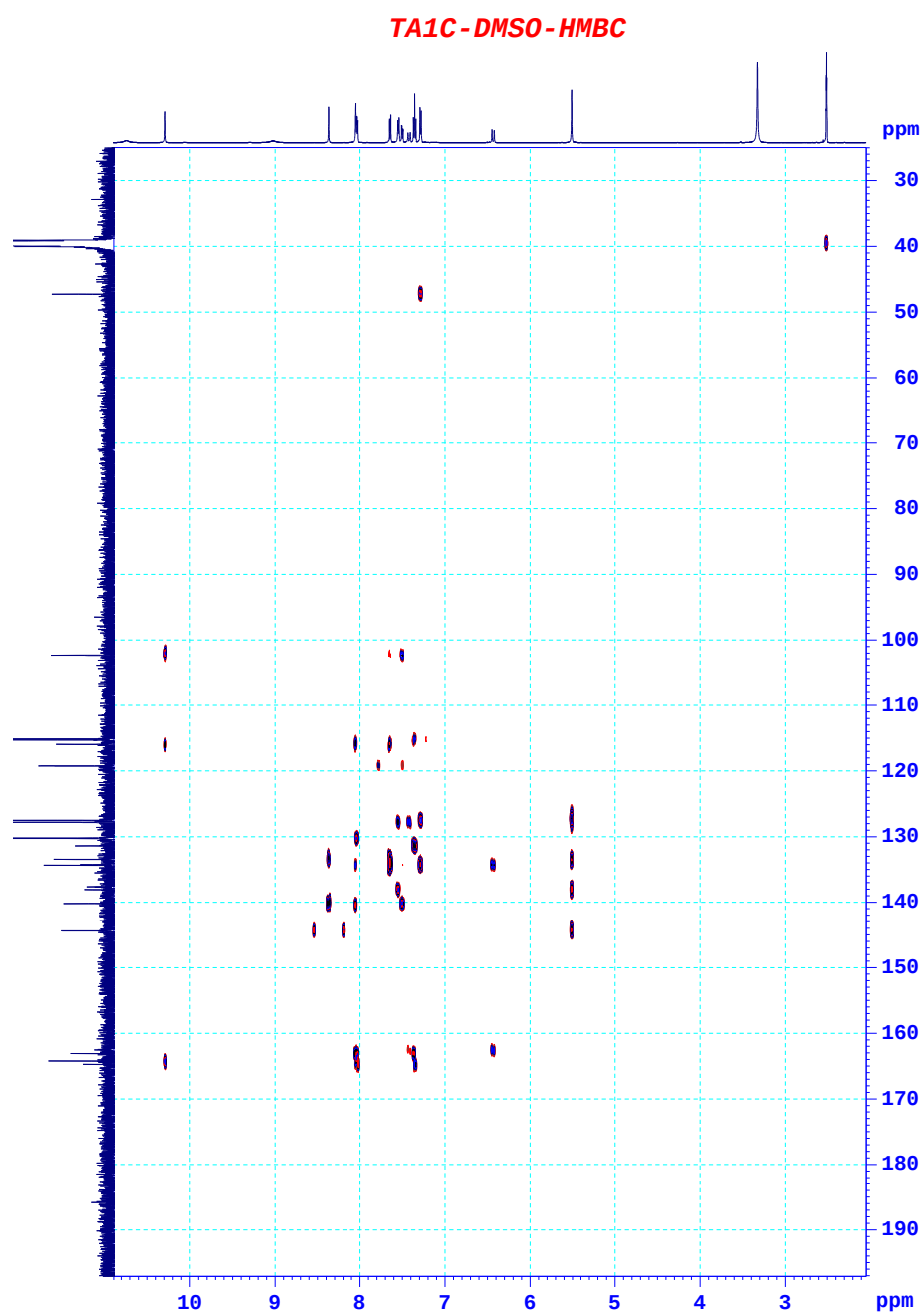


Figure S42. HMBC spectrum (600 MHz, DMSO- d_6) of IXf (sample code TA1-C): full spectrum.

TA1C-DMSO-HMBC

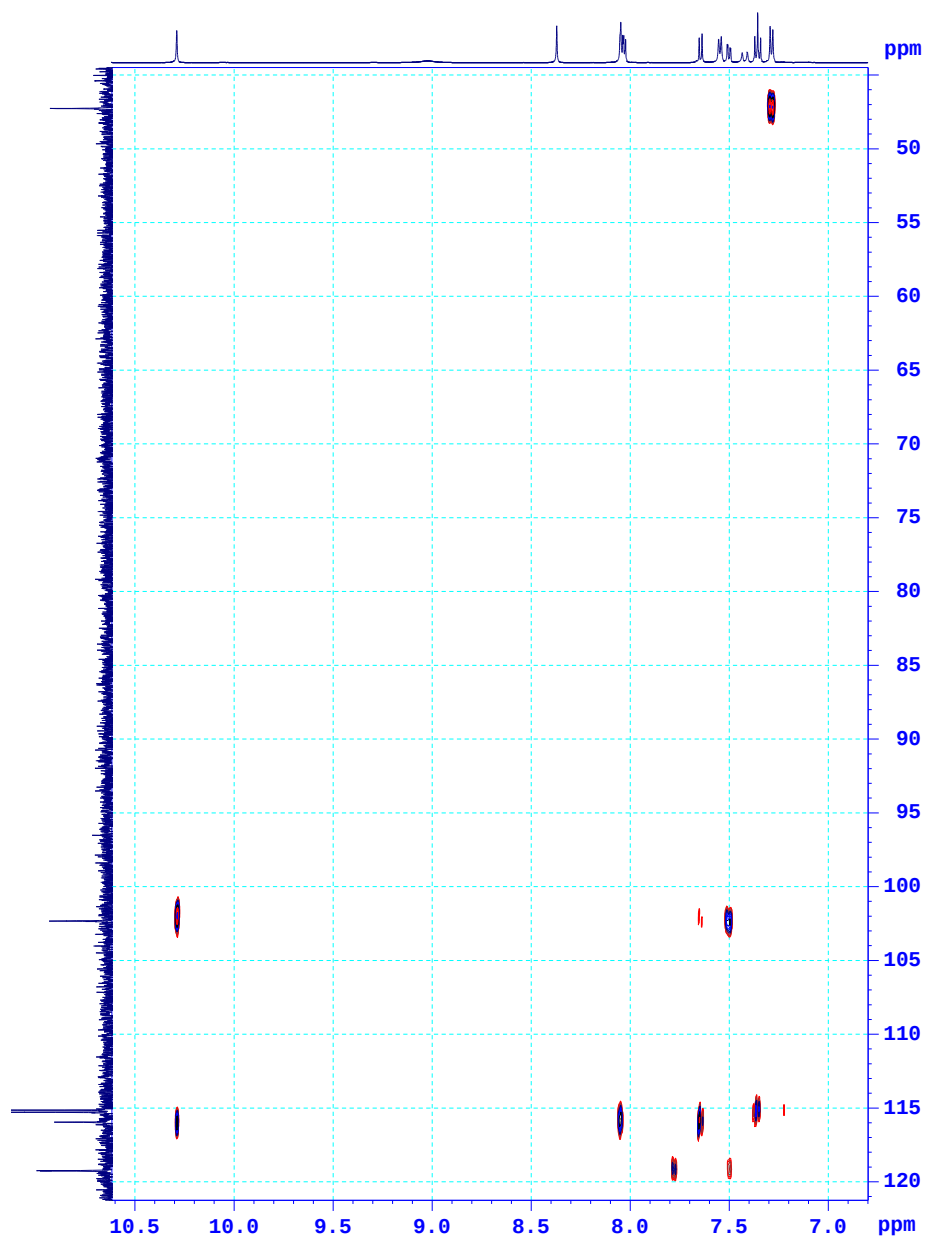


Figure S42. HMBC spectrum of IXf, continued: expansion.

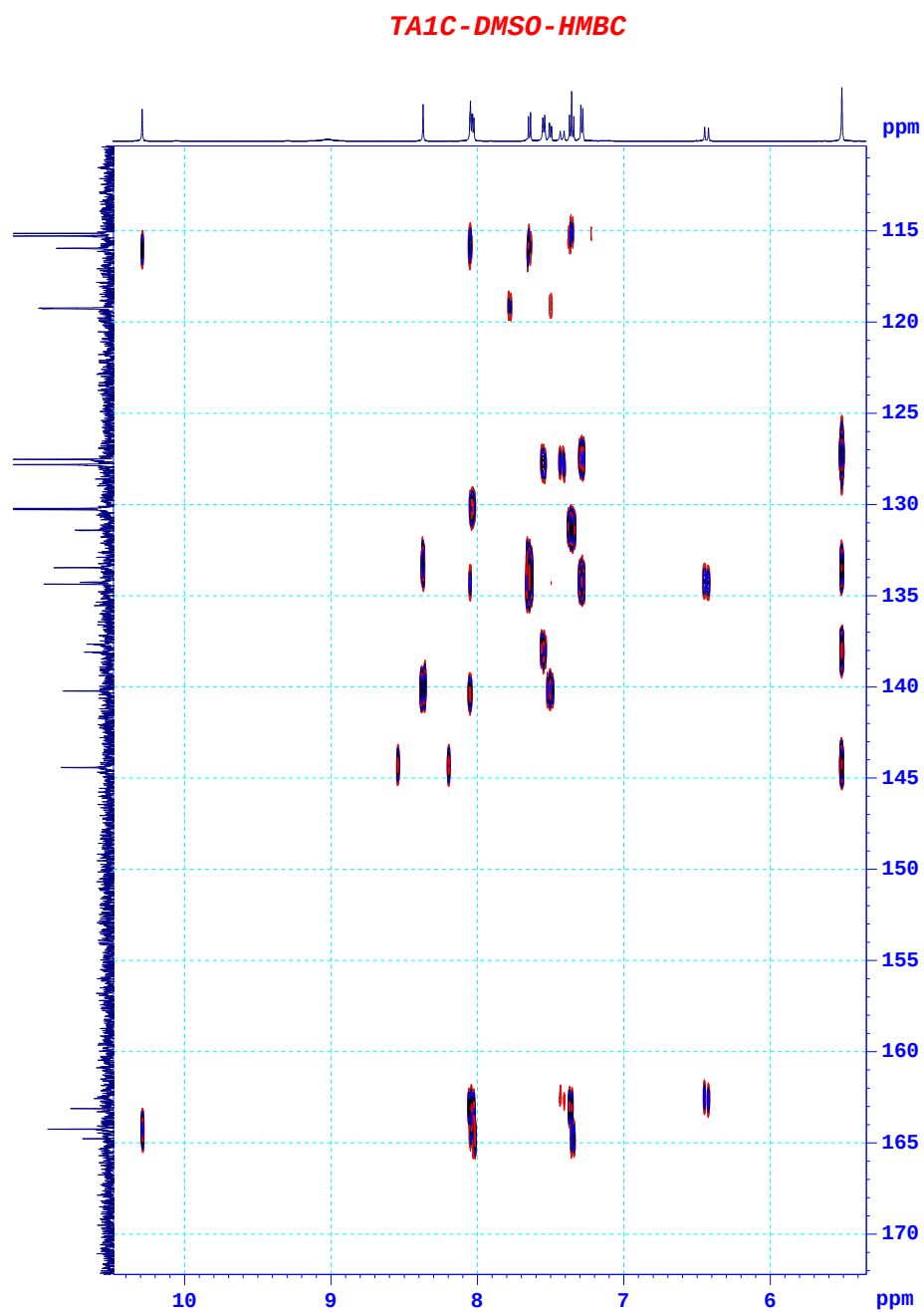


Figure S42. HMBC spectrum of IXf, continued: expansion.

TA1C-DMSO-HMBC

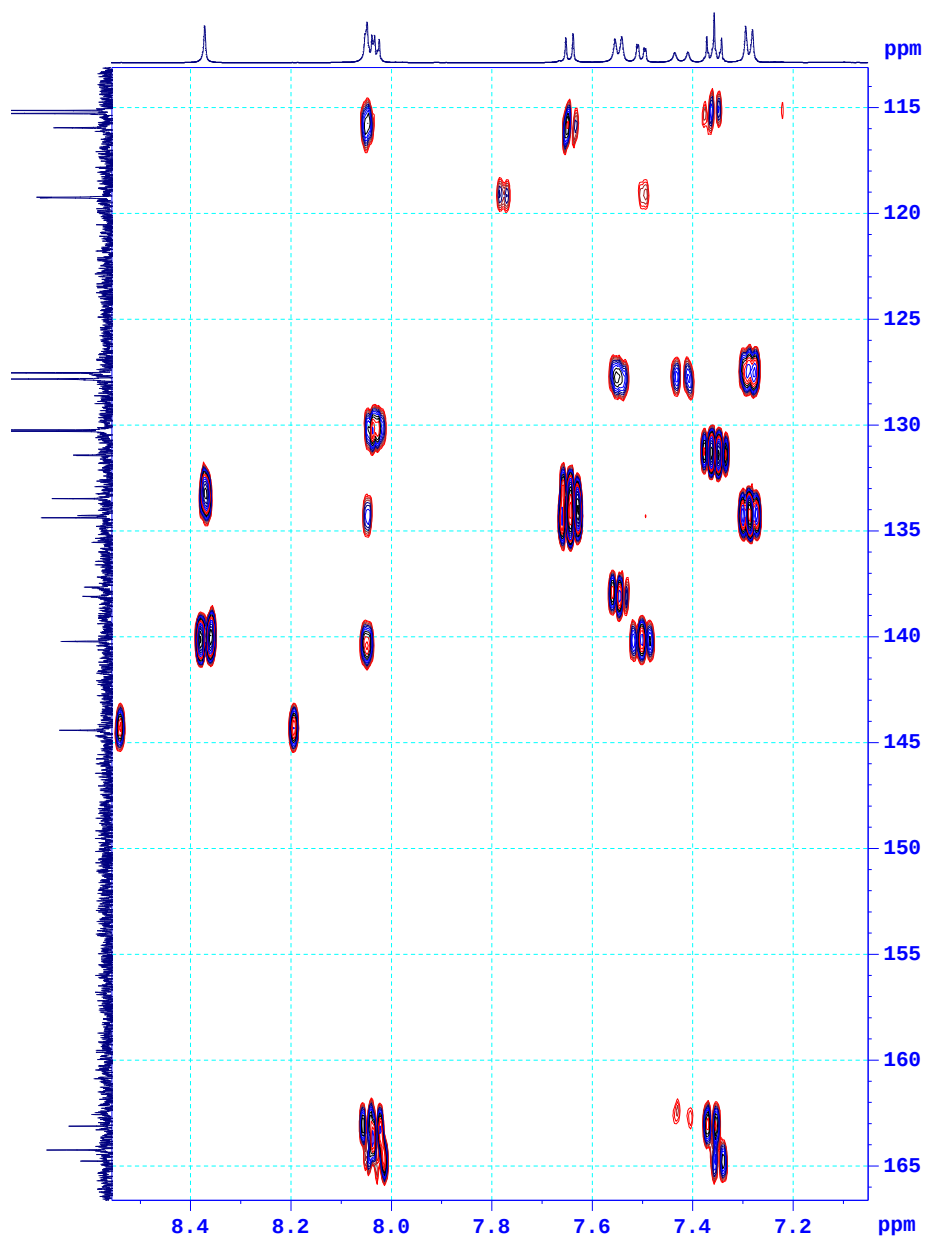


Figure S42. HMBC spectrum of IXf, continued: expansion.

S8 ROESY spectra

Phase-sensitive ROESY spectra (pulse programme `roesyphpp.2`, 200 ms spin-lock) were recorded in DMSO- d_6 at 500 MHz. The full spectrum of each compound is followed by its expansions. The $N1$ -methylene protons (δ 5.51) correlate with H-7, which is an ortho-coupled doublet at δ 7.46–7.48 in IXa–IXc and the isolated singlet at δ 8.04–8.05 in IXd–IXf.

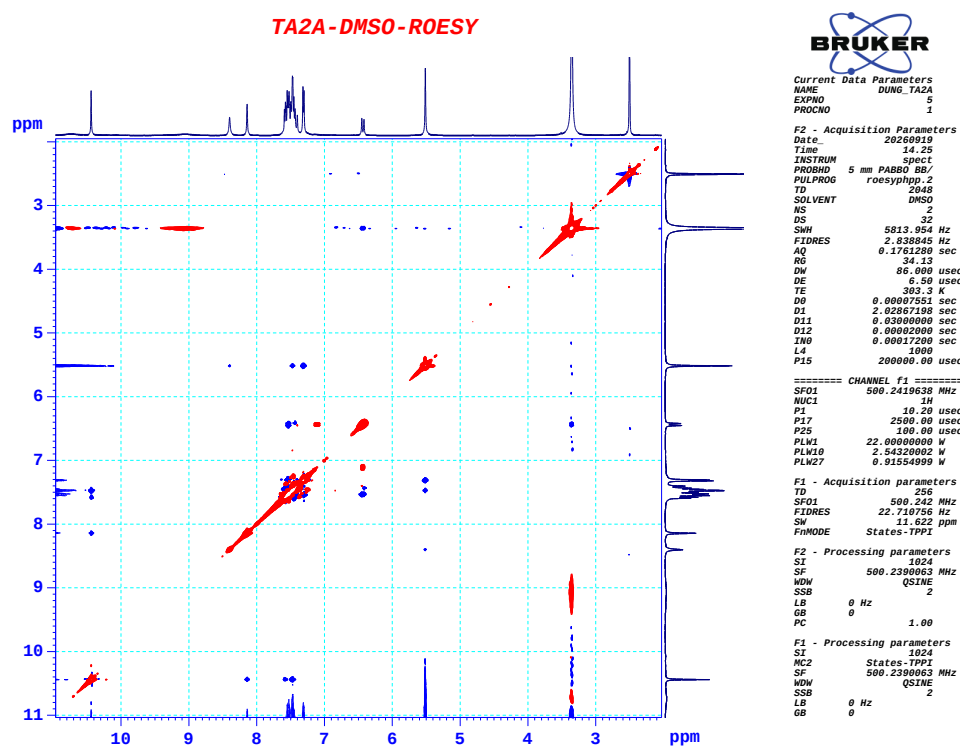


Figure S43. Phase-sensitive ROESY (200 ms spin-lock) spectrum (500 MHz, DMSO- d_6) of IXa (sample code TA2-A): full spectrum.

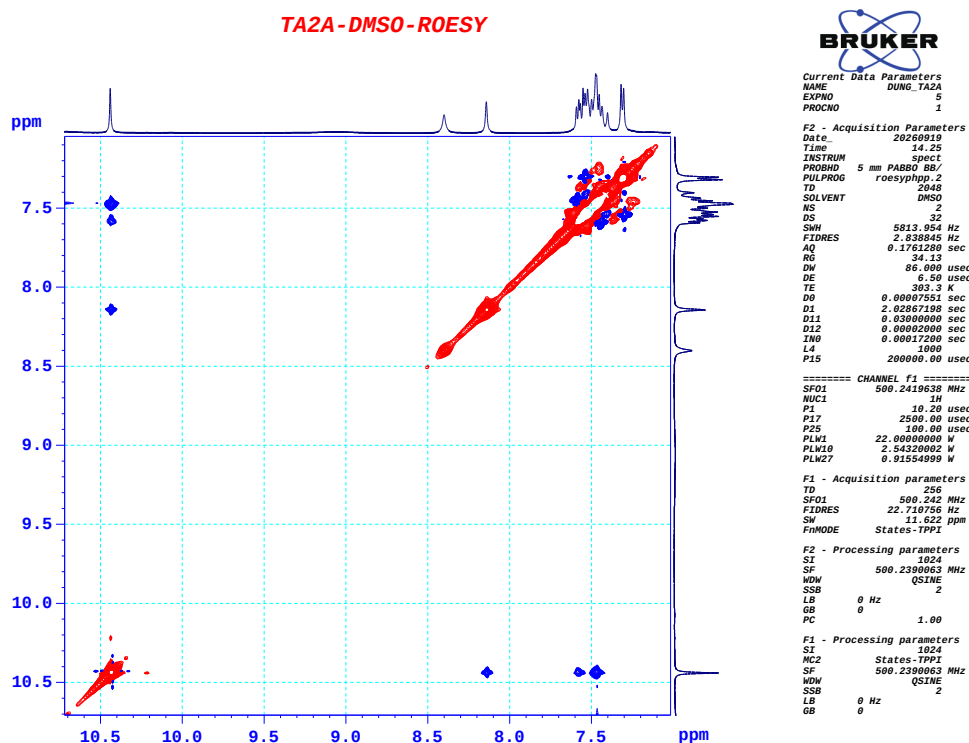


Figure S43. ROESY spectrum of IXa, continued: expansion.

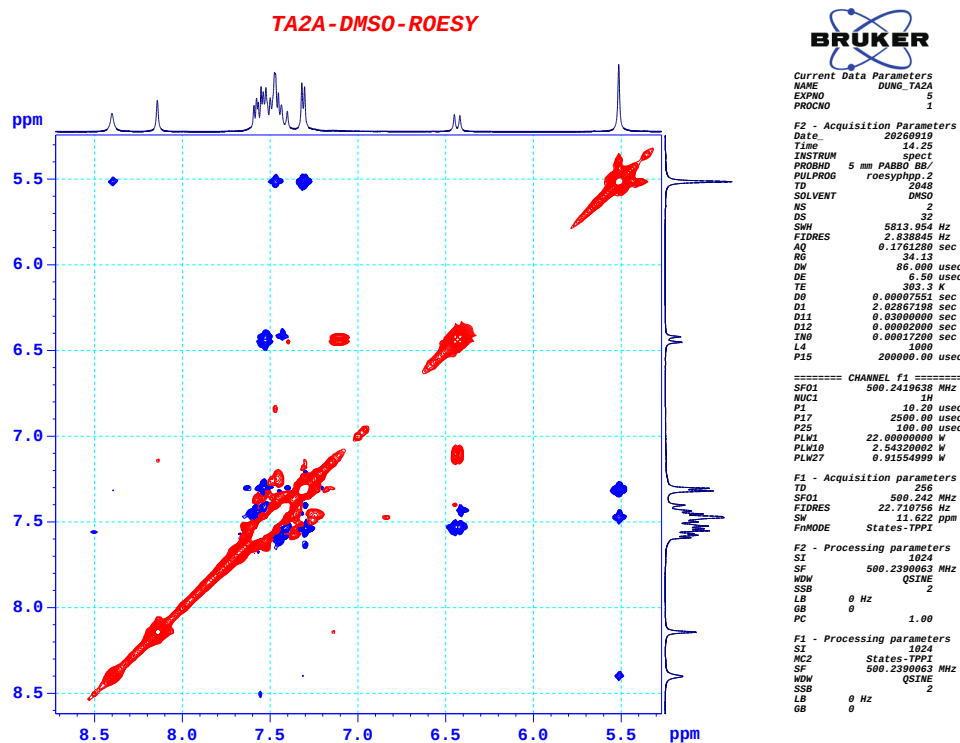


Figure S43. ROESY spectrum of IXa, continued: expansion.

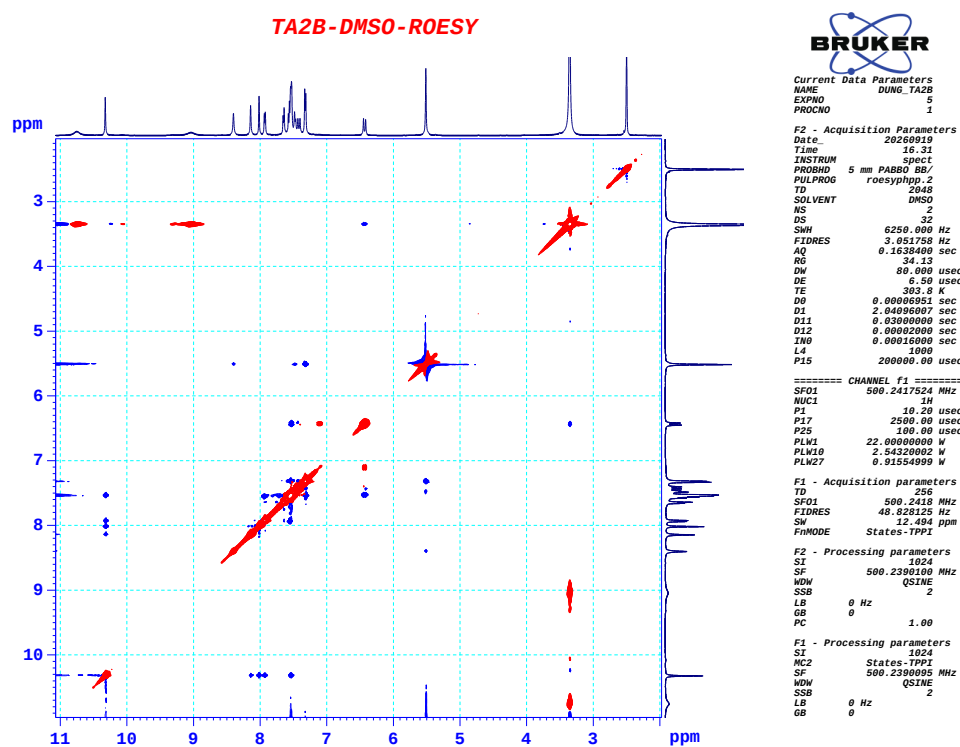


Figure S44. Phase-sensitive ROESY (200 ms spin-lock) spectrum (500 MHz, DMSO- d_6) of IXb (sample code TA2-B): full spectrum.

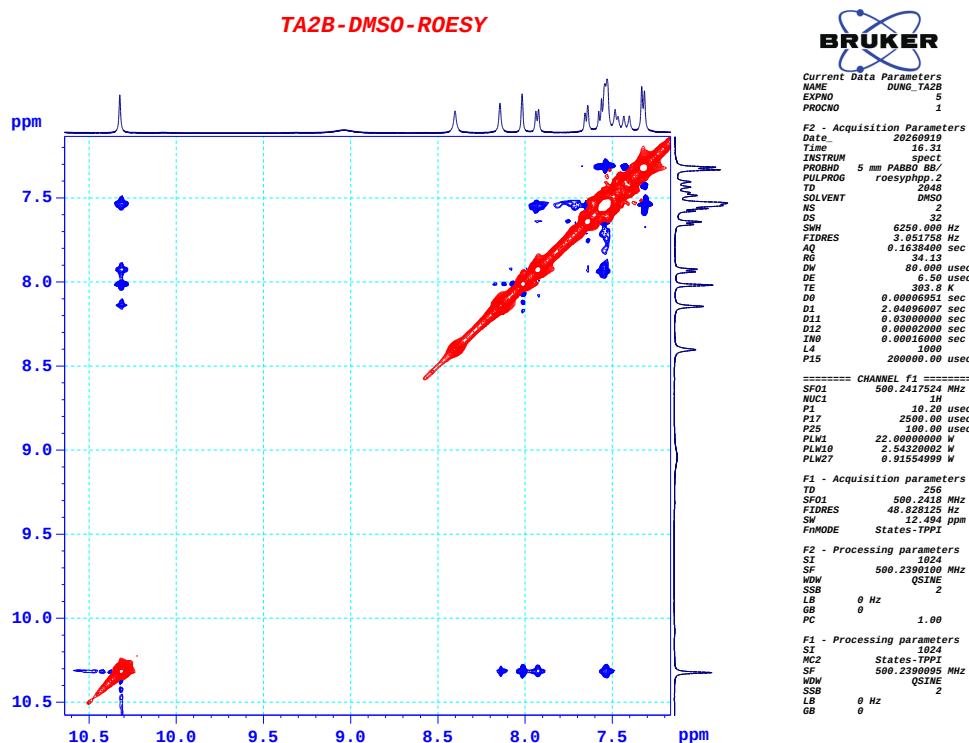


Figure S44. ROESY spectrum of IXb, continued: expansion.

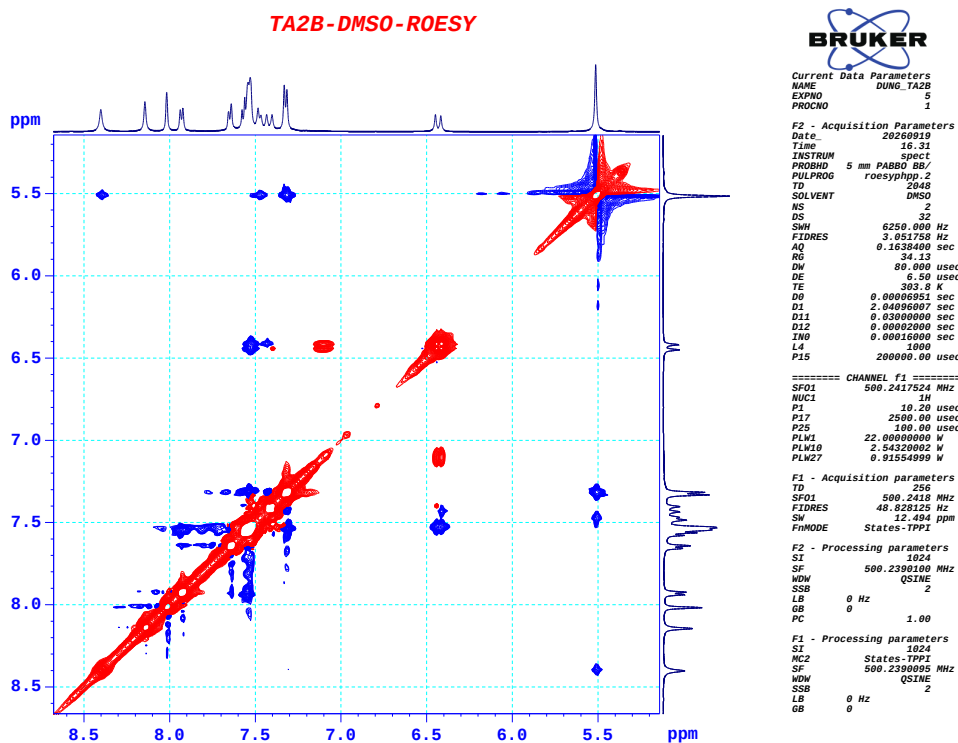


Figure S44. ROESY spectrum of IXb, continued: expansion.

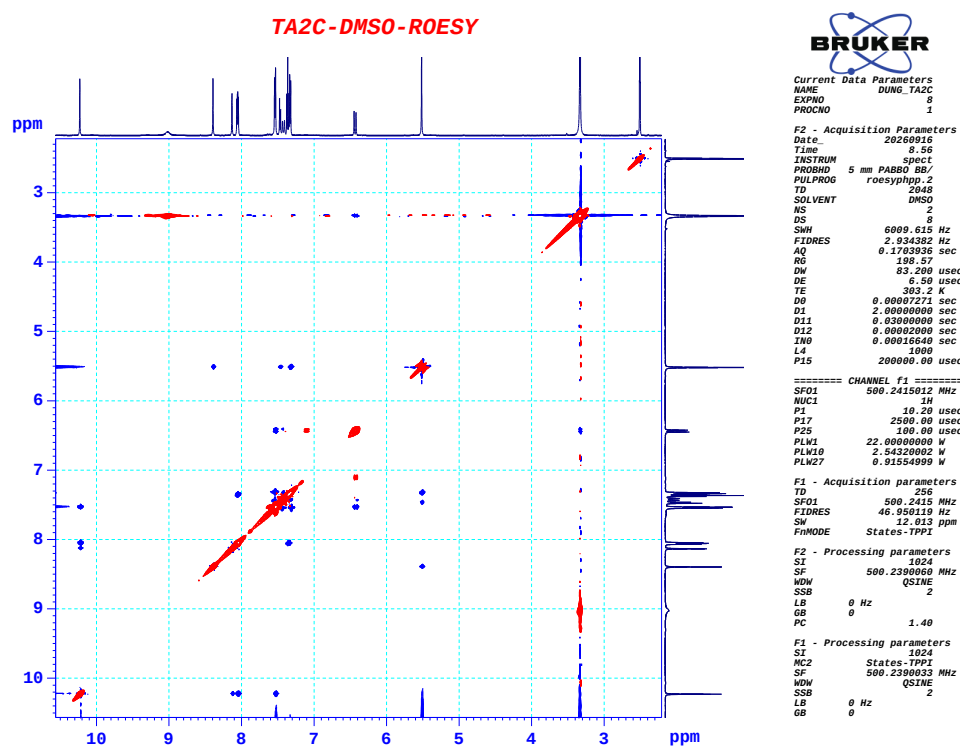


Figure S45. Phase-sensitive ROESY (200 ms spin-lock) spectrum (500 MHz, DMSO- d_6) of IXc (sample code TA2-C): full spectrum.

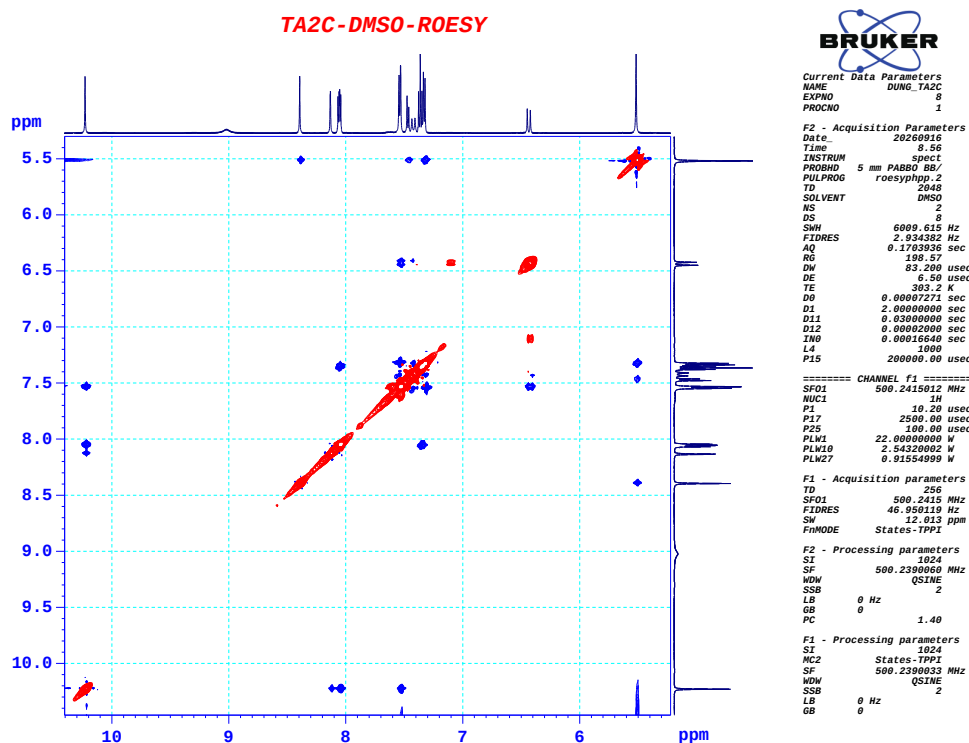


Figure S45. ROESY spectrum of IXc, continued: expansion.

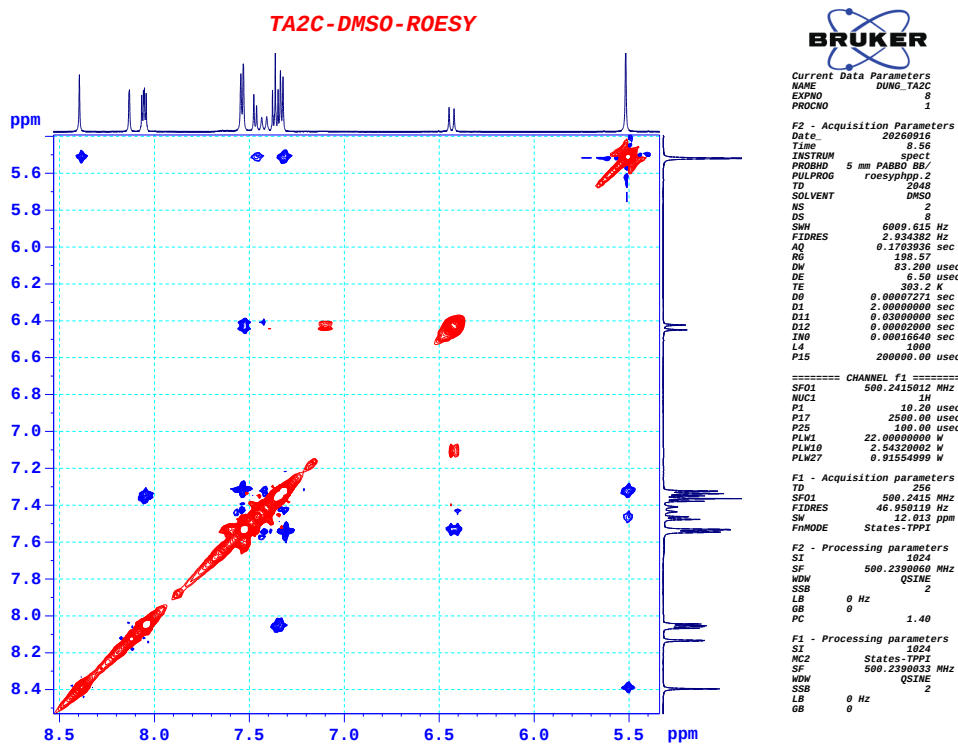


Figure S45. ROESY spectrum of IXc, continued: expansion.

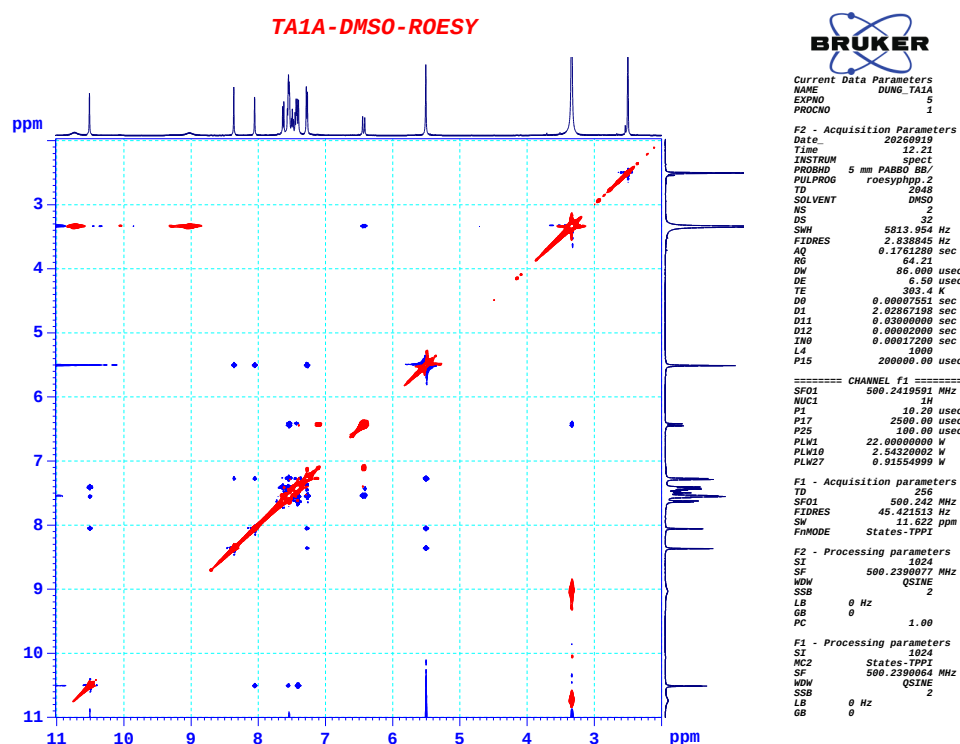


Figure S46. Phase-sensitive ROESY (200 ms spin-lock) spectrum (500 MHz, DMSO- d_6) of IXd (sample code TA1-A): full spectrum.

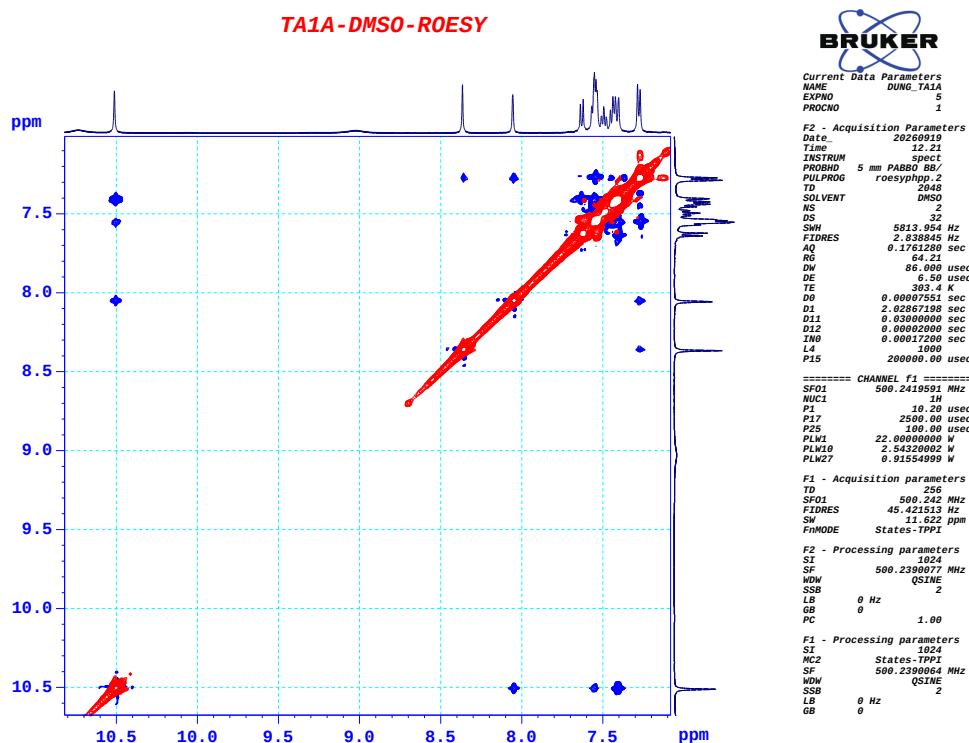


Figure S46. ROESY spectrum of IXd, continued: expansion.

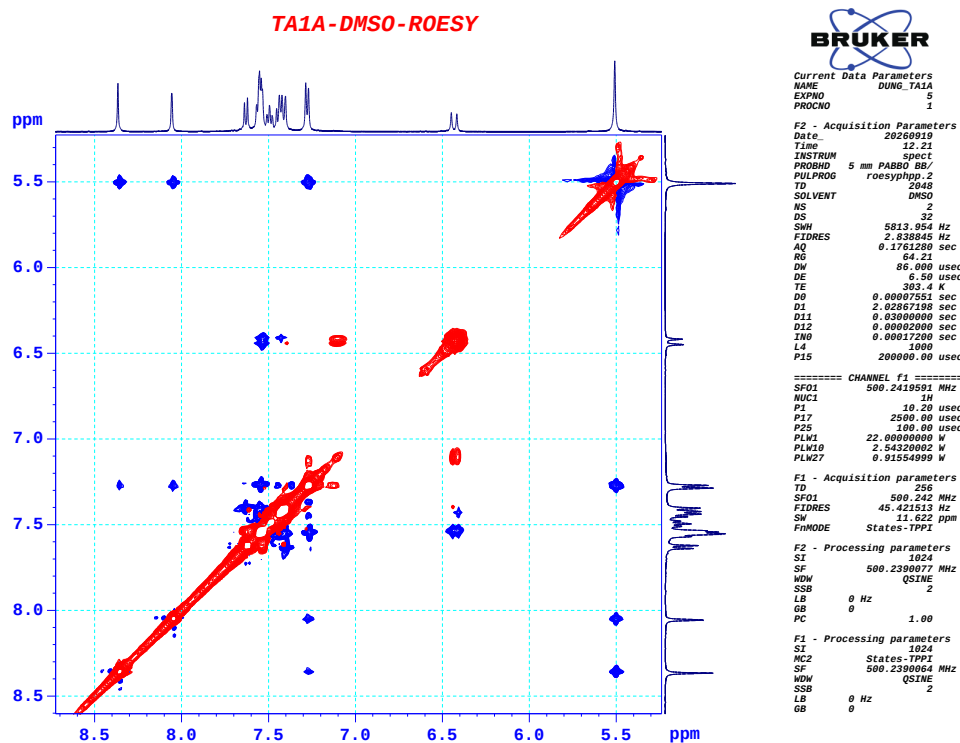


Figure S46. ROESY spectrum of IXd, continued: expansion.

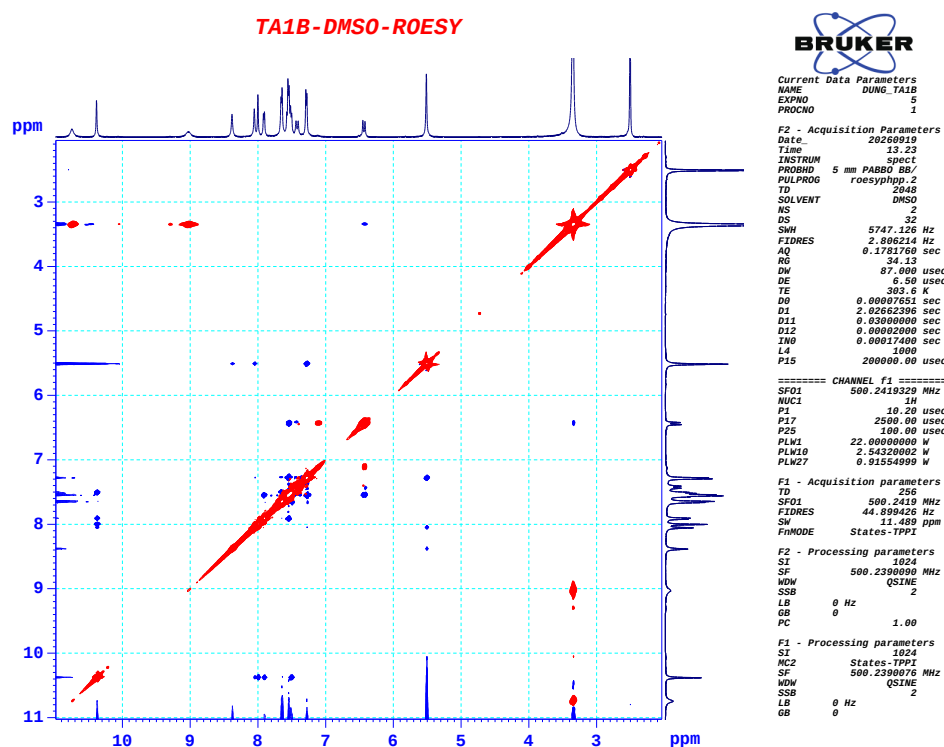


Figure S47. Phase-sensitive ROESY (200 ms spin-lock) spectrum (500 MHz, DMSO- d_6) of IXe (sample code TA1-B): full spectrum.

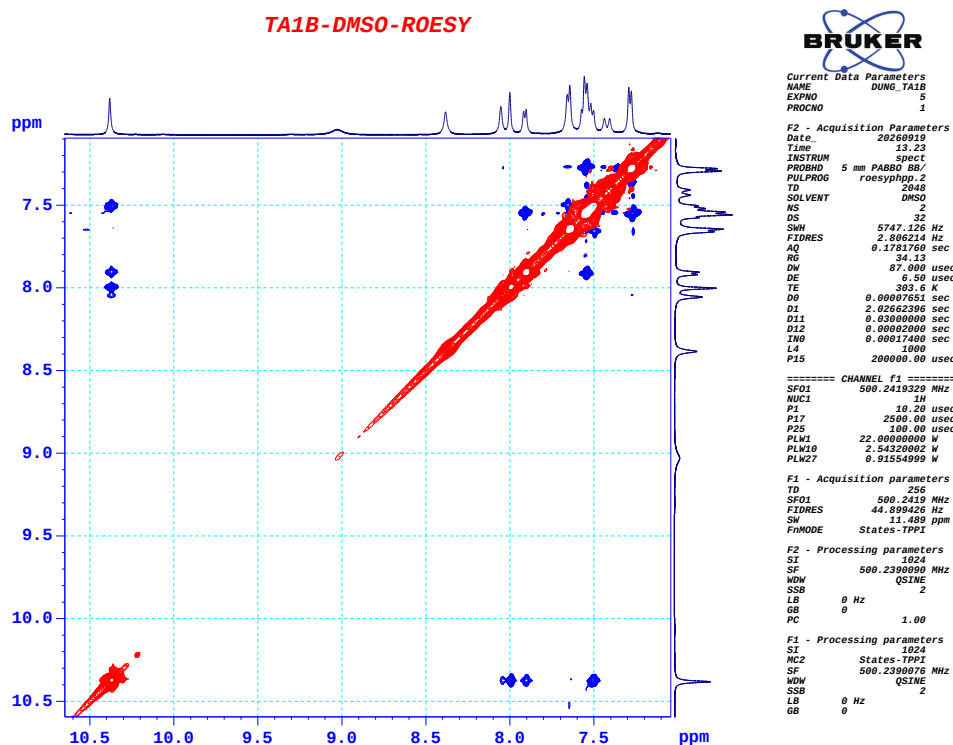


Figure S47. ROESY spectrum of IXe, continued: expansion.

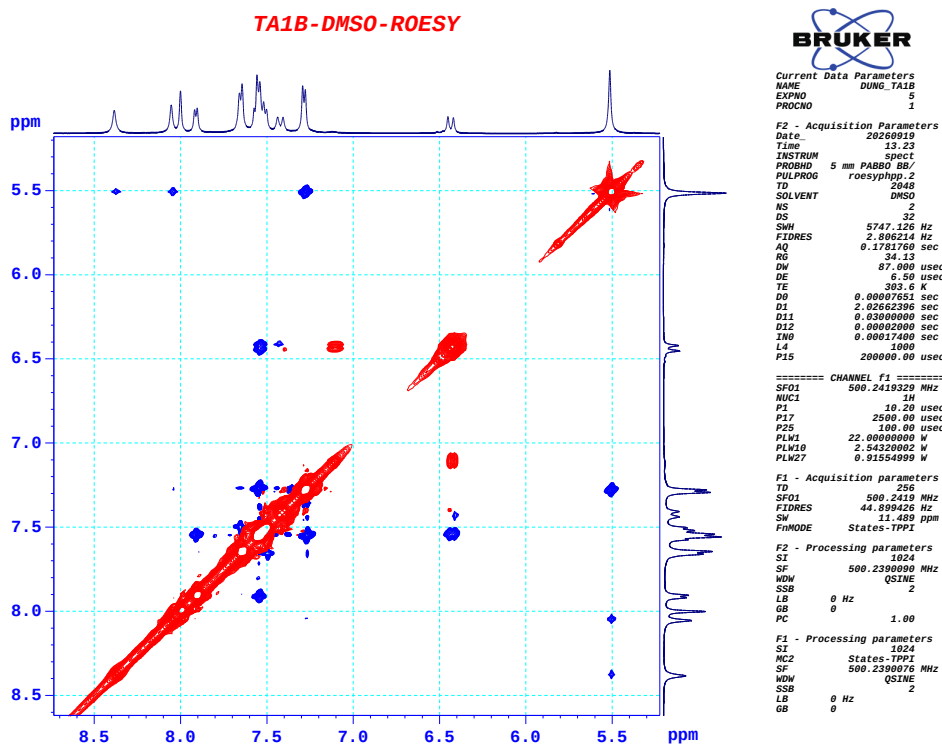


Figure S47. ROESY spectrum of IXe, continued: expansion.

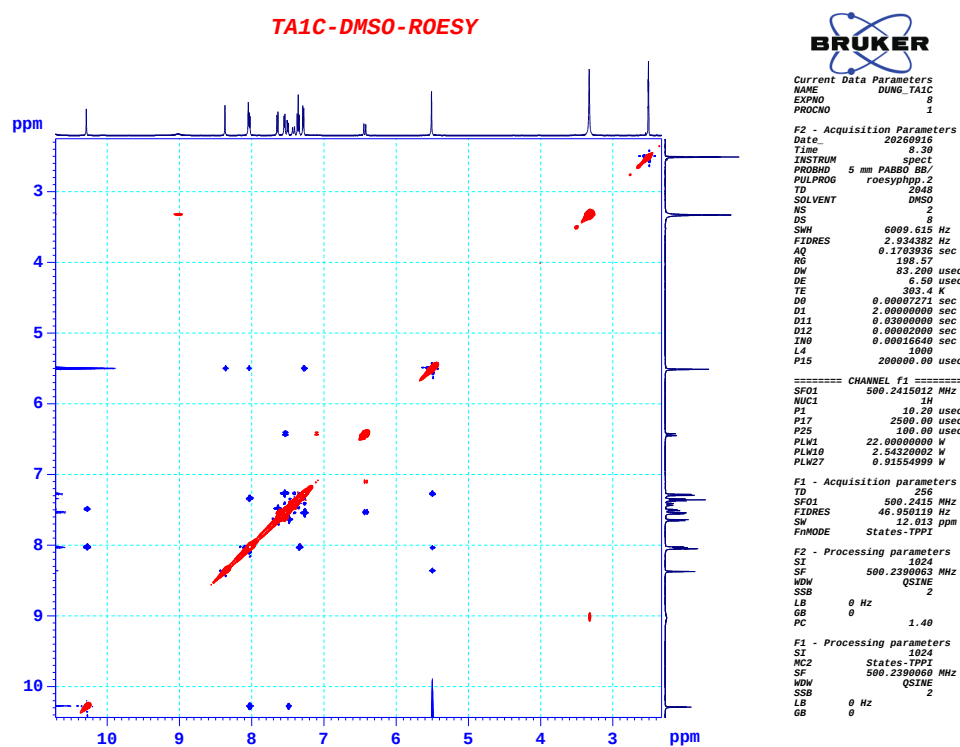


Figure S48. Phase-sensitive ROESY (200 ms spin-lock) spectrum (500 MHz, DMSO- d_6) of IXf (sample code TA1-C): full spectrum.

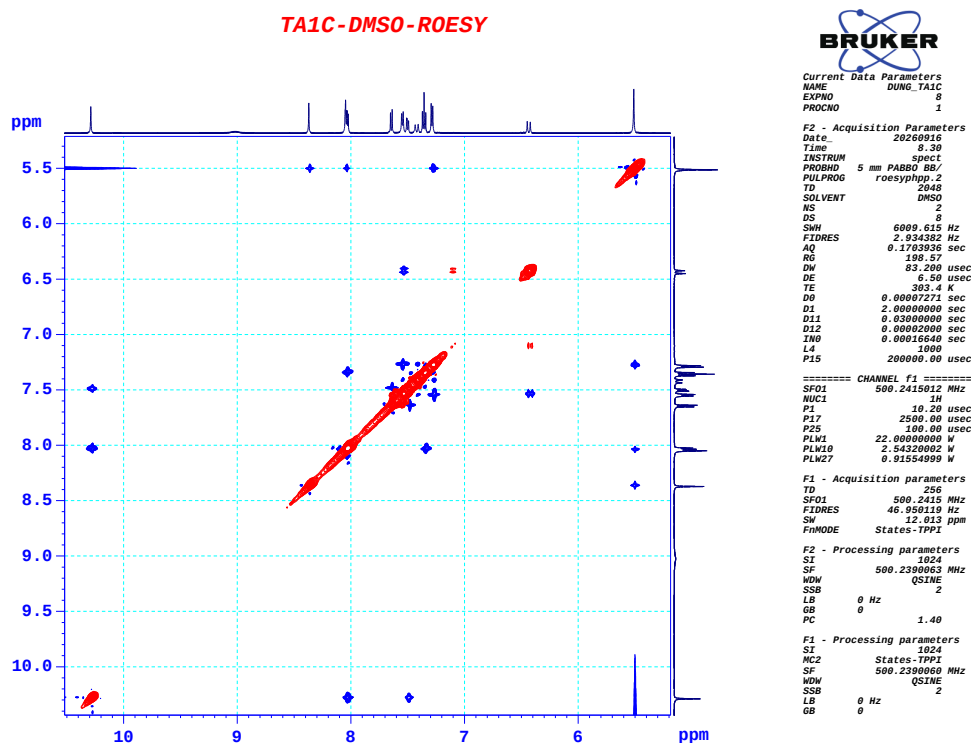


Figure S48. ROESY spectrum of IXf, continued: expansion.

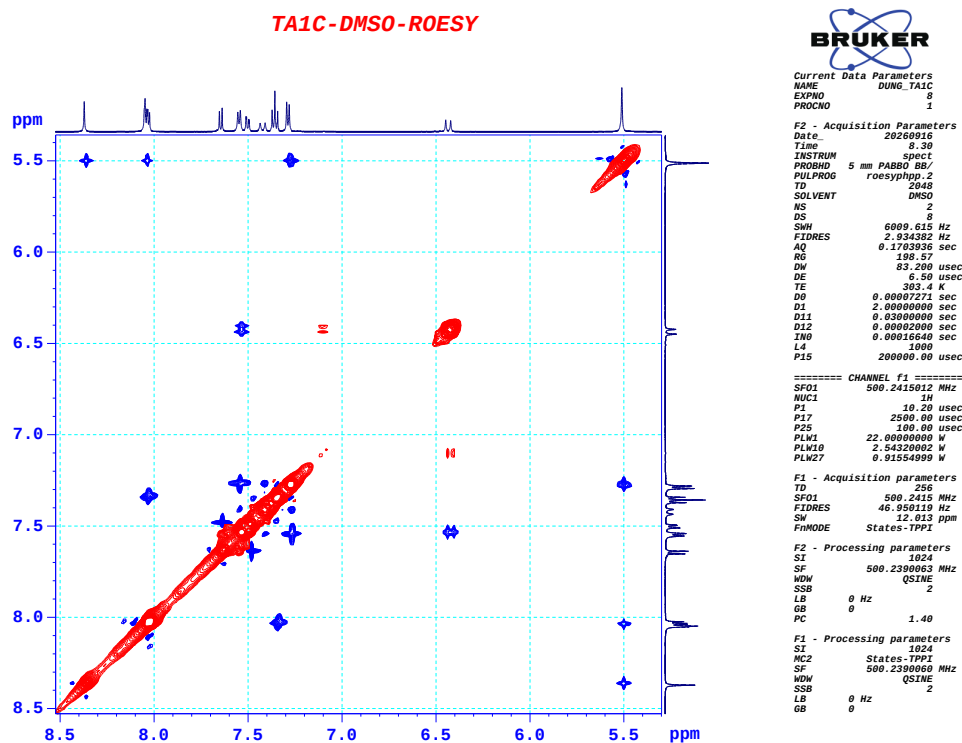


Figure S48. ROESY spectrum of IXf, continued: expansion.

S9 Molecular docking scores against five HDAC isoforms

The six series IX compounds and SAHA were docked with AutoDock Vina against the crystal structures of HDAC1 (PDB 5ICN), HDAC2 (4LXZ), HDAC3 (4A69), HDAC6 (5EDU) and HDAC8 (1T69).

Table S2. Docking scores (kcal mol^{-1}) of the six series IX compounds and SAHA against five zinc-dependent HDAC isoforms.

Cpd	HDAC1	HDAC2	HDAC3	HDAC6	HDAC8
IXa	-9.0	-9.7	-9.1	-10.5	-9.2
IXb	-9.2	-9.6	-9.1	-10.6	-9.3
IXc	-8.9	-9.6	-9.2	-10.7	-9.2
IXd	-9.0	-9.5	-9.2	-10.6	-9.3
IXe	-9.0	-9.2	-9.3	-11.1	-9.4
IXf	-9.0	-9.2	-9.2	-11.2	-9.5
SAHA	-7.1	-7.1	-7.4	-7.7	-7.6

S10 Per-residue interaction energies from the molecular dynamics simulations

Coulomb and van der Waals interaction energies between each ligand and the individual active-site residues, averaged over the five 700 ns trajectories of each system. Values are grouped by functional region; the regional sums are those reported in Table 7 of the main text. Residue numbering follows the HDAC2 active-site description of Bressi *et al.*, *Bioorg. Med. Chem. Lett.* **2010**, *20*, 3142–3145.

Table S3. Per-residue interaction energies (kcal mol⁻¹) at the HDAC2 active site.

Residue	SAHA	IXa	IXb	IXc	IXd	IXe	IXf
<i>Cap region (pocket rim)</i>							
Gln31	-0.04	-0.82	-1.67	-1.13	-1.04	-0.19	-0.34
Gly32	-0.40	-1.37	-1.94	-0.81	-0.73	-0.24	-1.08
His33	-1.12	-3.34	-3.49	-3.11	-3.15	-0.79	-3.36
Pro34	-1.69	-3.34	-3.10	-2.21	-2.70	-0.70	-2.51
<i>Sum</i>	<i>-3.25</i>	<i>-8.87</i>	<i>-10.20</i>	<i>-7.26</i>	<i>-7.62</i>	<i>-1.92</i>	<i>-7.29</i>
<i>Catalytic triad</i>							
His145	-2.99	-5.17	-5.80	-5.23	-4.14	-2.28	-5.59
His146	-1.86	-1.23	-1.41	-1.42	-0.97	-1.48	-0.83
Tyr308	-4.49	-2.76	-2.69	-2.59	-2.89	-2.82	-2.88
<i>Sum</i>	<i>-9.34</i>	<i>-9.16</i>	<i>-9.90</i>	<i>-9.24</i>	<i>-8.00</i>	<i>-6.58</i>	<i>-9.30</i>
<i>Lipophilic tube</i>							
Gly154	-9.67	-6.62	-6.66	-6.46	-8.06	-7.74	-8.62
Phe155	-5.77	-6.57	-6.70	-6.57	-5.66	-4.73	-5.81
His183	-1.30	-1.53	-1.37	-1.68	-0.67	-1.40	-0.81
Phe210	-3.56	-4.13	-4.19	-4.19	-5.14	-8.57	-4.43
Leu276	-2.44	-3.93	-3.61	-3.35	-4.15	-3.49	-3.24
<i>Sum</i>	<i>-22.74</i>	<i>-22.78</i>	<i>-22.53</i>	<i>-22.25</i>	<i>-23.68</i>	<i>-25.93</i>	<i>-22.91</i>
<i>Foot pocket</i>							
Tyr29	-0.04	-0.10	-0.14	-0.19	-0.13	-0.04	-0.11
Met35	-1.05	-0.72	-0.76	-0.67	-0.69	-0.44	-0.59
Leu144	-0.32	-0.23	-0.29	-0.28	-0.34	-0.27	-0.40
Gly305	-0.74	-0.50	-0.23	-0.08	+0.12	-0.25	+0.30
Gly306	-1.64	-2.30	-2.48	-2.13	-2.25	-2.03	-2.24
<i>Sum</i>	<i>-3.79</i>	<i>-3.85</i>	<i>-3.90</i>	<i>-3.35</i>	<i>-3.29</i>	<i>-3.03</i>	<i>-3.04</i>
<i>Outer channel</i>							
Asp104	-6.46	-4.29	-4.31	-4.72	-5.15	-4.65	-6.17
<i>Sum</i>	<i>-6.46</i>	<i>-4.29</i>	<i>-4.31</i>	<i>-4.72</i>	<i>-5.15</i>	<i>-4.65</i>	<i>-6.17</i>

S11 Ligand root-mean-square deviation during the molecular dynamics simulations

Figures [S49–S55](#) show the ligand root-mean-square deviation of each complex relative to the docked pose over the 700 ns production window, for each of the five independent trajectories.

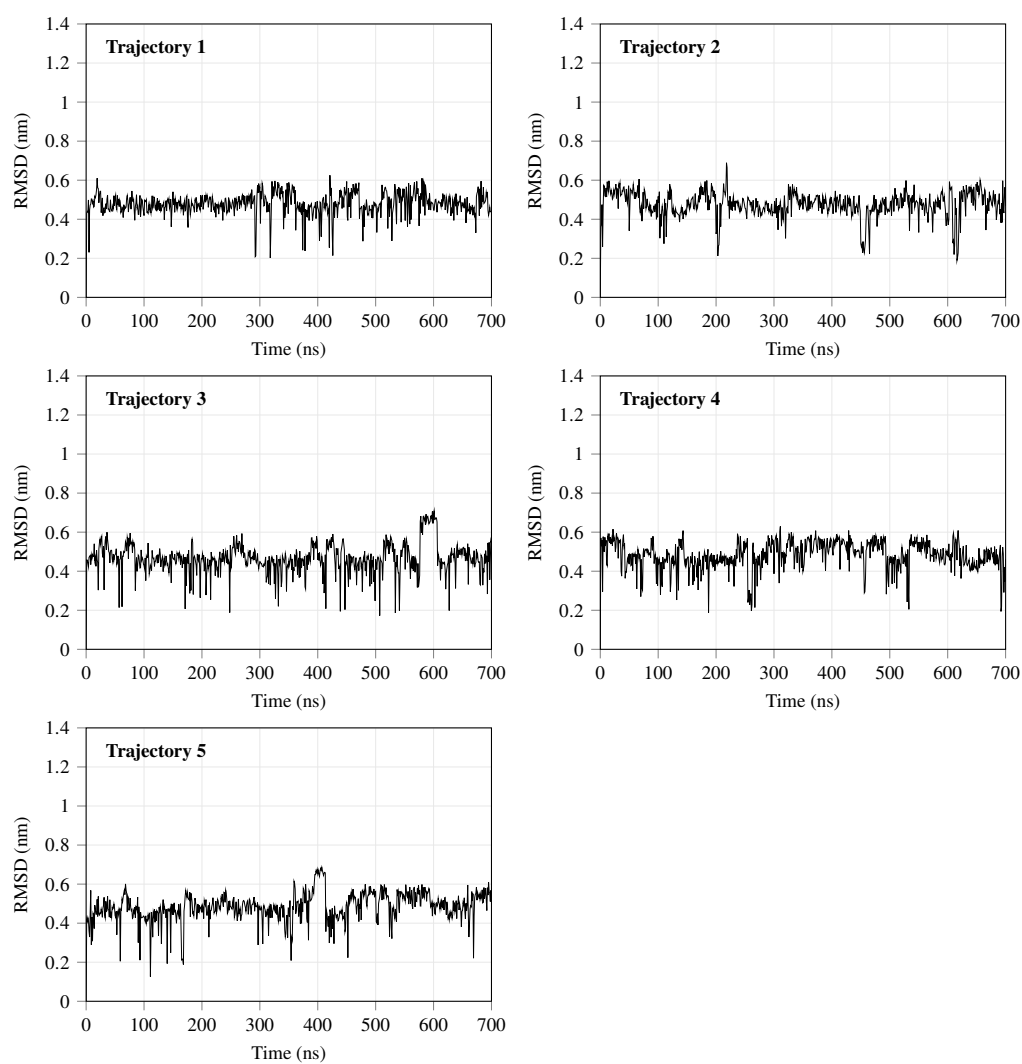


Figure S49. Ligand root-mean-square deviation of the SAHA-HDAC2 complex over 700 ns, shown for each of the five independent trajectories.

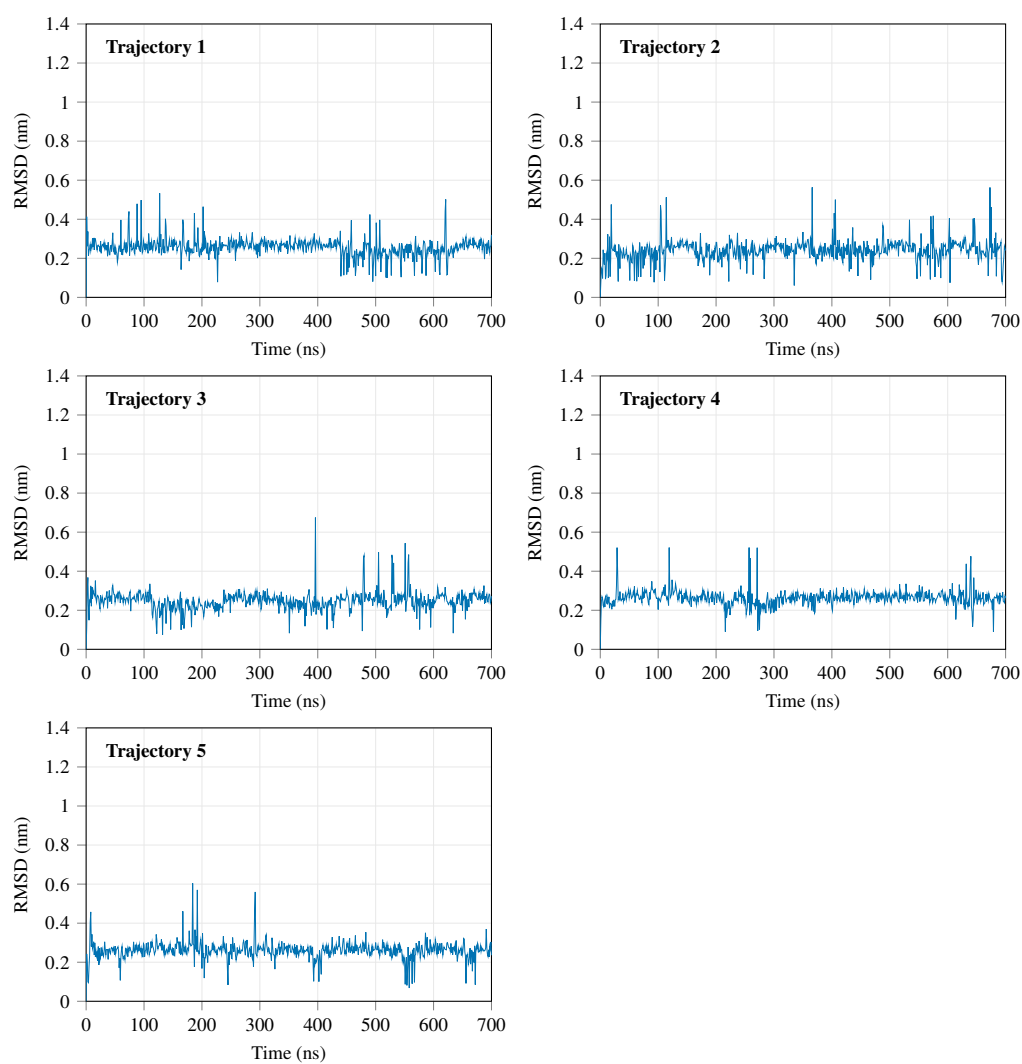


Figure S50. Ligand root-mean-square deviation of the **IXa**-HDAC2 complex over 700 ns, shown for each of the five independent trajectories.

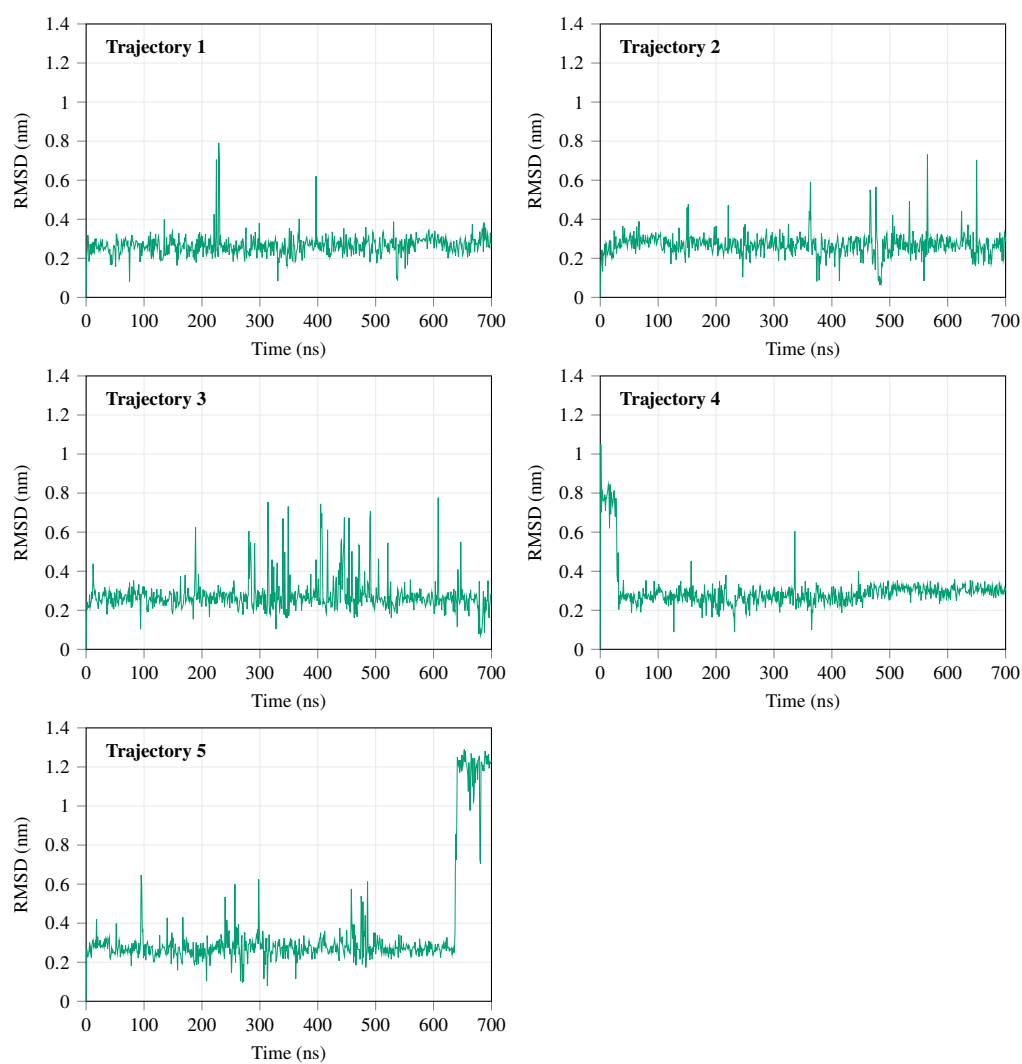


Figure S51. Ligand root-mean-square deviation of the **IXb**-HDAC2 complex over 700 ns, shown for each of the five independent trajectories.

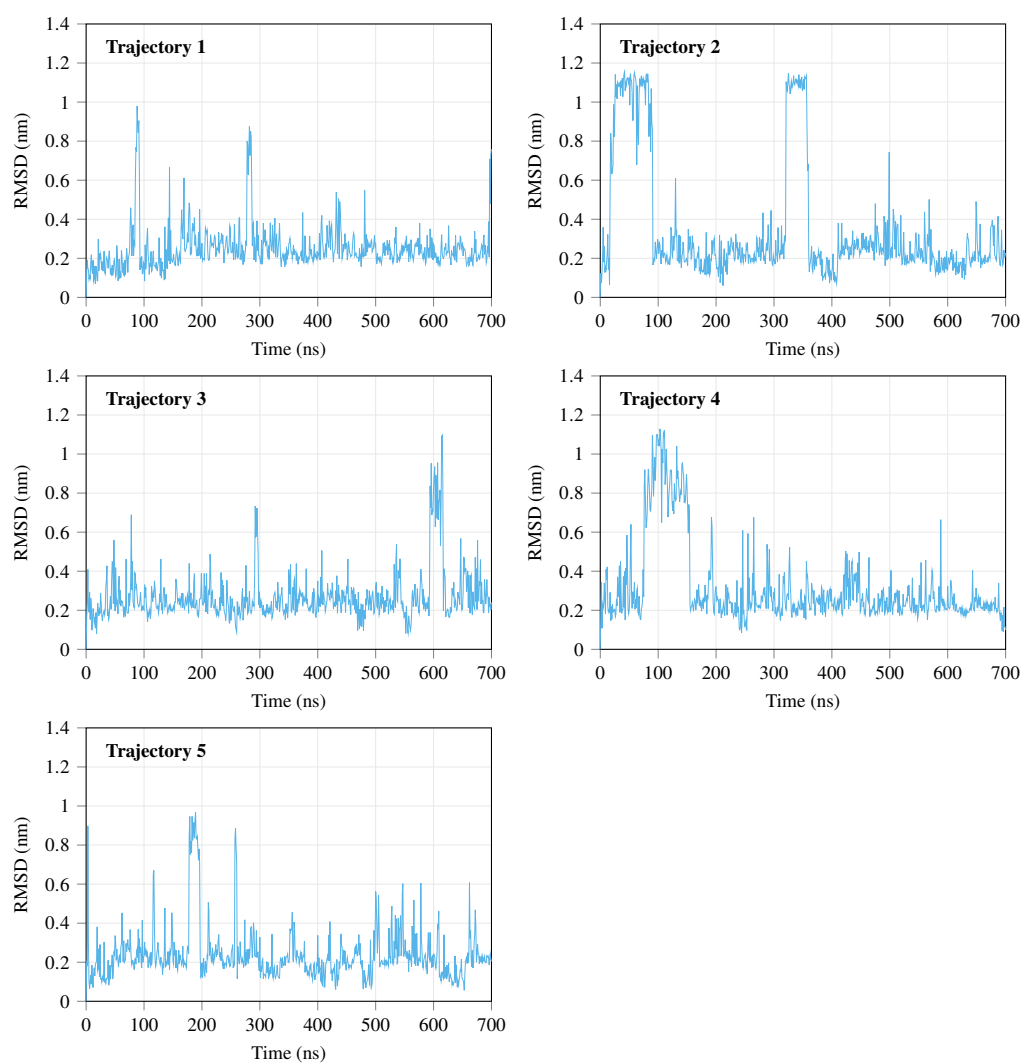


Figure S52. Ligand root-mean-square deviation of the **IXc**-HDAC2 complex over 700 ns, shown for each of the five independent trajectories.

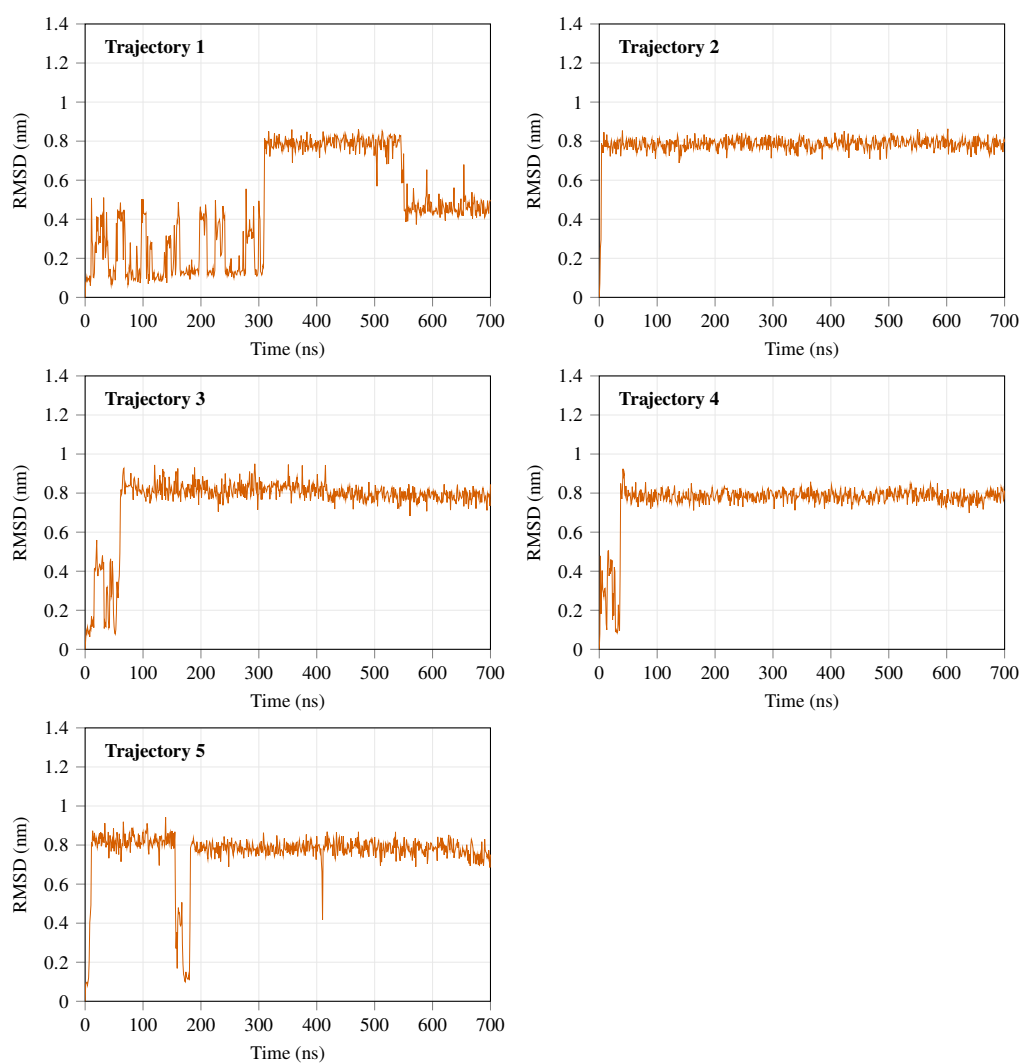


Figure S53. Ligand root-mean-square deviation of the **IXd**-HDAC2 complex over 700 ns, shown for each of the five independent trajectories.

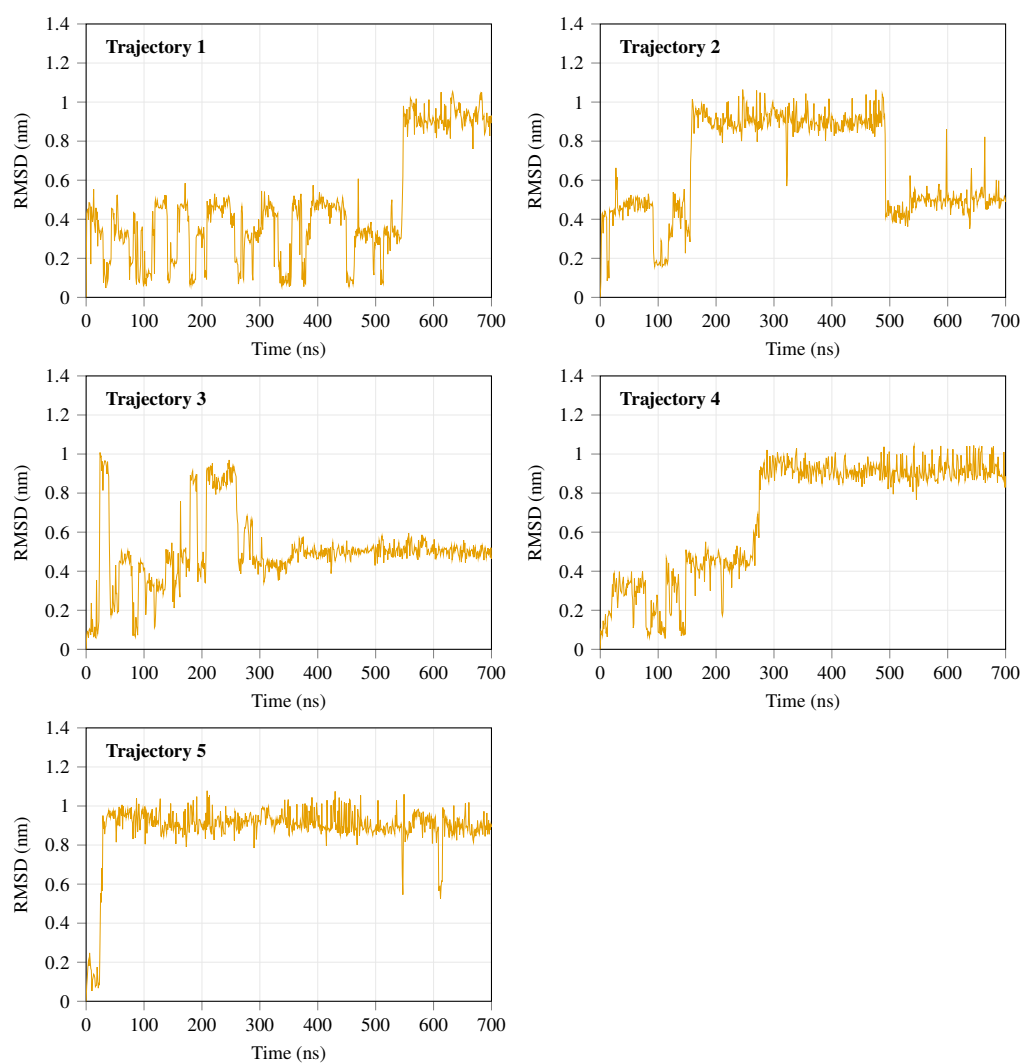


Figure S54. Ligand root-mean-square deviation of the **IXe**–HDAC2 complex over 700 ns, shown for each of the five independent trajectories.

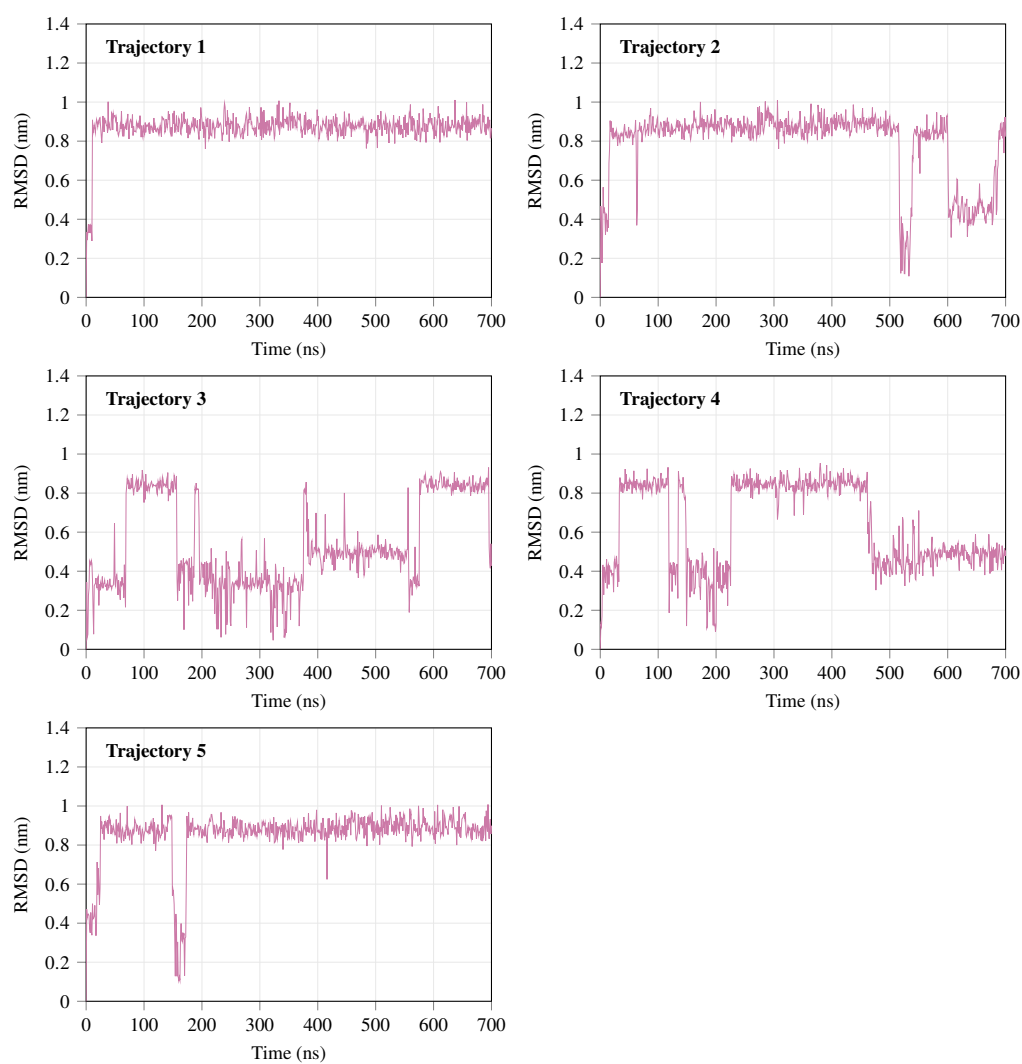


Figure S55. Ligand root-mean-square deviation of the **IXf**-HDAC2 complex over 700 ns, shown for each of the five independent trajectories.

S12 Zinc coordination distances during the molecular dynamics simulations

Figures [S56–S62](#) show the distance between the hydroxamate carbonyl oxygen and the catalytic zinc, and Figures [S63–S69](#) the corresponding hydroxyl–zinc distance, for each of the five independent trajectories.

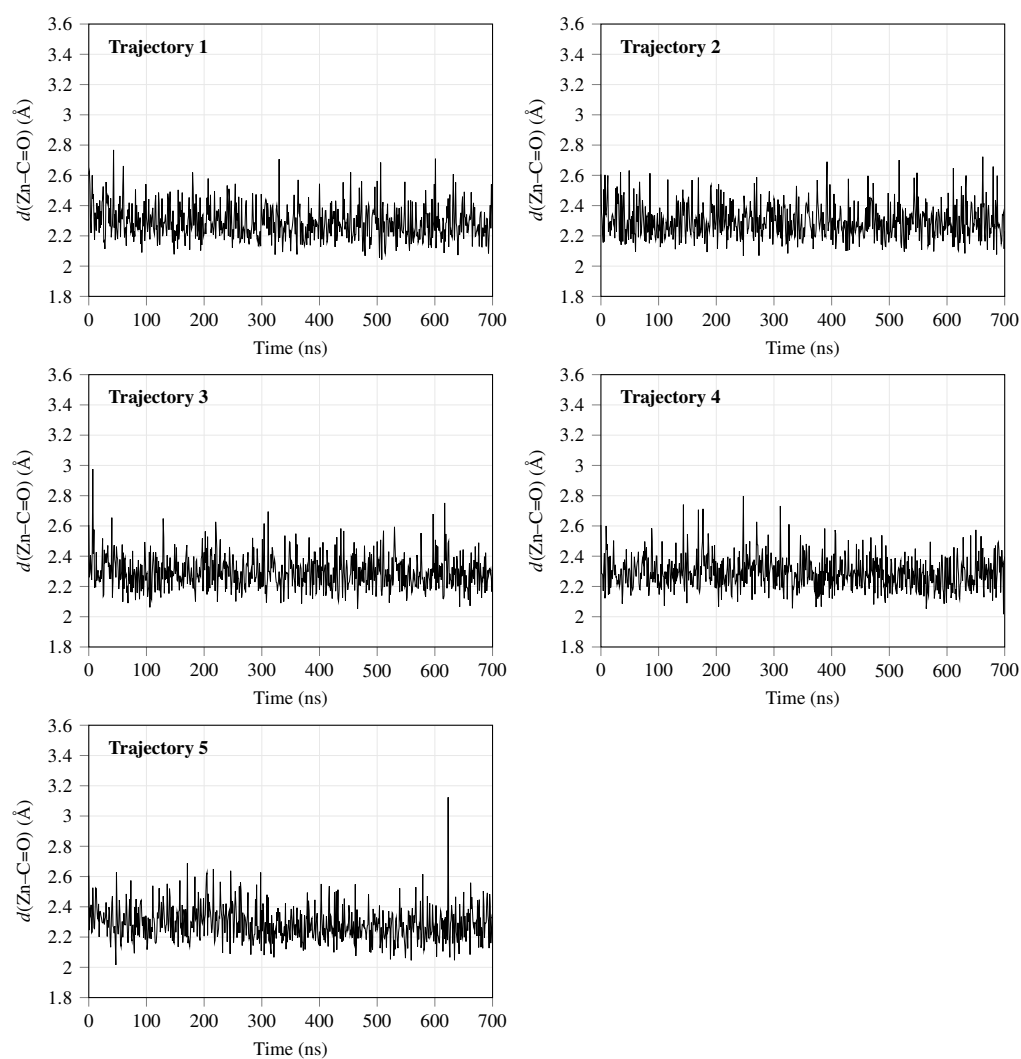


Figure S56. Hydroxamate carbonyl–zinc distance in the SAHA–HDAC2 complex over 700 ns, shown for each of the five independent trajectories.

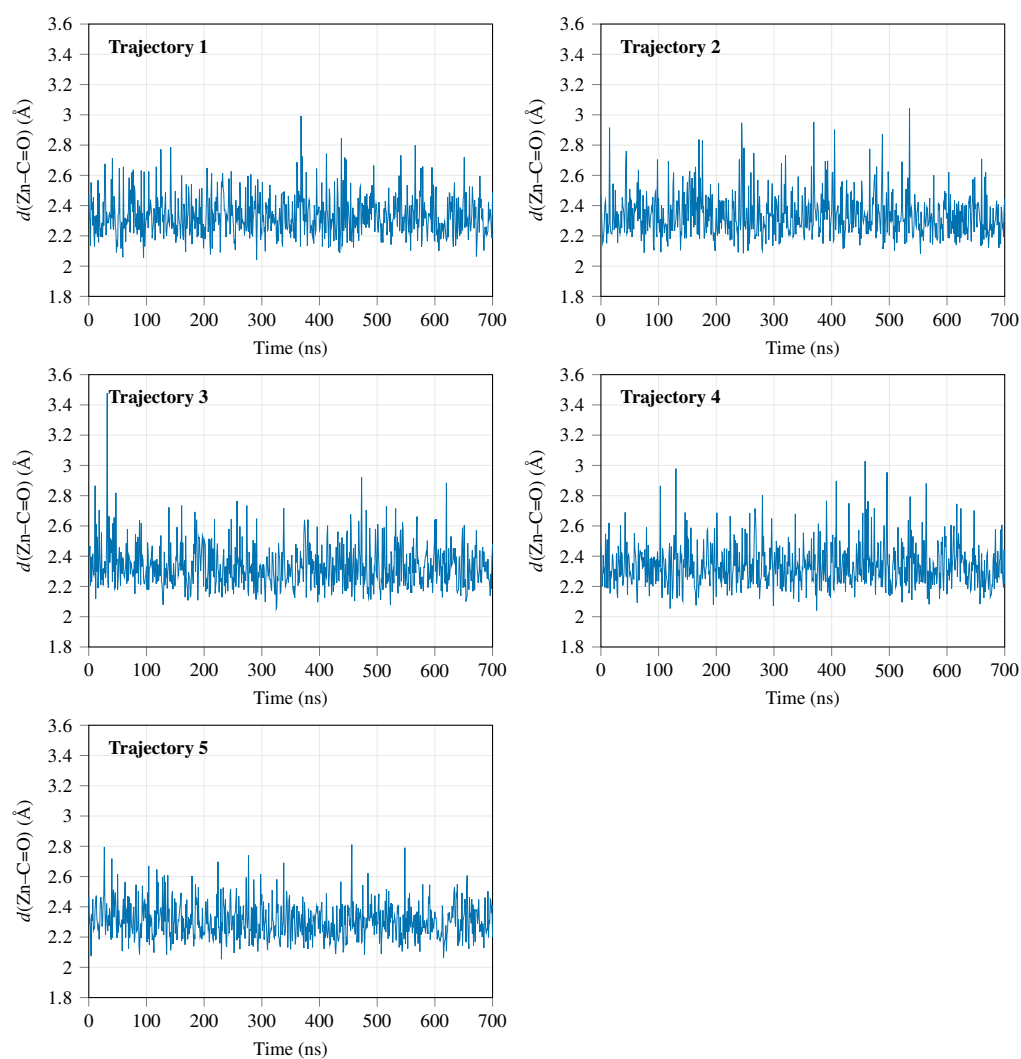


Figure S57. Hydroxamate carbonyl–zinc distance in the **IXa**–HDAC2 complex over 700 ns, shown for each of the five independent trajectories.

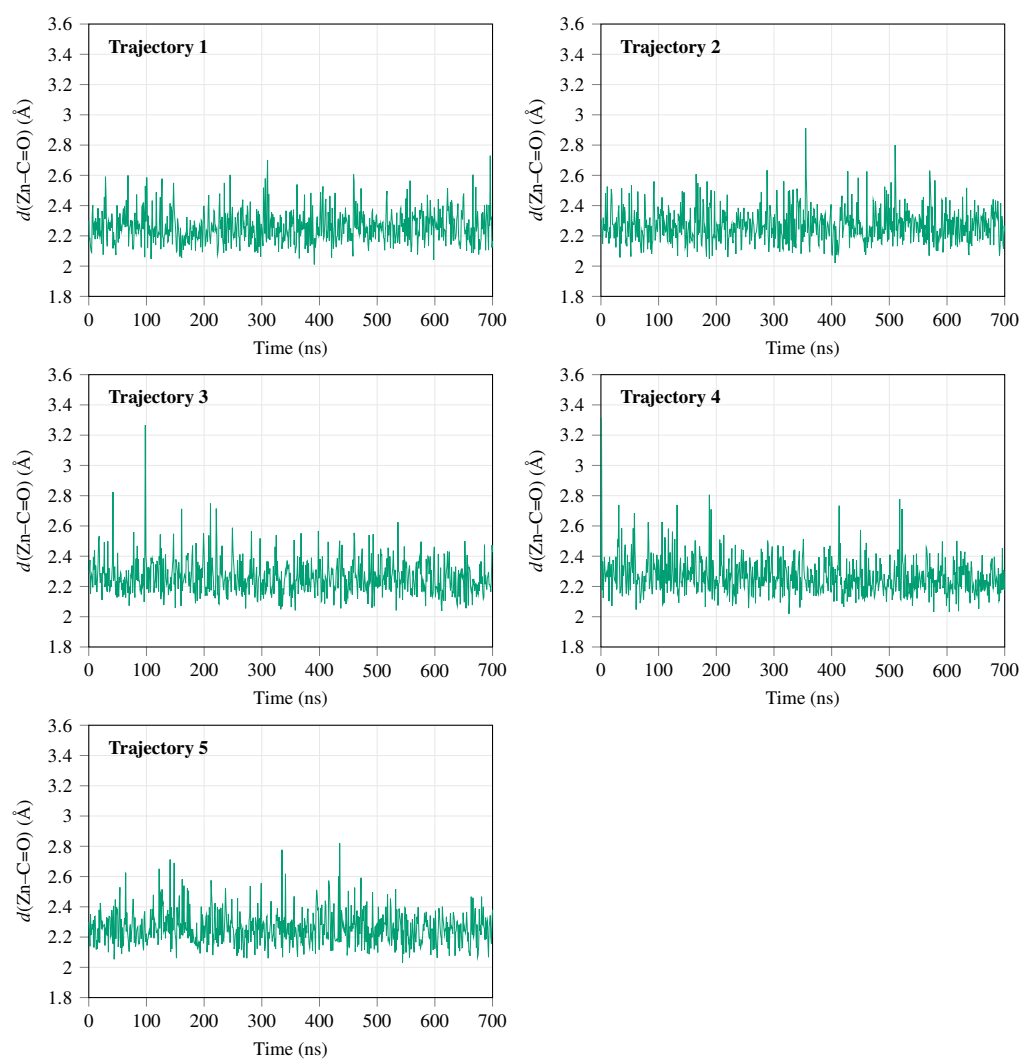


Figure S58. Hydroxamate carbonyl–zinc distance in the **IXb**–HDAC2 complex over 700 ns, shown for each of the five independent trajectories.

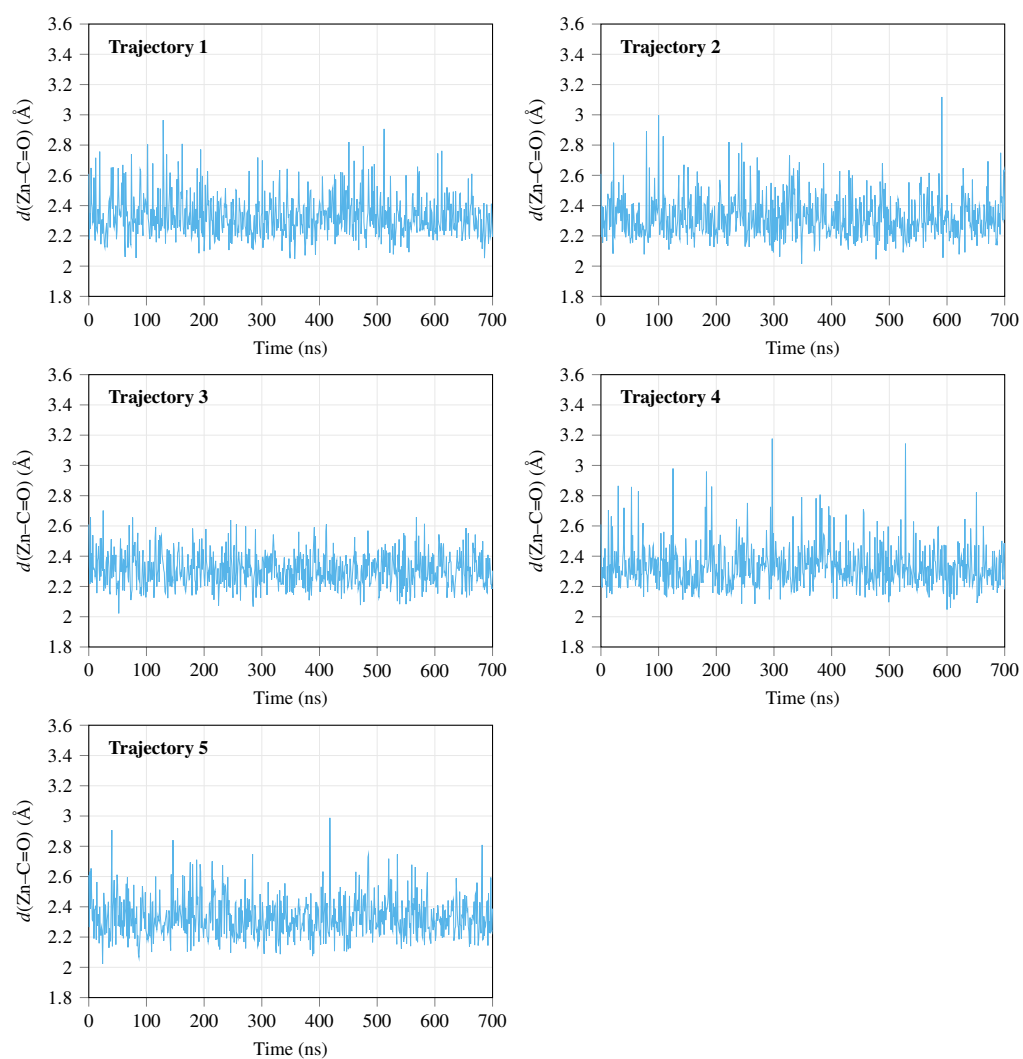


Figure S59. Hydroxamate carbonyl–zinc distance in the **IXc**–HDAC2 complex over 700 ns, shown for each of the five independent trajectories.

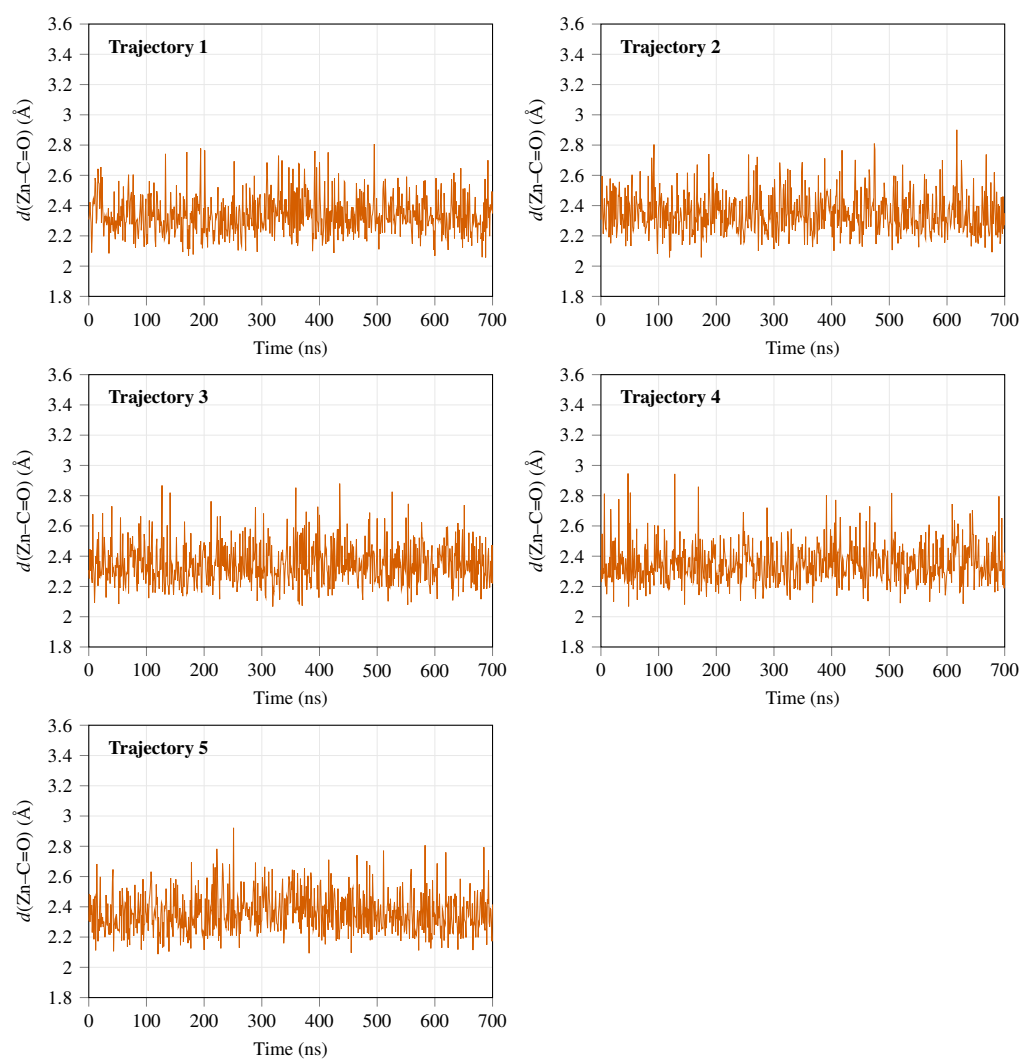


Figure S60. Hydroxamate carbonyl–zinc distance in the **IXd**–HDAC2 complex over 700 ns, shown for each of the five independent trajectories.

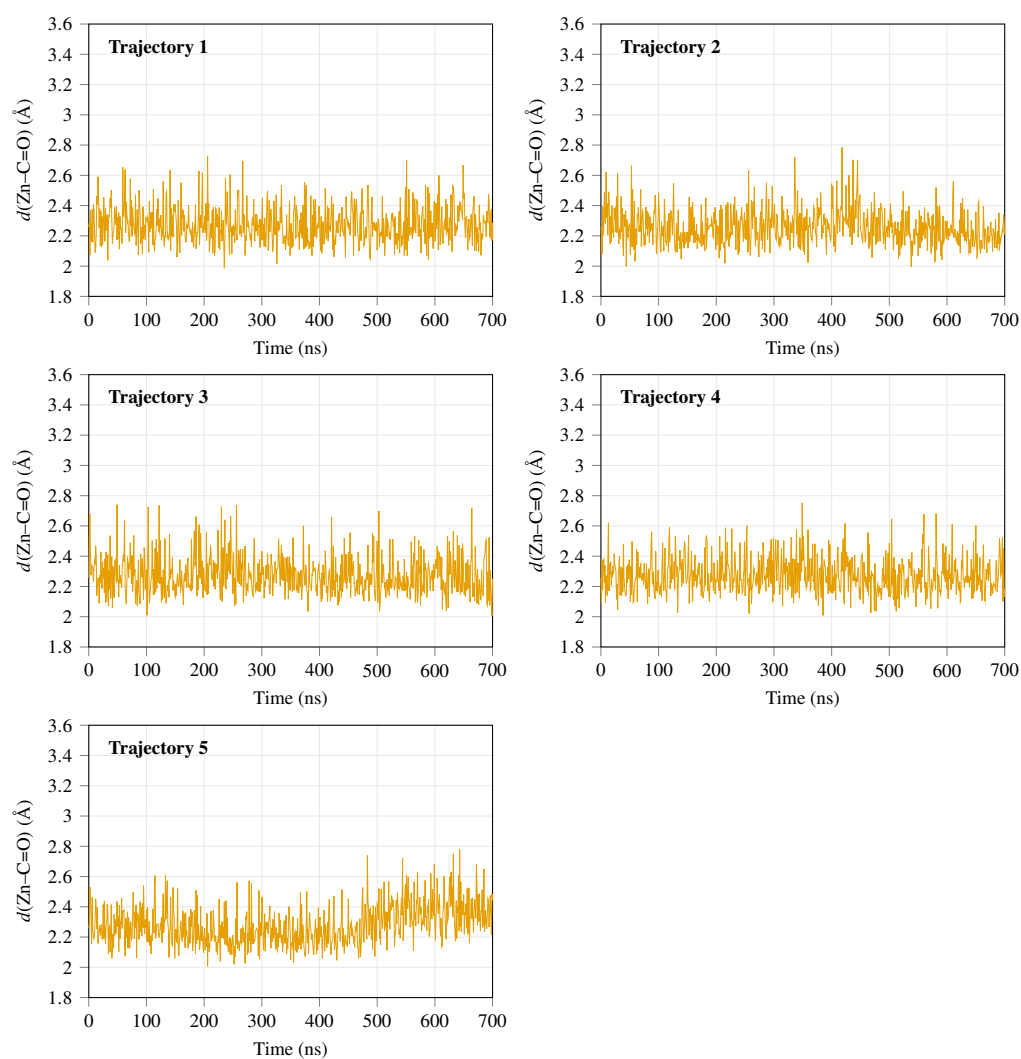


Figure S61. Hydroxamate carbonyl–zinc distance in the **IXe**–HDAC2 complex over 700 ns, shown for each of the five independent trajectories.

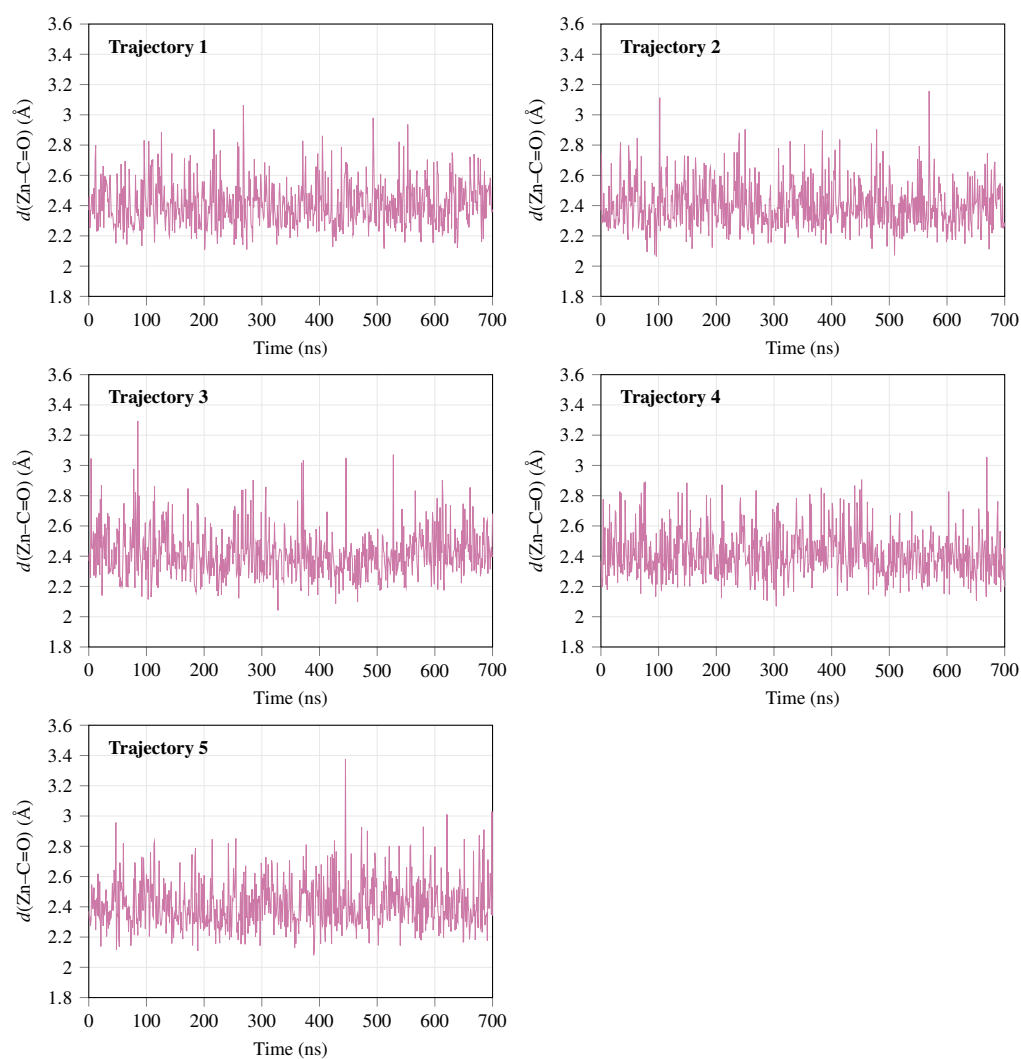


Figure S62. Hydroxamate carbonyl–zinc distance in the **IXf**–HDAC2 complex over 700 ns, shown for each of the five independent trajectories.

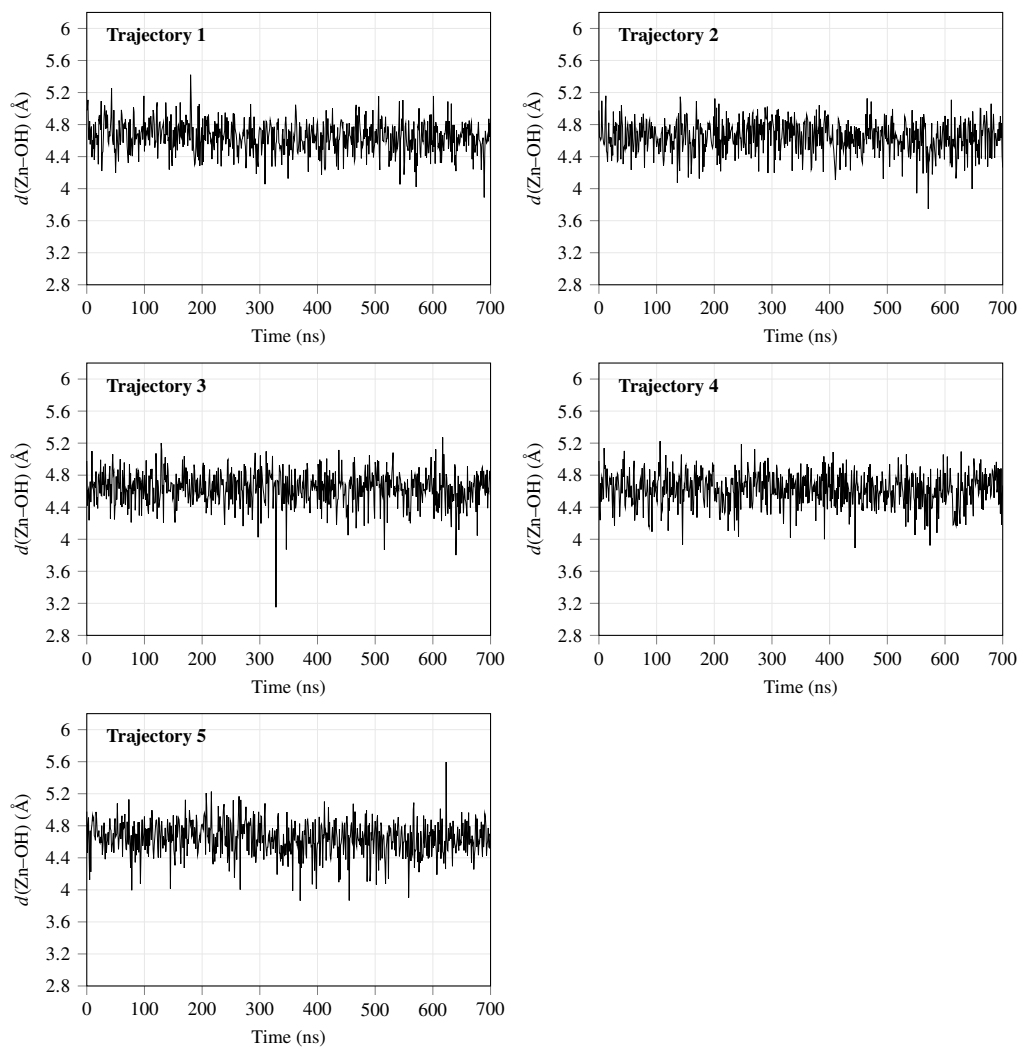


Figure S63. Hydroxamate hydroxyl-zinc distance in the SAHA-HDAC2 complex over 700 ns, shown for each of the five independent trajectories.

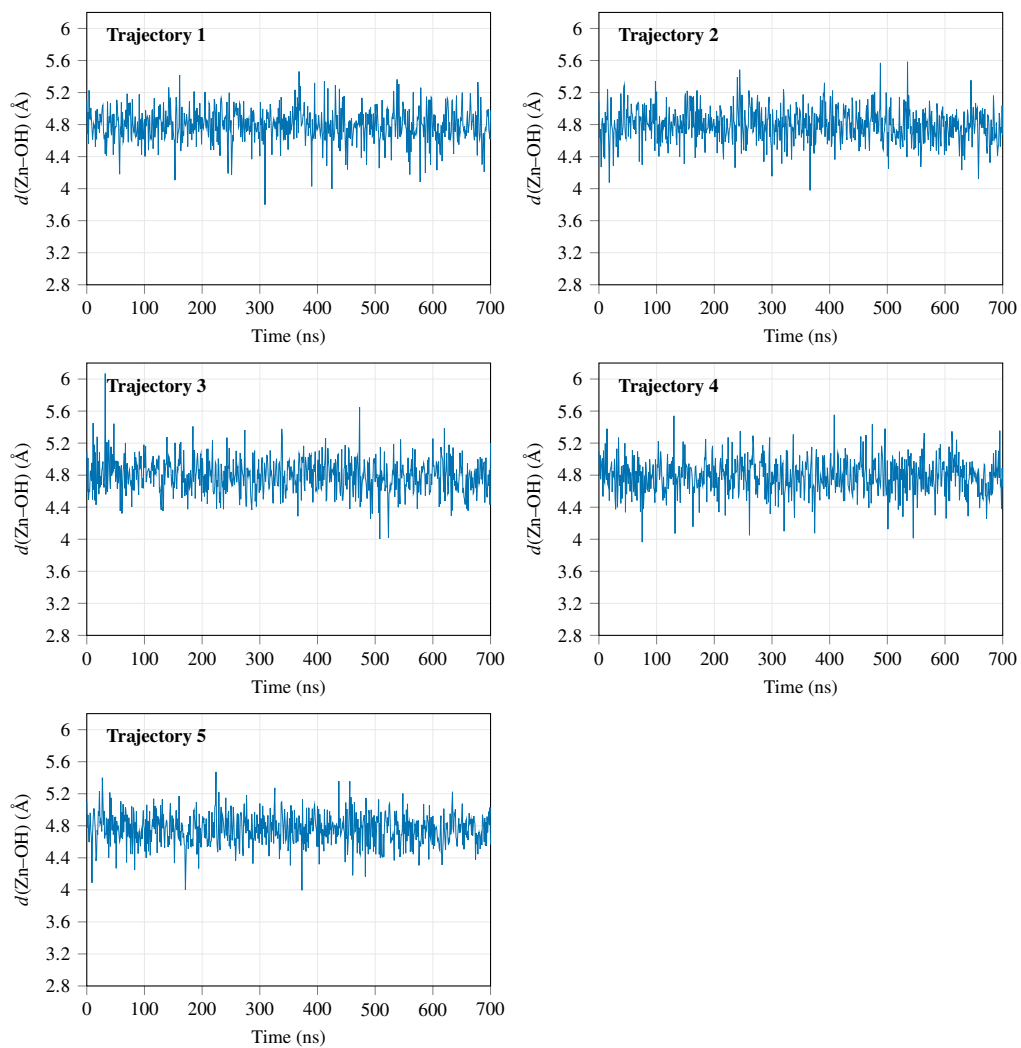


Figure S64. Hydroxamate hydroxyl-zinc distance in the **IXa**-HDAC2 complex over 700 ns, shown for each of the five independent trajectories.

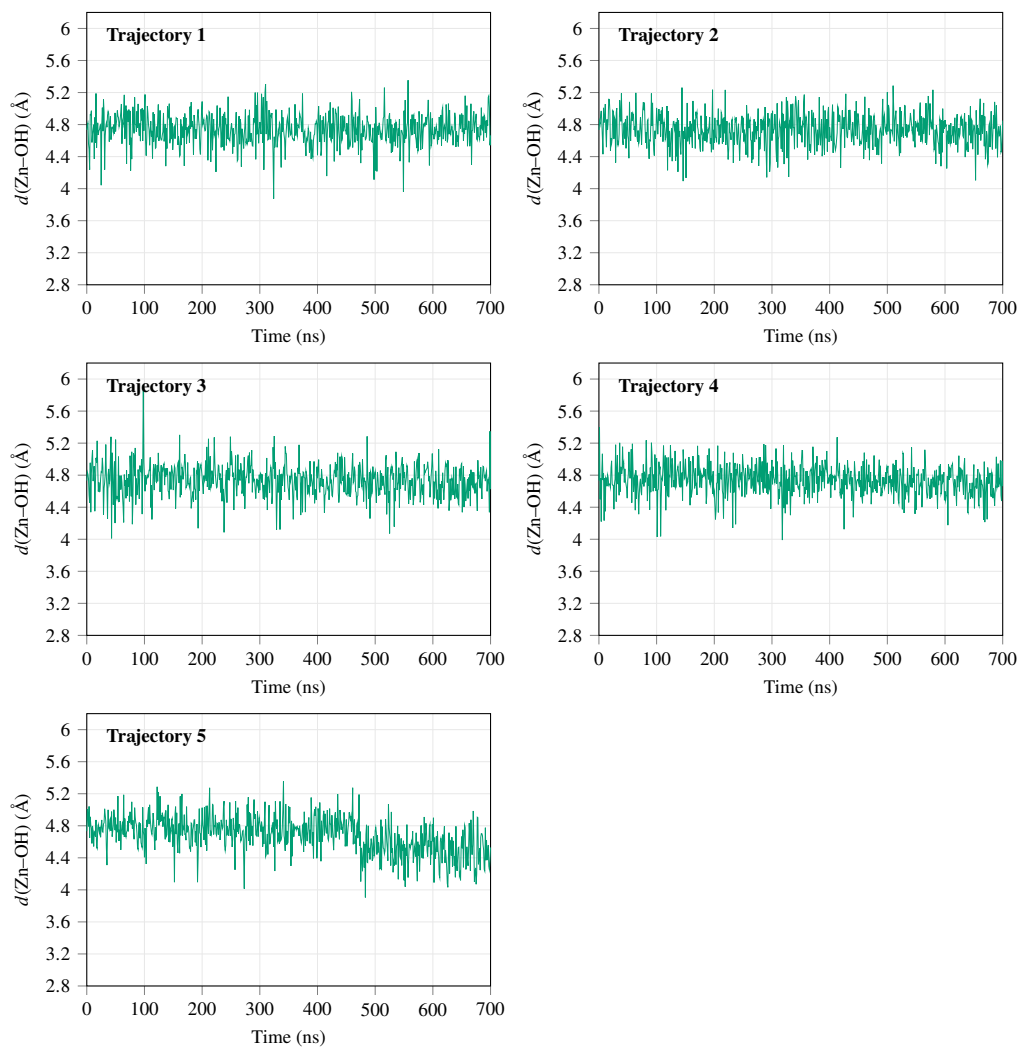


Figure S65. Hydroxamate hydroxyl–zinc distance in the **IXb**–HDAC2 complex over 700 ns, shown for each of the five independent trajectories.

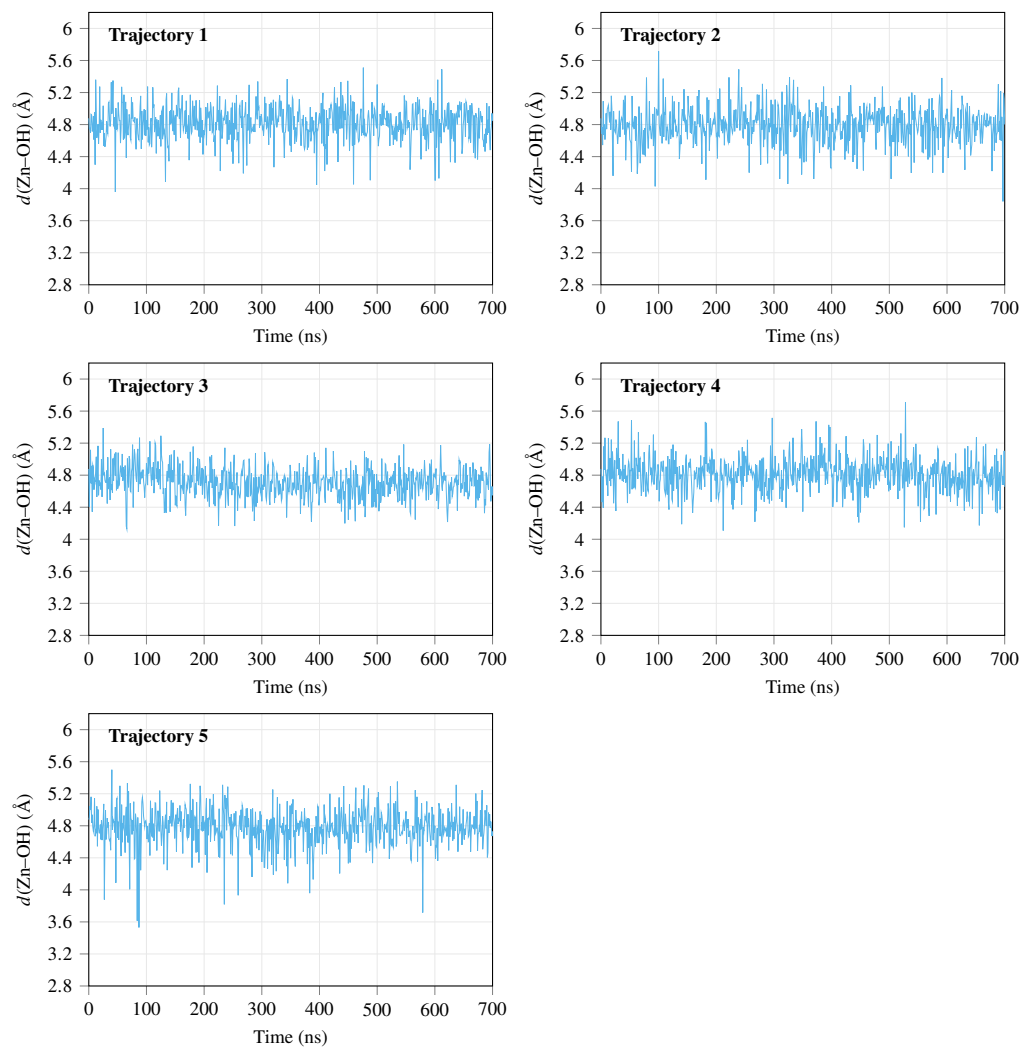


Figure S66. Hydroxamate hydroxyl-zinc distance in the **IXc**-HDAC2 complex over 700 ns, shown for each of the five independent trajectories.

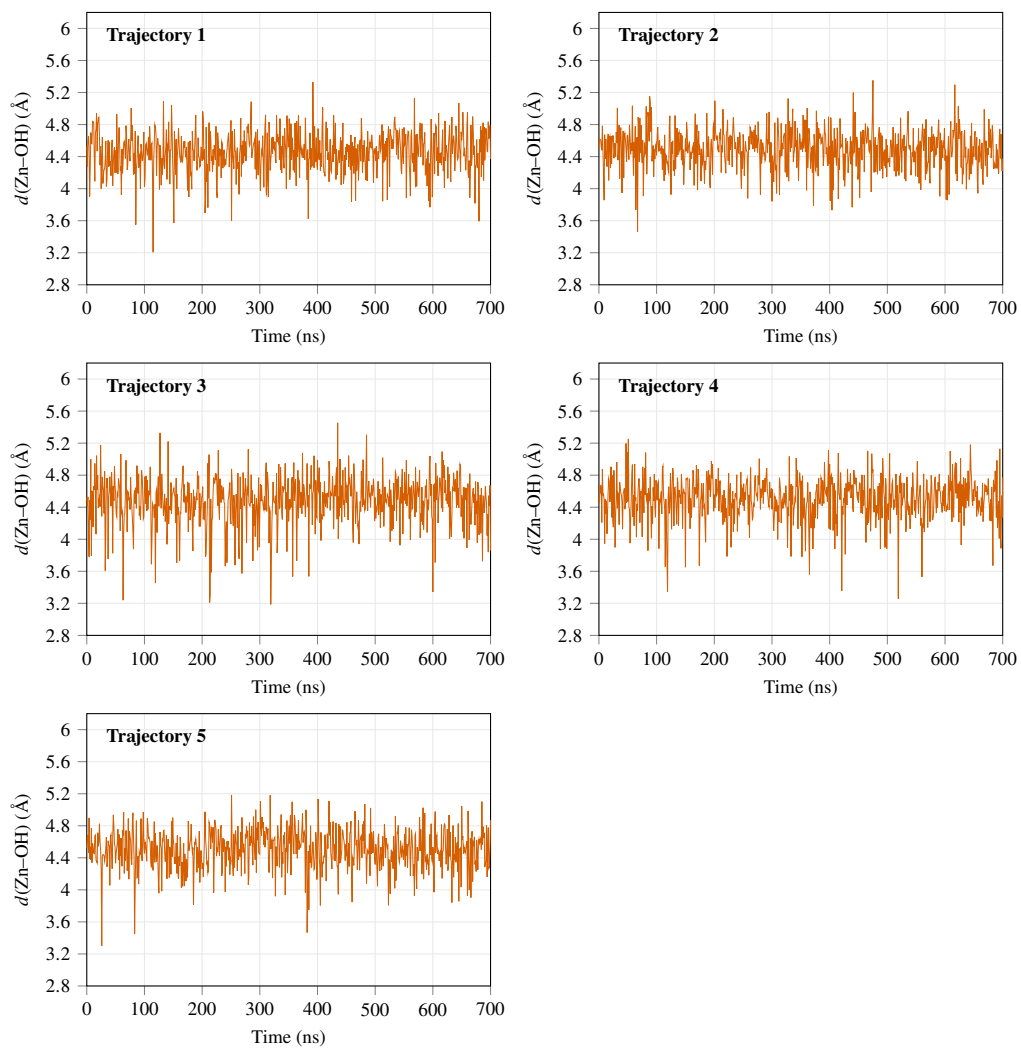


Figure S67. Hydroxamate hydroxyl–zinc distance in the **IXd**–HDAC2 complex over 700 ns, shown for each of the five independent trajectories.

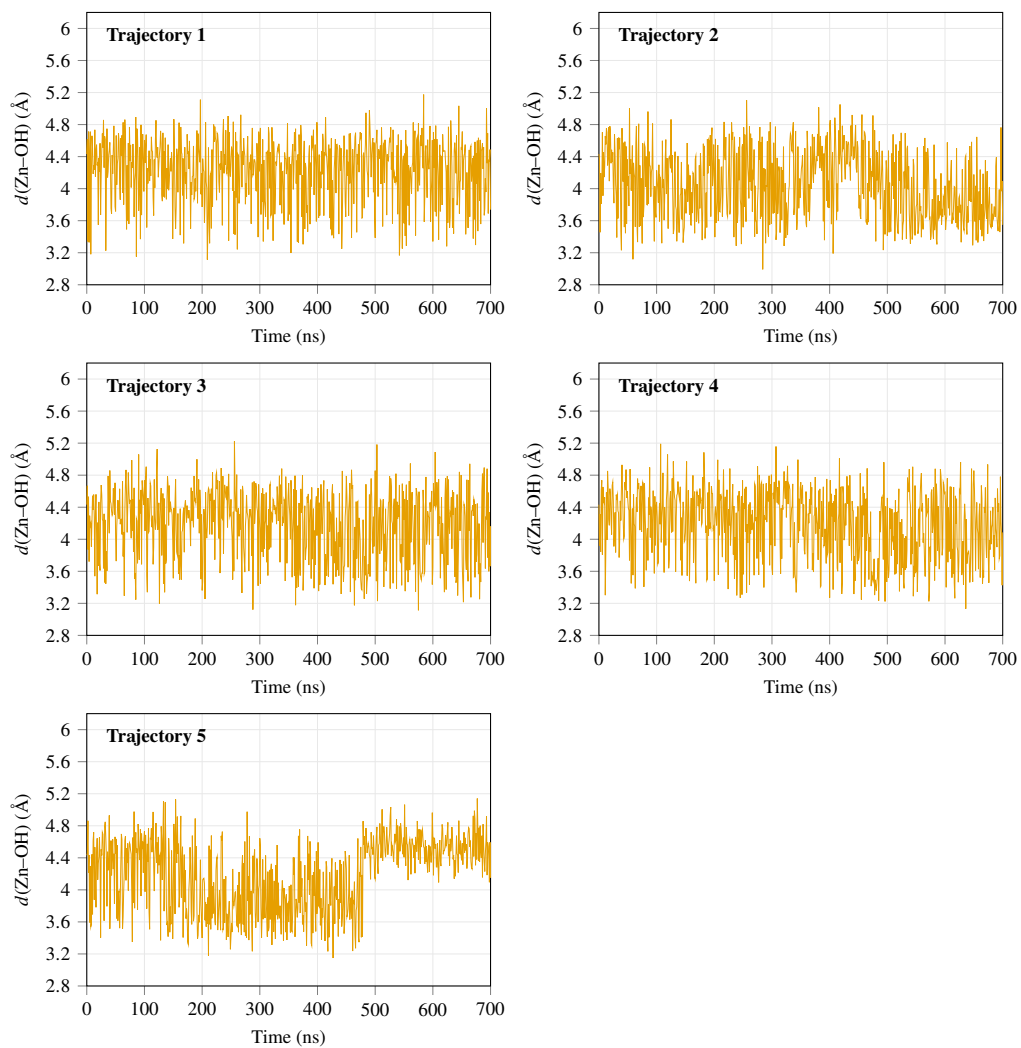


Figure S68. Hydroxamate hydroxyl-zinc distance in the **IXe**-HDAC2 complex over 700 ns, shown for each of the five independent trajectories.

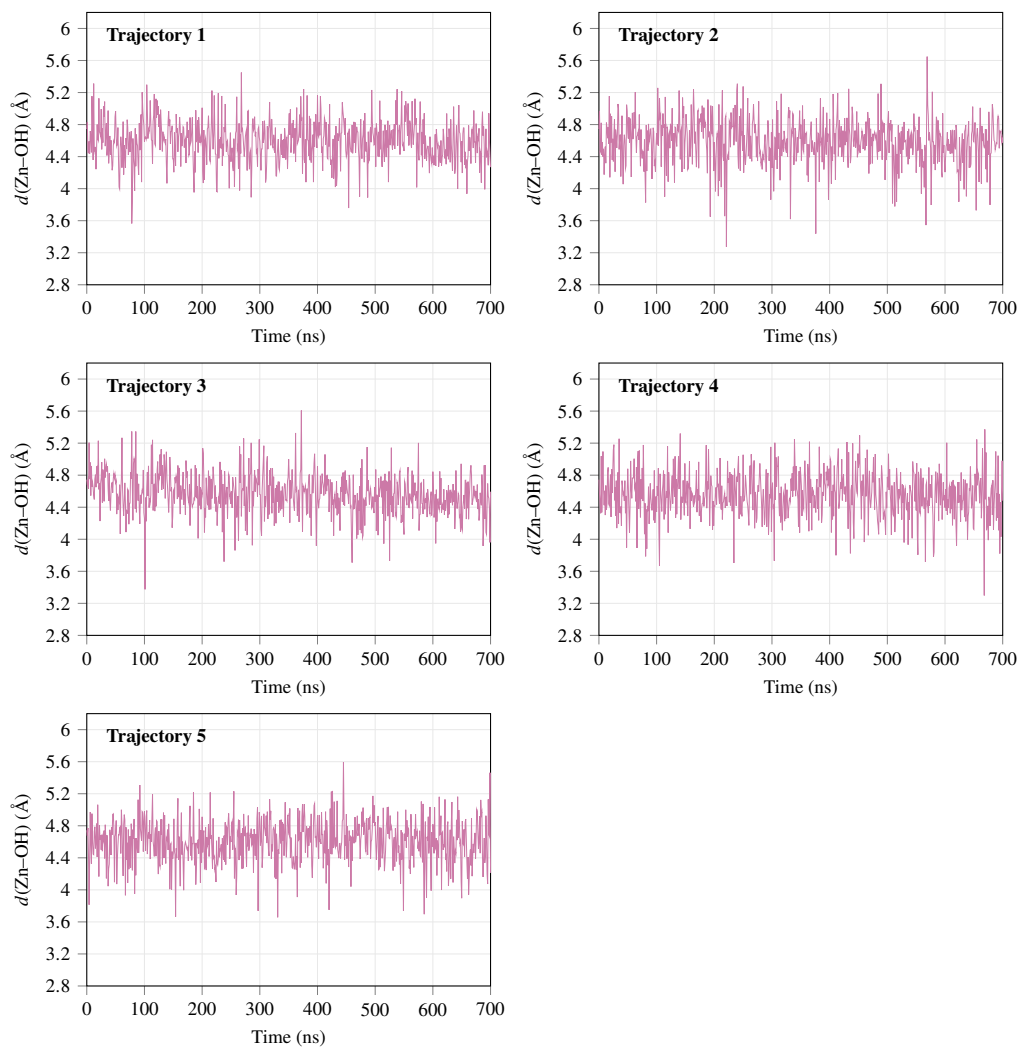


Figure S69. Hydroxamate hydroxyl-zinc distance in the **IXf**-HDAC2 complex over 700 ns, shown for each of the five independent trajectories.

S13 Ligand–protein hydrogen bonds during the molecular dynamics simulations

Figures [S70–S76](#) show the number of ligand–protein hydrogen bonds over the production window, for each of the five independent trajectories.

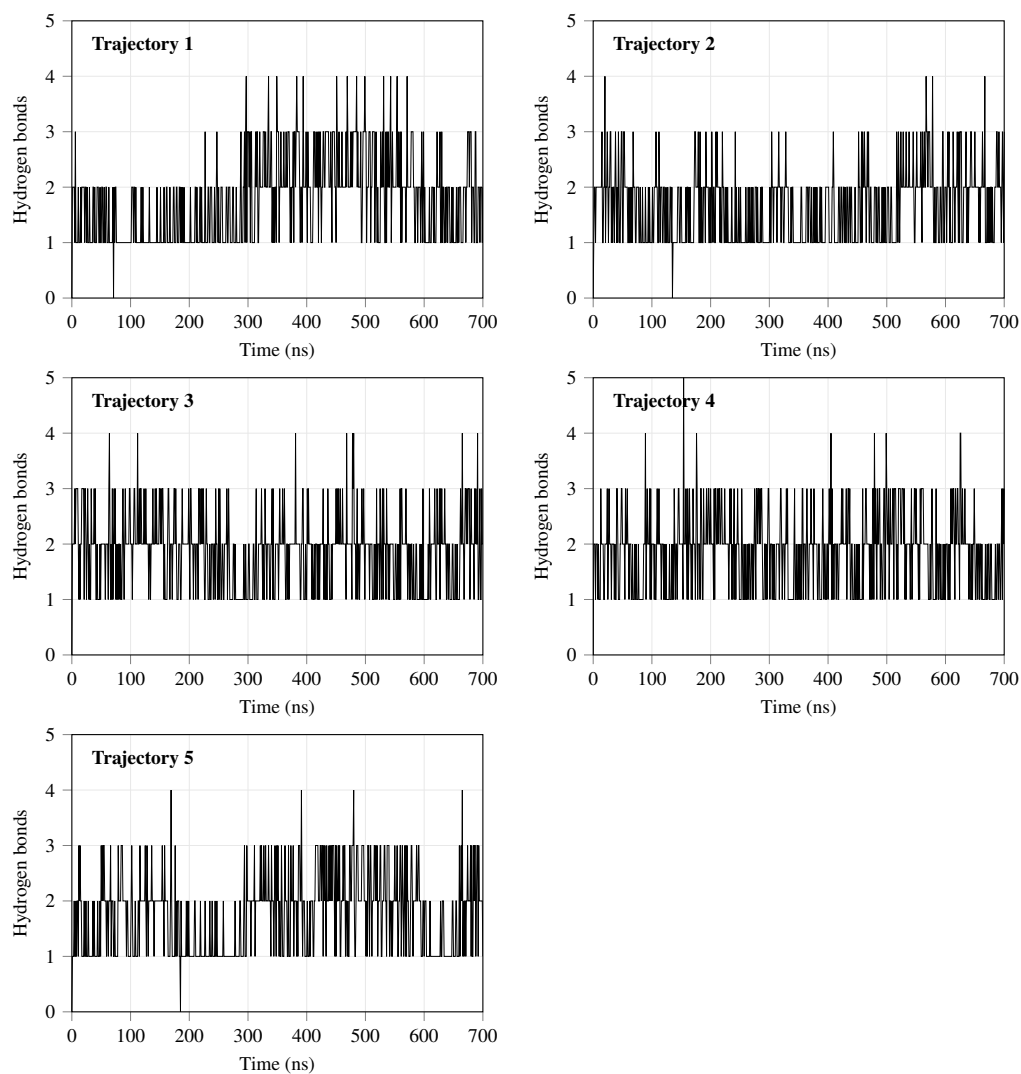


Figure S70. Ligand–protein hydrogen bonds in the SAHA–HDAC2 complex over 700 ns, shown for each of the five independent trajectories.

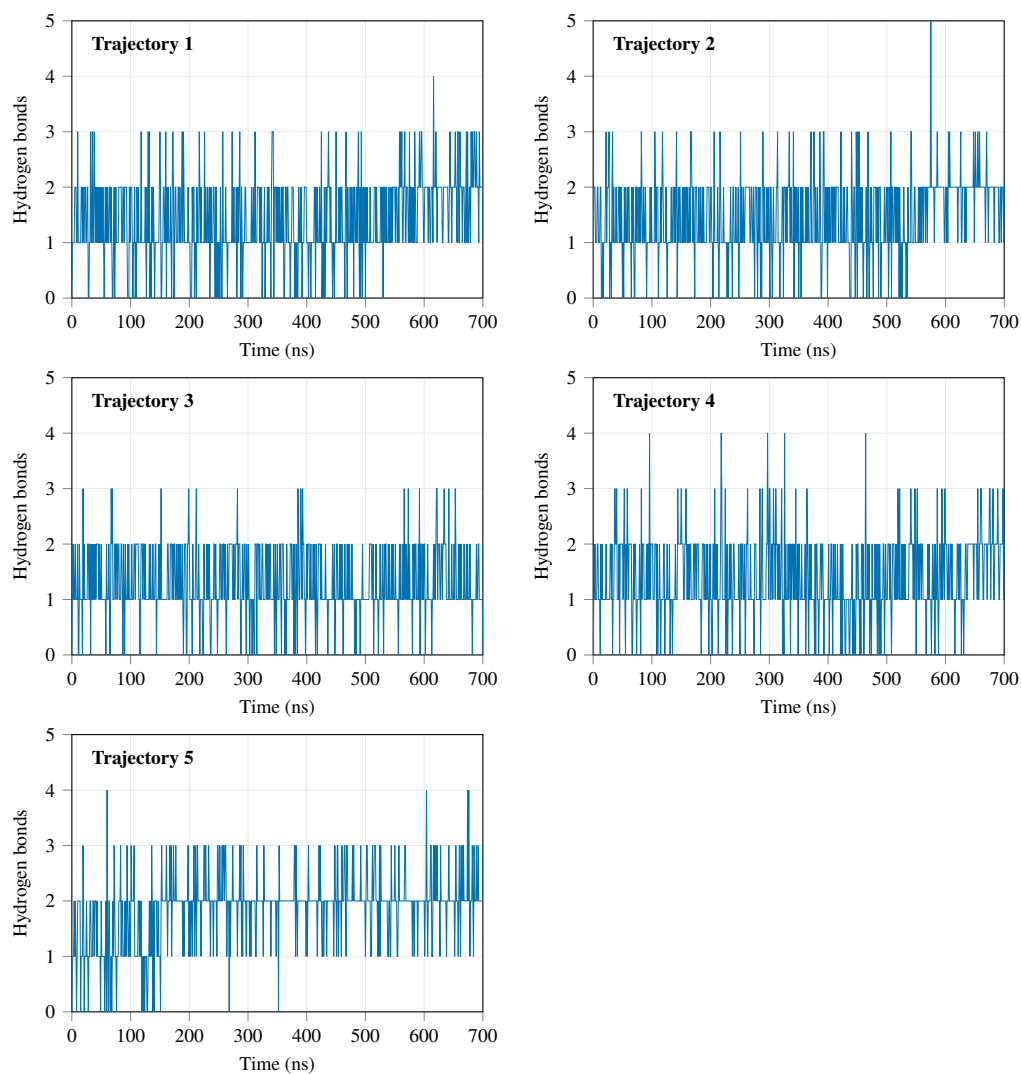


Figure S71. Ligand–protein hydrogen bonds in the **IXa**–HDAC2 complex over 700 ns, shown for each of the five independent trajectories.

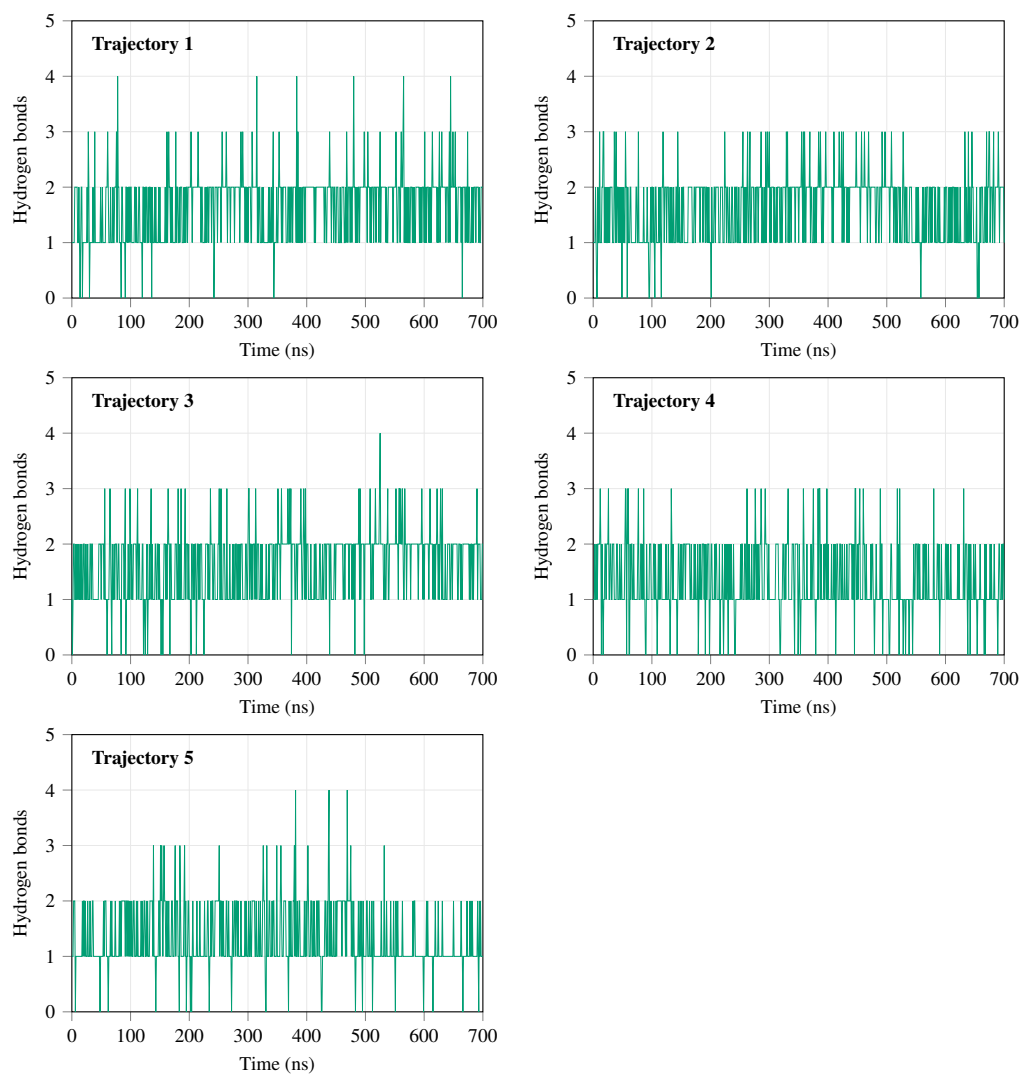


Figure S72. Ligand–protein hydrogen bonds in the **IXb**–HDAC2 complex over 700 ns, shown for each of the five independent trajectories.

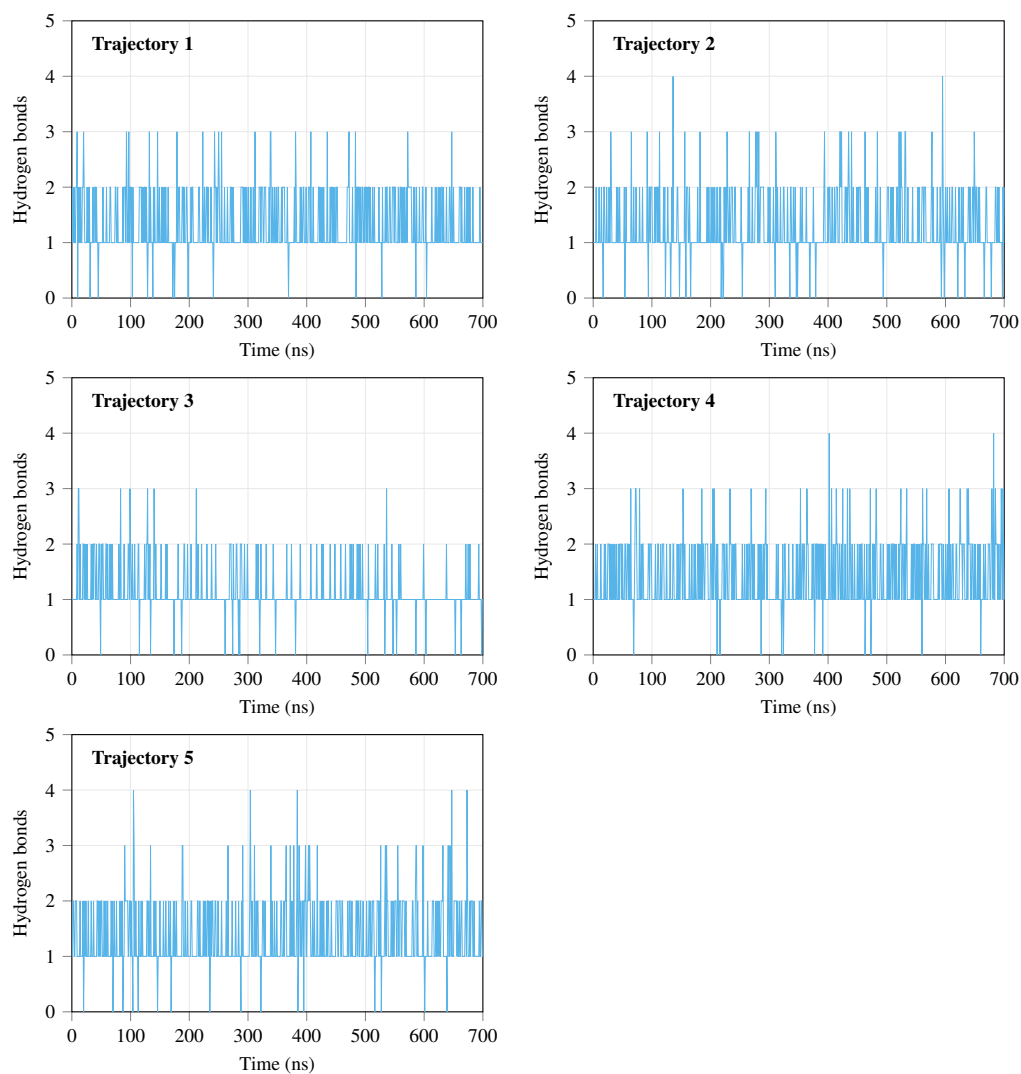


Figure S73. Ligand–protein hydrogen bonds in the **IXc**–HDAC2 complex over 700 ns, shown for each of the five independent trajectories.

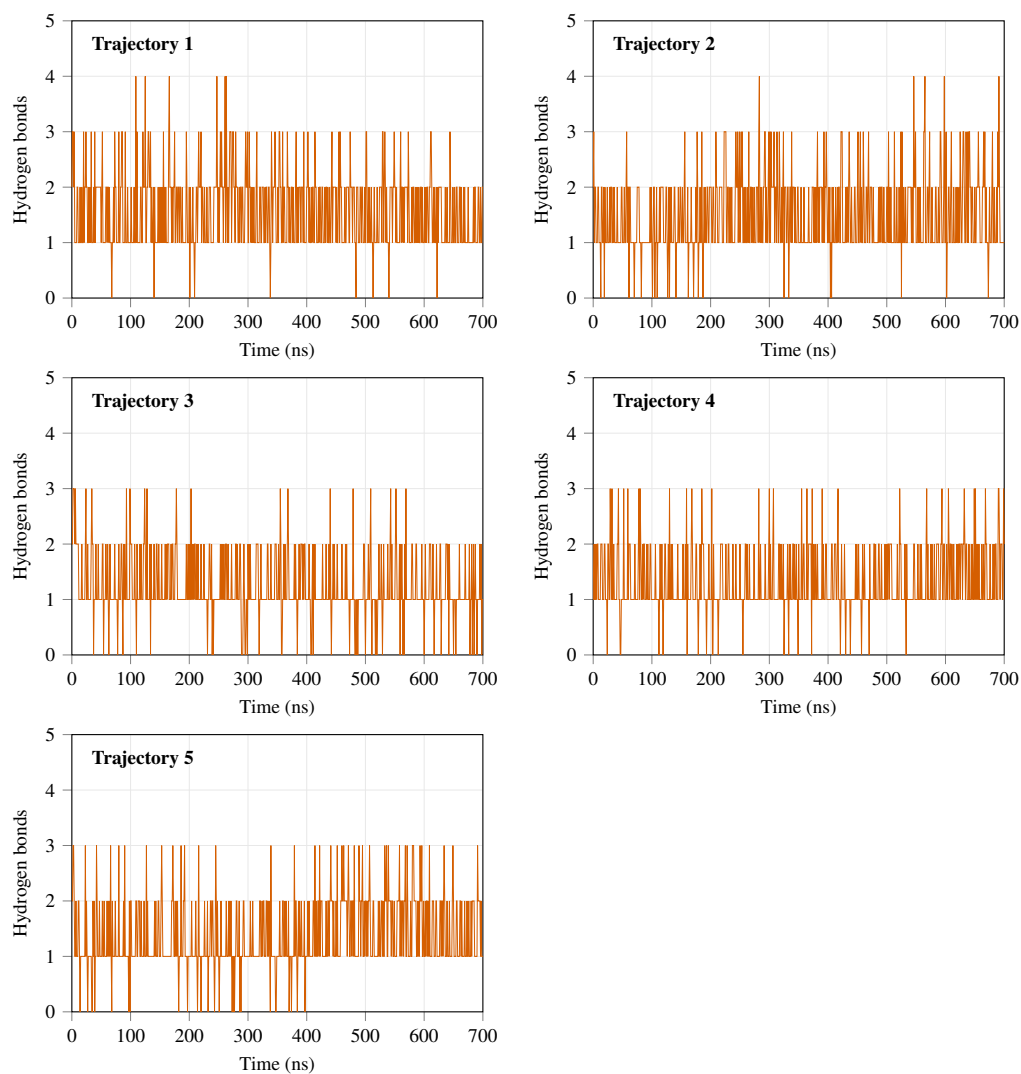


Figure S74. Ligand–protein hydrogen bonds in the **IXd**–HDAC2 complex over 700 ns, shown for each of the five independent trajectories.

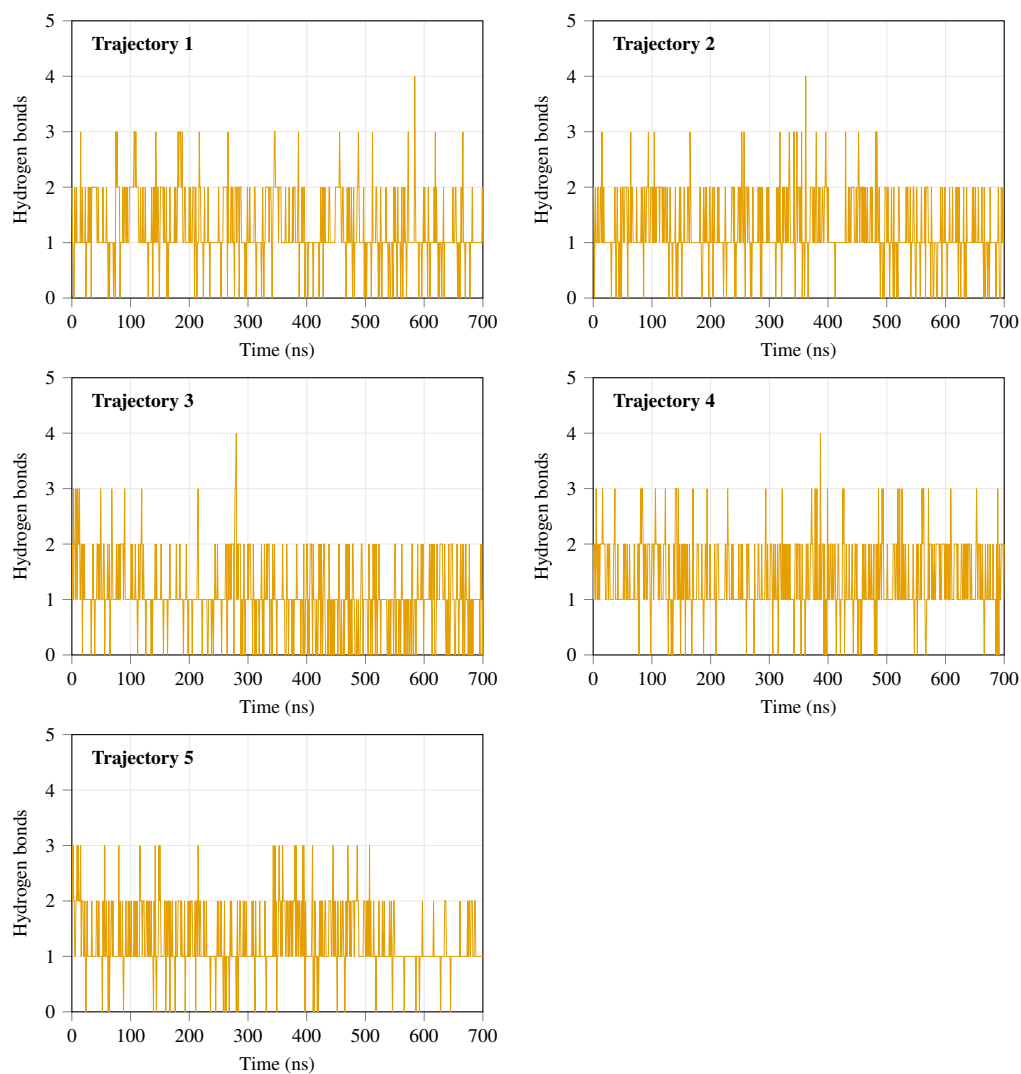


Figure S75. Ligand–protein hydrogen bonds in the **IXe**–HDAC2 complex over 700 ns, shown for each of the five independent trajectories.

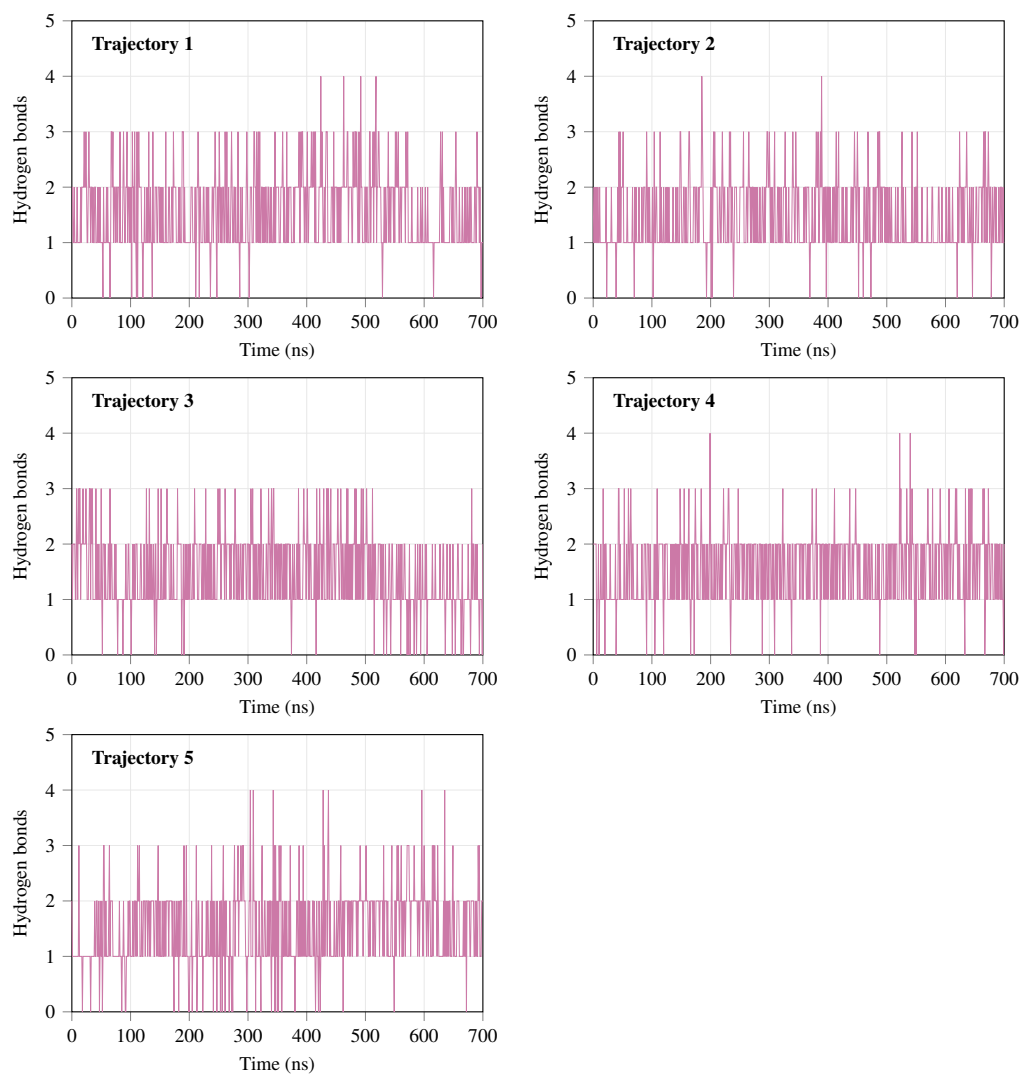


Figure S76. Ligand–protein hydrogen bonds in the **IXf**–HDAC2 complex over 700 ns, shown for each of the five independent trajectories.

S14 Free-energy surfaces and representative binding poses

For each system the distribution of the ligand–protein Coulomb and van der Waals interaction energies over the five 700 ns trajectories is shown as a free-energy surface, beside the representative binding pose taken from the populated region. The colour scale of each surface is set independently. SAHA and **IXf** are reproduced in the main text.

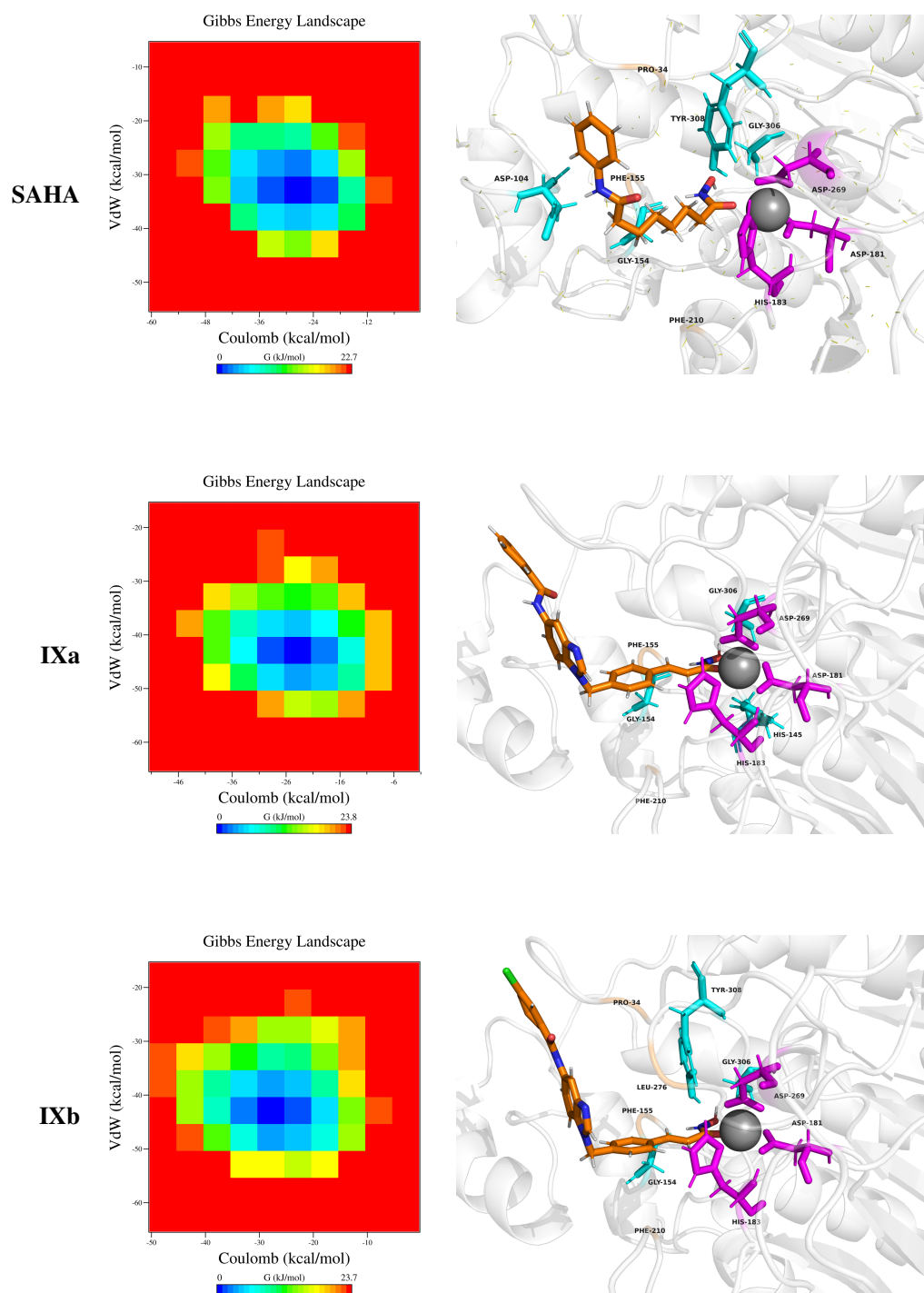


Figure S77. Free-energy surfaces (left) and representative binding poses (right) of SAHA, **IXa** and **IXb** against HDAC2.

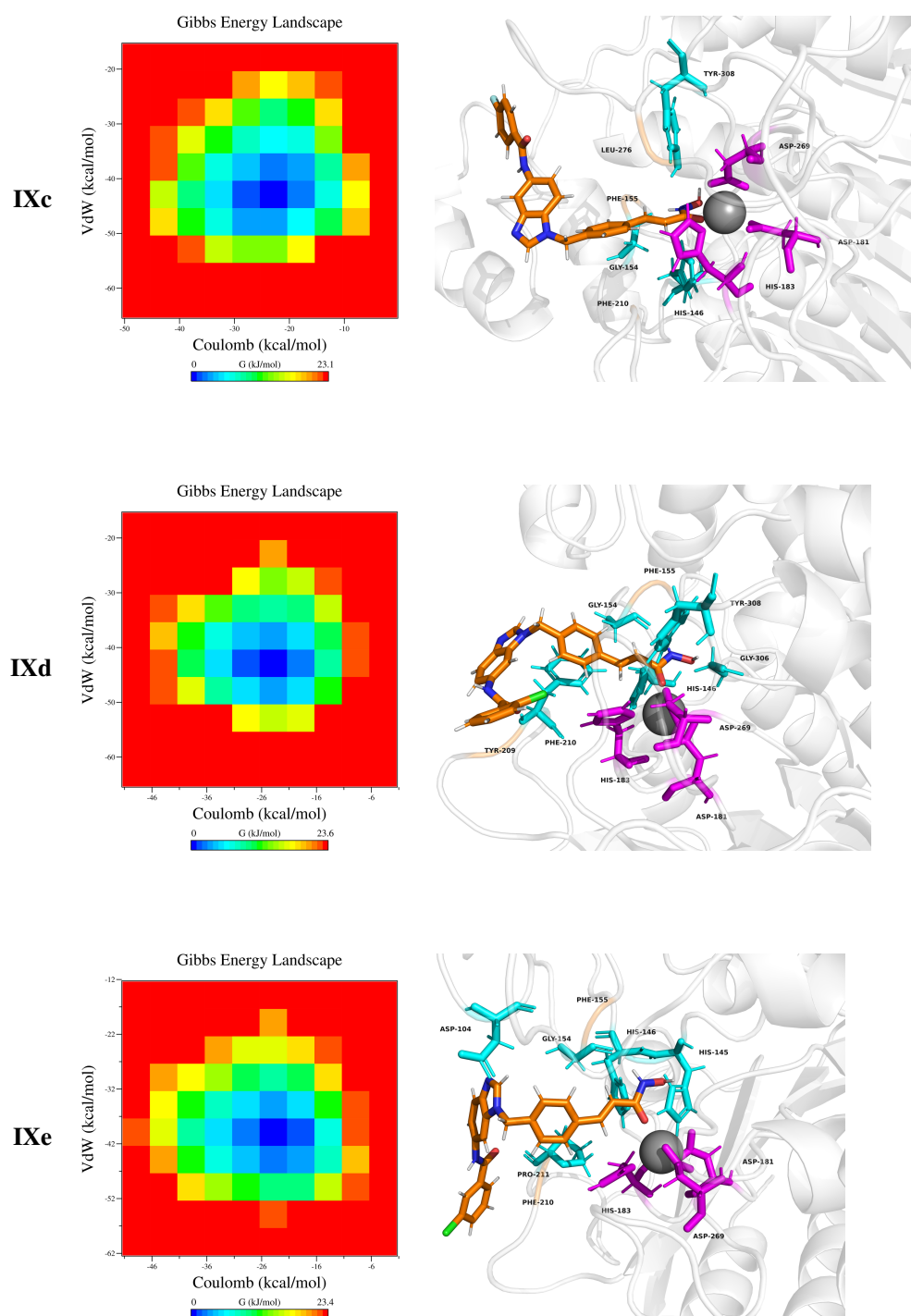


Figure S78. Free-energy surfaces (left) and representative binding poses (right) of **IXc**, **IXd** and **IXe** against HDAC2.

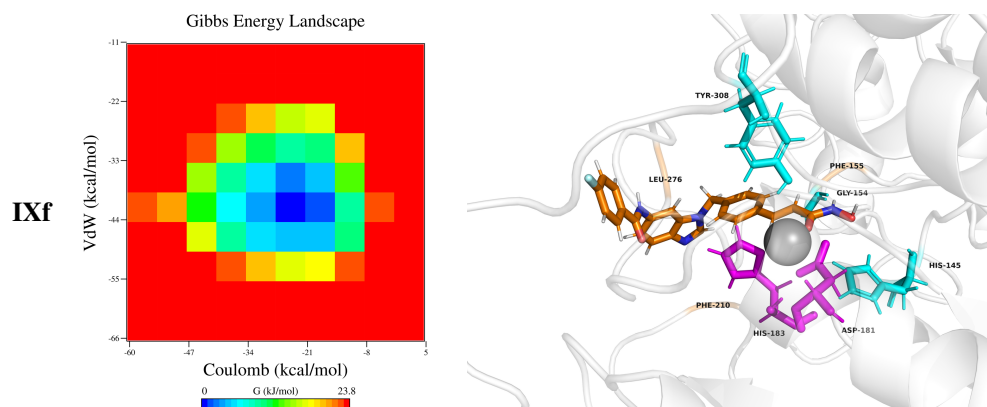


Figure S79. Free-energy surface (left) and representative binding pose (right) of **IXf** against HDAC2.